In-Context Learning for Robots

Methods and Applications

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General-purpose robots must infer what a new task requires and translate that understanding into appropriate physical action. In-context learning (ICL) for robots supports this process by using demonstrations and interaction to direct existing competence with neural parameters held fixed during deployment. We organize this literature review around the interfaces connecting contextual evidence to execution, distinguishing four families: context-conditioned policies, geometric demonstration transfer, world-model-based control, and skill- and agent-based execution. Comparing these interfaces clarifies their transfer assumptions and the roles of training, correspondence, and memory in making context useful. Across manipulation and navigation, we examine how these mechanisms preserve taught requirements as objects, environments, and execution conditions change. This analysis links method design to evaluation practices that distinguish responsiveness to teaching, physical transfer, and benefits from retained experience. The resulting agenda connects compositional task acquisition and faithful transfer with physical recursive self-improvement, in which experience improves the ability to learn subsequent tasks.

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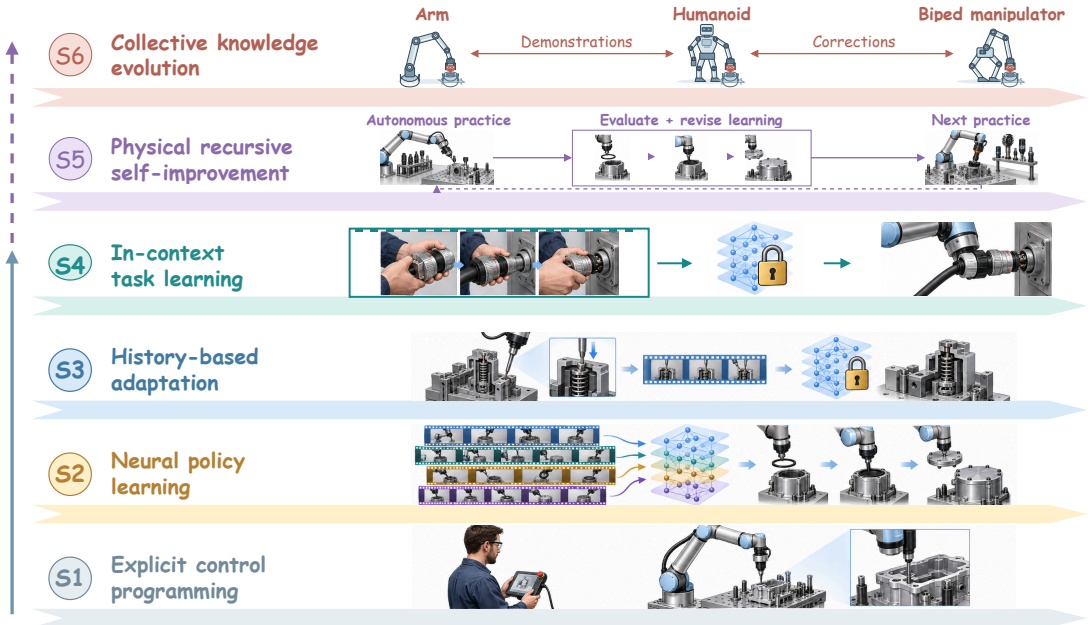

Figure 1 Six learning horizons. S1 encodes control explicitly; S2 learns reusable policies. With neural parameters fixed at deployment, S3 adapts to interaction outcomes, while S4 infers task requirements from teaching. S5 improves how subsequent tasks are learned; S6 extends this process through validated knowledge exchange across embodiments.

We welcome relevant work and corrections via email or GitHub.

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1 Introduction

A robot may know how to move yet still need evidence about what to do. Demonstrations specify a fold or assembly order; corrections revise the procedure. Interaction reveals friction or misalignment. In navigation, earlier visits reveal locations outside the current view [203]. In-context learning (ICL) for robots studies how such evidence changes deployed behavior without another task-specific update to neural parameters.

Broad pretraining supplies reusable perception and control [118–120] from multi-task and multi-embodiment collections [57, 60]. Yet the current scene can leave several procedures feasible, conceal an earlier event, or reveal little about an unfamiliar material's response. These are information gaps that broader motor competence alone cannot resolve. Fine-tuning incorporates new evidence into weights; ICL makes it available at the decision. Cross-object transfer tests whether teaching remains useful after replacing either or both interacting objects [345]. The task relation must survive while its physical realization changes.

Two historical roots motivate a broad notion of context: one-shot imitation infers intended behavior, while meta-reinforcement learning infers tasks or dynamics from outcomes [1, 2]. Sequence models retain trajectories and learning histories [5, 6]; multimodal prompting combines task specifications and sensorimotor examples [13, 100]. With broader priors, teaching can direct generalist actions [98, 99], predicted futures [24], or executable procedures [31]. Physical experiments supply evidence for revising execution [88, 89]. Language participates throughout as instructions, examples, corrections, and retained summaries. Figure 1 situates these capabilities within complementary horizons of adaptation and longer-term learning.

Context use depends on the relationships learned during training. GPT-3 demonstrated few-shot performance after autoregressive pretraining [121]; controlled studies identify distributional conditions that promote using examples [111]. For robots, the critical relationship links earlier teaching or interaction to the later action it changes. Retaining a correction can extend that relationship across attempts [47, 49], provided its conditions remain valid. Broader motor competence, more informative teaching, and selective experience reuse therefore address different limits of adaptation.

Existing surveys organize learning from demonstration by teaching interfaces and learned policies, rewards, or plans [122, 123]. ICL and in-context RL reviews examine adaptation through examples and interaction [124, 125], including the validity of context under environmental change [127]. In robotics, human-video reviews compare the information transferred from observation to control [126]; manipulation ICL reviews organize context content, inference targets, adaptation mechanisms, and transfer [371]. VLM-based VLA reviews distinguish monolithic and hierarchical integration of planning and action generation [372]. Complementary reviews emphasize data and evaluation [129], cross-embodiment adaptation [130], predictive control [128, 178], and the acquisition and improvement of executable skills [369].

Our organizing question is *how new evidence resolves what existing competence leaves undetermined, and how that resolution survives physical execution.* The four families place this burden in different intermediates; correspondence, training, and memory determine whether those intermediates preserve the needed information. This connects the reason for using context to the mechanism and limits of transfer. Table C1 in Appendix C compares related reviews, including recent world-action taxonomies [370].

Placing an object at the correct destination can still violate a required handle grasp. We therefore trace both the intended effect and prescribed order or contact through execution, developing three contributions:

- A taxonomy of context-conditioned policies, geometric demonstration transfer, world-model-based control, and skill- and agent-based execution, with shared correspondence and memory.
- An account of how training relationships establish context use and how object substitution, unfamiliar environments, and execution conditions limit transfer, separating broader motor competence from broader ability to learn through teaching.
- A synthesis of reported comparisons and evaluation controls that separates context dependence, transfer, and retained-experience benefits, motivating compositional learning and improved teachability.

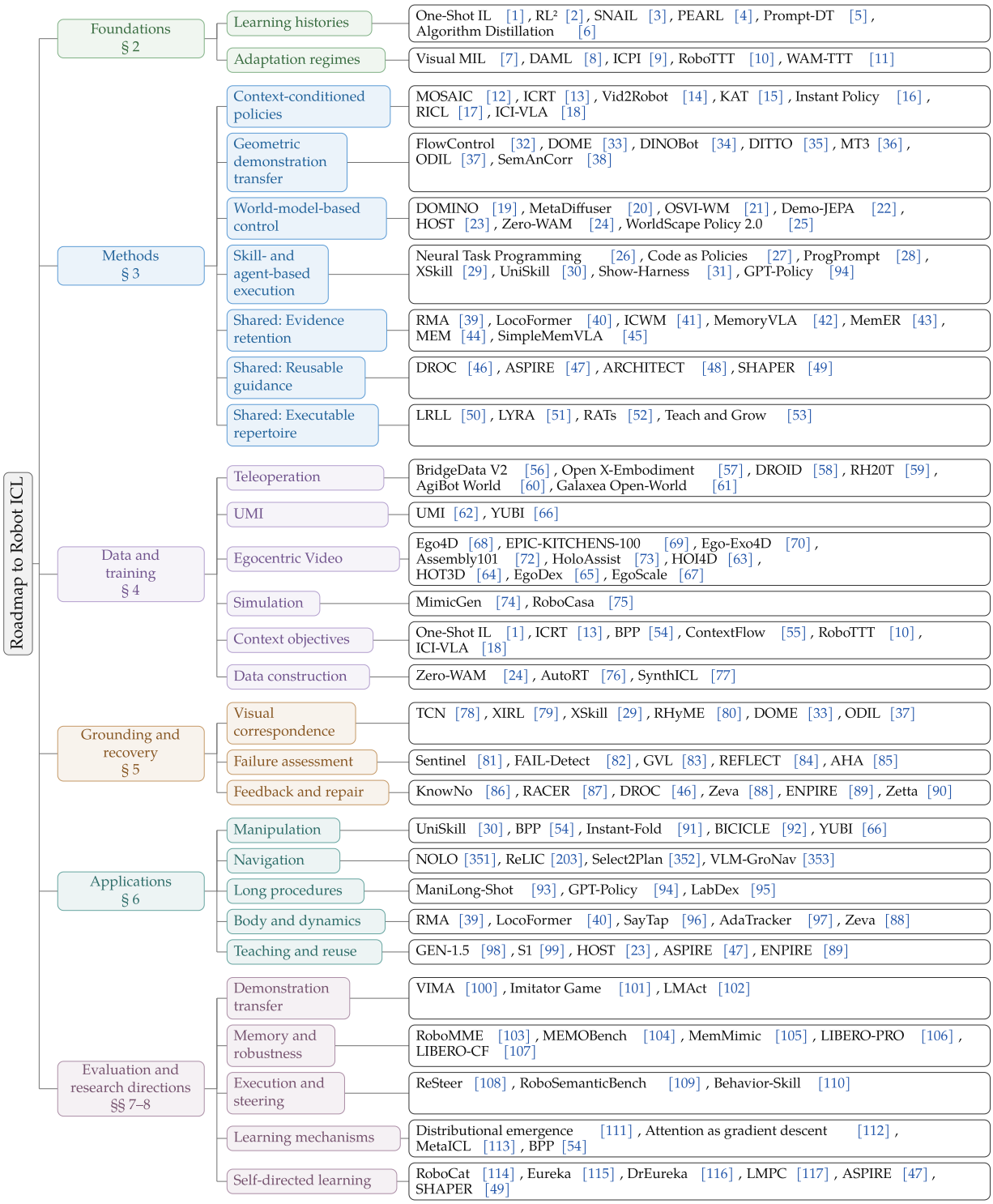

Figure 2 Organization of the literature on ICL for robots. Four method families connect contextual evidence to behavior; the other branches organize foundations, data, grounding, applications, evaluation, and research directions. Correspondence and memory mechanisms connect methods across families.

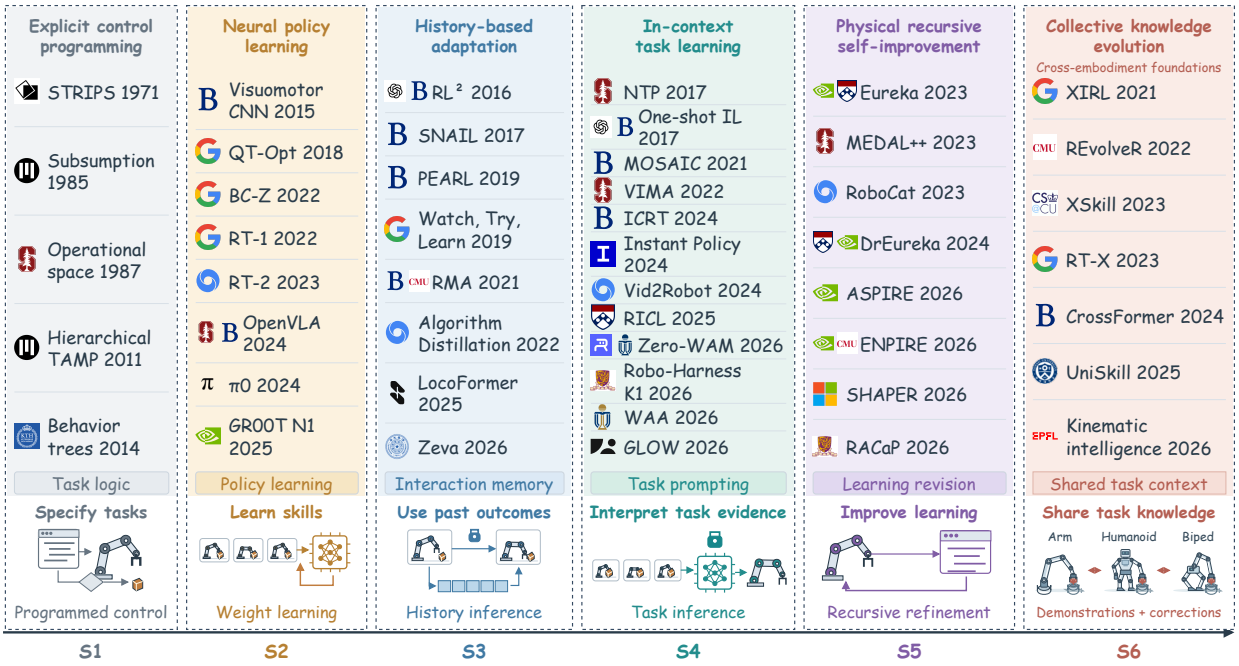

Figure 3 Technical foundations for six learning horizons. S1–S2 establish programmed and learned competence. S3 and S4 adapt execution through interaction history and supplied teaching, respectively, with fixed neural parameters. The S5 and S6 columns collect mechanisms supporting improvements in subsequent learning and knowledge exchange across embodiments, including reusable procedures, shared representations, and policy transfer.

Figure 2 follows this argument: why a robot needs context (Section 2), how context changes action (Section 3), and how that dependence is learned (Section 4). Recovery and applications expose its physical limits (Sections 5–6); evaluation tests its benefit (Section 7); the research agenda asks which limits must be overcome for broader transfer and improved learning (Section 8).

2 Foundations: Why Do Robots Need Context?

A packing robot may already grasp every part yet need a demonstration to establish their order, history to identify completed placements, and contact evidence to accommodate a tighter fit. The same visible scene can therefore require different actions. Useful context resolves this decision-relevant ambiguity through actions the robot can already execute. This distinction motivates the learning horizons, adaptation regimes, and four control interfaces below. Their history traces how robots acquired broader ways to interpret and act on new evidence.

2.1 From competence to contextual adaptation

We use six learning horizons to organize the relationship between prior competence, contextual adaptation, and learning from retained experience. S3 and S4 describe complementary sources of evidence; S5 and S6 express research objectives whose realization can combine several earlier capabilities. Together, the horizons connect distinct adaptation mechanisms to individual and collective learning objectives.

The six horizons distinguish where learning changes the system (Figure 3). Explicit control programming (S1) encodes task logic, control laws, and planning models [134–138]; neural policy learning (S2) acquires reusable behavior through imitation and reinforcement [139, 140]. History-based adaptation (S3) uses interaction memory to infer task or dynamics information. In-context task learning (S4) infers new task requirements from supplied teaching, including instructions, demonstrations, textual examples, and corrections. Both forms of ICL for robots redirect execution with deployed neural parameters fixed.

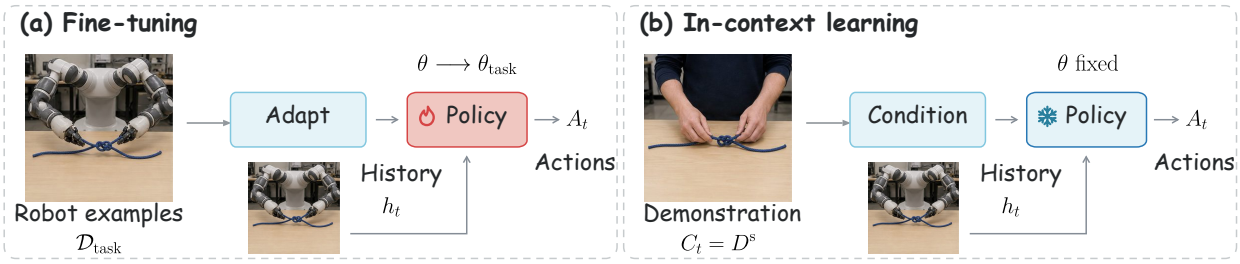

Figure 4 Fine-tuning and in-context learning. (a) Task data $\mathcal{D}_{\text{task}}$ update θ to θ_{task}. (b) A demonstration supplies $C_t = D^{\text{s}}$ with θ fixed. Both policies use observation–action history h_t to predict the next action block A_t.

In assembly, contact outcomes can reveal a tighter fit (S3), while a demonstration or correction can specify a new part order (S4). A retained correction may guide both interpretation and execution; these roles can share a model and input modality.

Automated practice [114, 141], reward and experiment design [89, 115, 116], and retained guidance [47, 49] support a further objective: physical recursive self-improvement (S5), in which experience improves how the next task is learned. Collective knowledge evolution (S6) extends this learning cycle to a group of robots. Its premise is that an execution contains knowledge another body can use, even when the original motion cannot be copied. Shared progress and skill representations preserve what should be accomplished [29, 30, 79]; pooled policy learning supplies competence across sensing and action spaces [57, 132]; morphology-aware transfer changes the physical realization [131, 133]. These are complementary foundations for exchanging demonstrations and corrections. Whether such exchanges improve the group's subsequent learning is the S6 research objective in Figure 1.

Physical self-improvement describes the feedback loop through which interaction changes later behavior. It can operate through retained context, executable programs, or neural updates. This is a cross-cutting process: S3 identifies history-based adaptation, S5 concerns improvements in subsequent learning, and S6 adds knowledge exchange between robots. Section 8.5 compares these update objects and their evaluation.

New evidence changes either policy parameters or inference-time inputs (Figure 4). Fine-tuning uses task data $\mathcal{D}_{\text{task}}$ to update pretrained parameters θ to task-adapted parameters θ_{task}. The illustrated ICL case supplies a support demonstration D^{s} as context C_t, keeping θ fixed. Both use existing perception and motor competence; the storage of task information determines acquisition cost, persistence, and reset behavior.

What context enables a robot to learn. Fixed-parameter adaptation through supplied teaching or interaction is the review's central subject. Parameter-adaptation methods provide comparisons; pretrained policies and control mechanisms supply the competence on which adaptation operates. New evidence can specify a command, identify a familiar routine, compose known relations, or teach an unfamiliar rule. These behaviors differ in the information acquired from context and the prior competence used to realize it. Section 8.1 develops these distinctions. The initial test is whether changing task-defining evidence changes behavior appropriately under matched execution conditions.

2.2 Contextual and parameter adaptation

Adaptation can reside in contextual state, a retained external artifact, or neural parameters. A common policy interface makes these alternatives comparable; their storage and reset behavior determine how an evaluation can identify the source of change. Table 1 collects the shared notation, separating supplied evidence, execution intermediates, and retained state.

At decision step t, the current attempt has history $h_t = (o_1, a_1, \ldots, a_{t-1}, o_t)$. Observation o_t includes the available images, proprioception, force, tactile or inertial measurements, and feedback messages. Action a_t is the applied command; its measured physical effect appears in subsequent observations. Task-relevant context C_t may contain instructions, goal references, multimodal support demonstrations D^{s}, textual input–output examples, corrections, retrieved episodes, or memory of interaction. Superscript s denotes support evidence;

Table 1 Core notation used throughout the survey. Groups follow the argument from contextual evidence to execution, retained knowledge, and evaluation. Superscripts s and q distinguish support and query; other superscripts identify component roles.

Notation	Meaning
	Interaction and contextual evidence (Section 2)
t, n	Decision step; complete-attempt index.
i, j, ℓ, b	Demonstration element; memory record; skill; motion segment.
o_t, a_t, h_t	Observation; executed command; observation–action history.
A_t, $\widehat{a}_{t+\omega\|t}$, H	Proposed action block; action at offset ω; block horizon.
D^{s}, L, d_i^{s}	Support demonstration; its length; its ith element.
C_t, C_t^{task}, r_t	Policy context; supplied task evidence; interpreted representation.
θ, θ_0	Deployed neural parameters; initialization of the adaptation phase.
$\bar{\theta}$, ϕ_t	Frozen backbone; adaptive parameter subset in test-time training.
	Context-to-action interfaces (Sections 3.1–3.4)
π_θ	Conditional distribution of the proposed action block.
E_θ^{c}, E_θ^{m}	Context interpreter; demonstration motion-descriptor extractor.
q_θ^{a}, q_θ^{v}	Action generators conditioned on context or a predicted future.
v^+, p_θ^{v}, p_θ^{d}	Anticipated future; context-conditioned predictor; action-conditioned dynamics.
g, f_θ, p_θ^{g}	Execution specification; deterministic or stochastic specification generator.
κ, k_θ, κ_{B_n}	Deterministic, stochastic, and artifact-dependent executors.
z_ℓ, N_g, ℓ_t	Skill specification; number of skills; active skill index.
ζ_i, $\mathcal{B}$	Motion descriptor; robot kinematics and control description.
	Memory and external revision (Sections 2.2, 3.5–3.6)
U_θ	Update of a self-contained context from an observed action transition.
m_t, M_t, R_t	Recurrent state; external archive; retrieved records.
x_t, ξ_j, J_t, N_t	Executed transition; archive record; selected indices; archive size.
B_n, $\widetilde{B}_{n+1}$	Retained external artifacts; proposed revision.
$\mathcal{T}_n$, χ_n	Attempt trace with feedback; revision-acceptance indicator.
	Training and evaluation (Sections 4, 7)
ψ, τ^{q}, P_{pair}	Task specification; query trajectory; support–query pair distribution.
I^{q}, ℓ_θ, $\mathcal{L}$	Supervised query positions; per-target action loss; expected training loss.
S_n, $\Delta_n^{\mathrm{reuse}}$	Success/adherence rate; gain from retained evidence at attempt n.

the demonstration-only case in Figure 4 sets $C_t = D^{\mathrm{s}}$. Context can summarize the current history or retain information from earlier attempts. The policy proposes an action block $A_t = (\widehat{a}_{t|t}, \ldots, \widehat{a}_{t+H-1|t})$ of horizon H, where $\widehat{a}_{t+\omega|t}$ is the action proposed at decision t for offset ω. With deployed neural parameters θ,

$$A_t \sim \pi_\theta(\cdot \mid h_t, C_t)\,. \tag{1}$$

Only executed actions enter the subsequent history; new observations can replace the unexecuted part of a proposed block. When C_t itself contains the state needed for future decisions, a memory update incorporates an action and its observed consequence:

$$C_{t+1} = U_\theta(C_t, o_t, a_t, o_{t+1}). \tag{2}$$

Equation (2) updates a self-contained context state. For context retrieved or compressed from experience, Equation (17) distinguishes the stored memory from the evidence supplied to the policy.

The policy interface is compatible with several architectures. Throughout the survey, θ denotes the collection of deployed neural parameters, including context interpretation, action generation, and any intervening

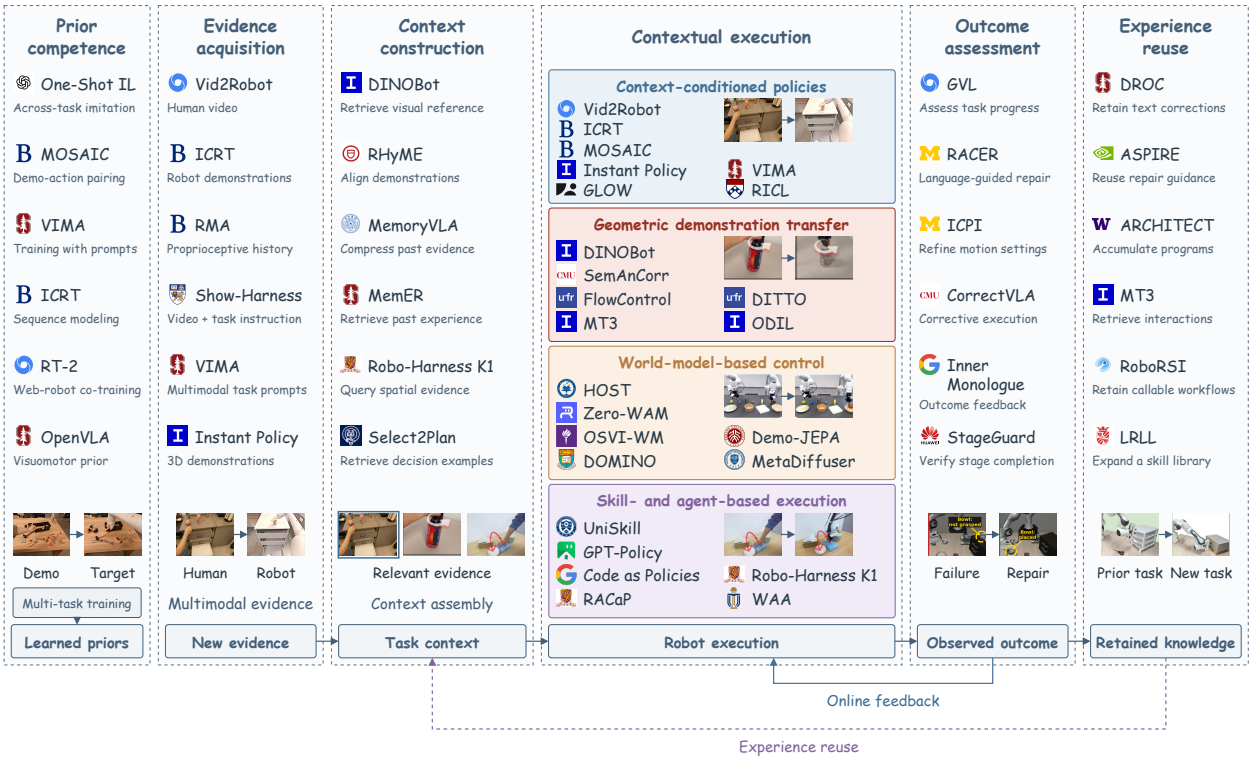

Figure 5 The workflow of ICL for robots. Prior training supplies competence; deployment acquires evidence, constructs context, and generates actions through four method families. Outcome assessment provides online feedback to execution, while retained experience informs subsequent context construction.

learned model. A subscript θ on a component means that it uses the relevant subset of this collection; different components need not share weights. During fixed-parameter adaptation,

$$\theta_{t+1} = \theta_t = \theta_0, \qquad h_{t+1} = (h_t, a_t, o_{t+1}). \tag{3}$$

The subscript 0 denotes the start of the deployment adaptation phase. Observations, retrieved examples, recurrent state, or stored experience change instead. Equations (2) and (3) use one executed action per update; if several actions of A_t are executed before replanning, their observed transitions are incorporated in order. A reward or other feedback signal can be included when available. Updating a neural component at deployment constitutes parameter adaptation, even when the rest of the system stays frozen.

Figure 5 traces a packing task. Prior competence supplies grasps and motions; teaching specifies part order; context construction binds its steps to the scene; execution produces placements; assessment checks them; and reuse retains verified corrections. The four families differ at contextual execution. The surrounding operations determine which evidence reaches it and whether its requirement is physically realized.

A corrected order changes the required actions; recorded placements identify the remaining work. Both can affect a fixed policy, but the retained information and its reset rules determine how long the change persists.

A fixed model can revise a goal, skill sequence, or program after feedback; an executor turns that specification into actions. For example, a language model can use prior errors to adjust a motion primitive's settings [9]. Retaining the revised program lets later attempts reuse the correction without training neural weights. This stores task information in executable content, whereas fine-tuning stores it in model parameters.

The surrounding execution system, or *harness*, assembles context, exposes robot tools, executes requests, and returns feedback. Normally, this machinery remains fixed while the supplied evidence and generated requests change. A learning system can also revise its retrieval rules, context-construction code, or executable controller. We call this *model-external adaptation*: experience changes the machinery connecting a model to

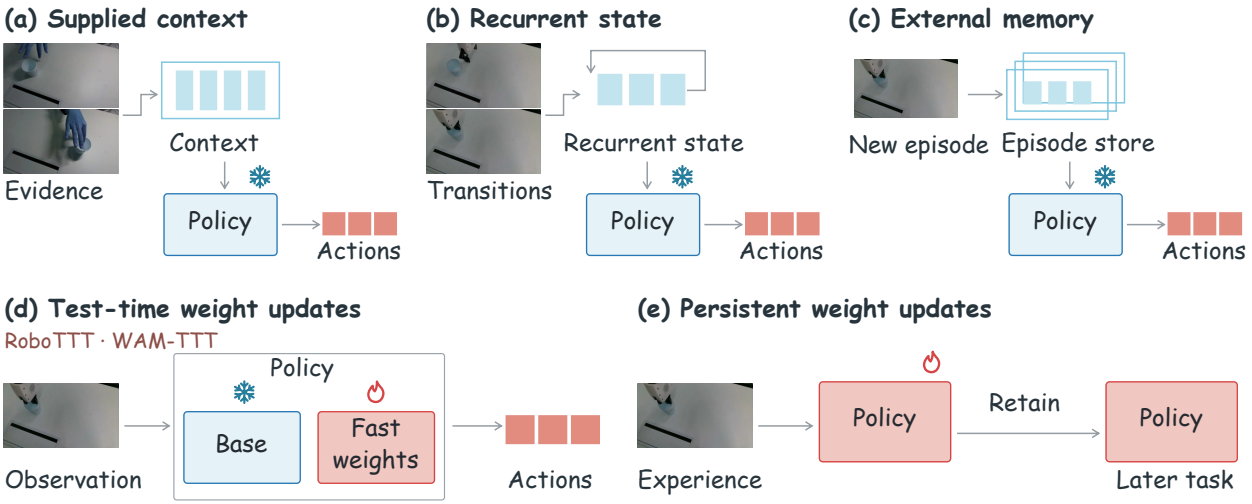

Figure 6 Five locations and timescales of adaptation. In (a–c), supplied context, recurrent state, or retrieved episodes condition a policy with fixed neural parameters. In (d), fast weights change while the base remains fixed; (e) retains updated weights for later tasks. Blue borders and snowflakes mark fixed parameters; red borders mark updated parameters, with flames indicating the update stage.

the robot. Identifying this revised component distinguishes changes in the execution system from changes in the neural model.

Figure 6 separates supplied evidence (a), recurrent state (b), and retrieved episodes (c) from fast-weight adaptation (d) and persistent weight updates (e). Storage and retention timescale are independent: external memory can persist without training, while test-time updates can be reset or retained. Continual learning examines acquisition and retention across changing experience [143].

These distinctions determine the evaluation controls. Removing supplied evidence tests (a); clearing recurrent state tests (b); suppressing episode retrieval tests (c). Restoring pre-adaptation weights tests the contribution of parameter updates in (d–e). Section 7 develops these interventions under matched task conditions, so gains can be attributed to the information or parameters that actually changed.

One-shot specifies the amount of teaching, not its storage mechanism. Visual meta-imitation uses one demonstration for task-specific gradient updates [7]. RoboTTT restricts updates to a neural subset during long-context execution [10]; WAM-TTT adapts video-side fast weights from target-domain human video before rollout [11]. Both follow the selective-update principle in (d), despite different update timing. Thus one demonstration can support either fixed-weight conditioning or parameter adaptation. The next taxonomy distinguishes how the resulting context reaches robot actions.

2.3 Four method families and shared design dimensions

The primary taxonomy follows the *deployed path from context to action*. The four families below are alternative computational interfaces, distinguished by the intermediate that execution consumes. We then separate this classification from the cross-cutting roles, modalities, and storage of evidence, before resolving boundaries between hybrid implementations.

- **Context-conditioned policies** infer actions from task evidence, including demonstrations, language, and interaction history as supported, through a learned generator or local action retrieval.
- **Geometric demonstration transfer** reconstructs a demonstrated motion or contact reference, aligns or retargets it to the current scene, and executes that reference through tracking or replay.
- **World-model-based control** predicts future observations, states, or transitions and uses those predictions to decode, plan, or select actions.

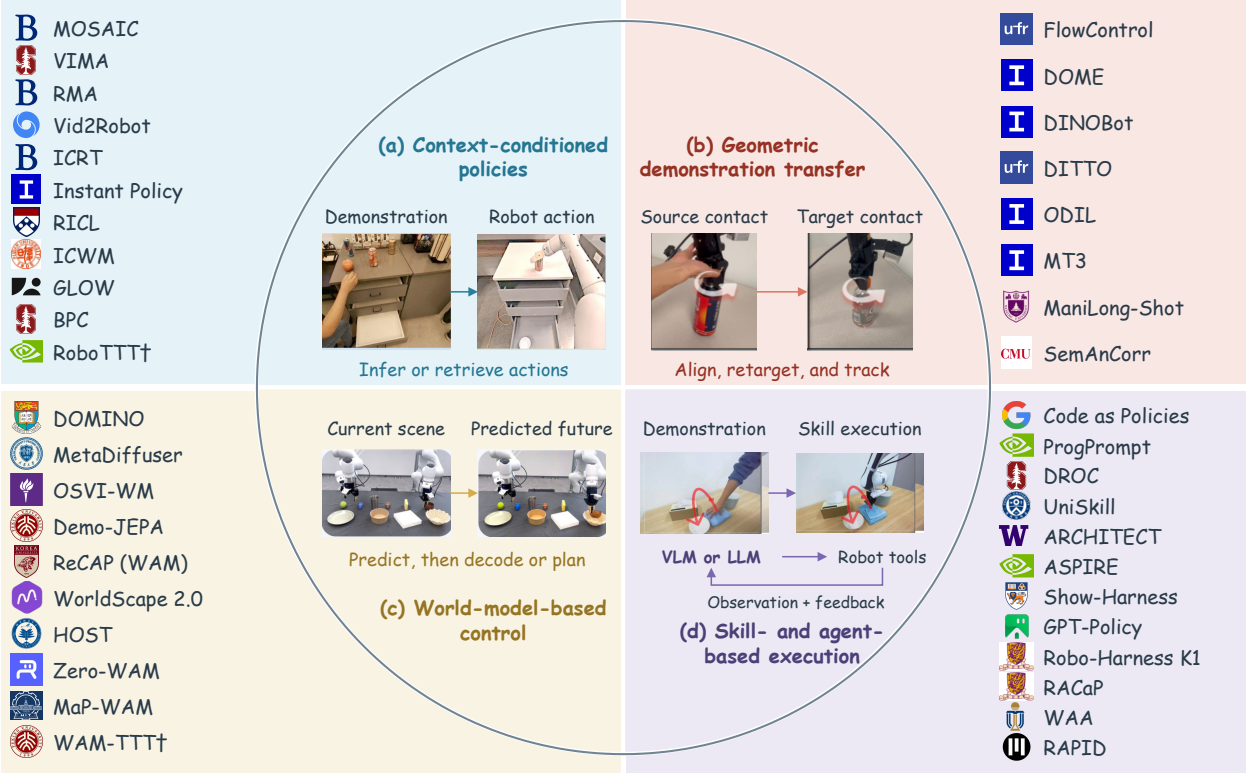

Figure 7 Four context-to-control mechanisms: (a) action inference, (b) motion transfer, (c) future prediction, and (d) skill/tool selection. Surrounding works locate representative methods; † marks fast-weight adaptation in RoboTTT and WAM-TTT [10, 11].

- **Skill- and agent-based execution** infers skill choices, programs, or robot-tool requests for a separate executor; VLM and LLM agents form one branch.

These interfaces extend the mappings, system models, and plans distinguished in learning from demonstration [122], independently of the representation, optimization, and training objective [144]. Transformers, diffusion models, and VLMs can each implement several interfaces; hybrids can combine predicted futures with skills or programs with reference tracking.

Four information roles explain what this context contributes. *Task specification* identifies the intended goal, relation, or procedure. *Correspondence* relates demonstrated objects, contacts, phases, or commands to their executable counterparts. *Physical response* describes how the current body and environment respond to action. *Execution state* records the scene, relevant hidden events, and completed prerequisites. A demonstration can specify a procedure and its contact geometry; an interaction can update both physical response and execution state. These roles guide evidence selection within every method family.

Language, images, and sensor records can carry the same information roles. Modality describes the encoding; the learning mechanism explains how its information changes a decision. Numerical states serialized as text retain their physical meaning.

Acquisition describes collection through teleoperation, UMI, egocentric video, or simulation (Section 4.1); sensorimotor channels describe recorded contents. Language can accompany any source. Collection choices and neural update rules remain independent of the control family and application demand.

The same demonstration can condition actions, define a motion reference, constrain a future, or specify a skill request (Figure 7). The intermediate required by execution distinguishes the families; acquisition, retrieval, and storage can support several interfaces.

Table 2 Two levels of the taxonomy. (A) Four method families differ in their execution intermediate and a key requirement for transfer. (B) Shared mechanisms establish correspondence and manage evidence within or across these interfaces. A system can combine several interfaces and shared mechanisms.

A. Control interfaces: how context determines actions		
Method family	**Intermediate**	**Transfer requirement**
Context-conditioned policies	Action distribution	Action inference preserves the taught distinction
Geometric demonstration transfer	Motion or contact reference	Matched interaction remains applicable
World-model-based control	Predicted consequences	Predicted evolution is realizable
Skill- and agent-based execution	Execution specification	Available skills preserve task constraints
B. Shared mechanisms: how evidence supports those interfaces		
Mechanism	**Information resolved**	**Example**
Geometric and functional correspondence	Where to interact	Handle contact on a new object
Semantic correspondence	Which roles and order	Cup as source; bowl as receiver
Temporal alignment	Which demonstration phase	Current grasp phase
Retrieval	Which experience applies	Relevant past demonstration
Evidence retention	What remains available	Earlier observation or outcome
External knowledge reuse	What guides later attempts	Repair guidance or reusable skill

Storage adds another independent choice. A context window, recurrent state m_t, or external archive M_t can supply the same task information with different access costs and persistence. Static API documentation establishes the command interface; task-specific examples and interaction supply changing evidence. For later comparisons with test-time training, write $\theta_t = (\bar{\theta}, \phi_t)$: $\bar{\theta}$ is the frozen backbone and ϕ_t the adaptive neural subset. This follows the update-regime distinction in Section 2.2.

The boundary between policy conditioning and geometric transfer is the object being executed. Matching points to condition a diffusion policy remains context-conditioned policy inference; transferring a recorded object-relative trajectory and tracking it is geometric demonstration transfer. Similarly, estimating current dynamics for an action policy differs from predicting future consequences for control. RMA and the physical-context inference in ICWM illustrate the former [39, 41]; DOMINO and Zero-WAM illustrate the latter [19, 24]. An auxiliary future-prediction head contributes to training; a deployed future predictor contributes to world-model-based control when its output guides action generation or selection.

A shared VLM can support different interfaces. Motor-action tokens implement policy inference; robot-tool selection, arguments, and returned feedback implement agent-based execution. GPT-Policy's Astra configuration follows the latter route [94], using demonstrations and interaction to determine its calls.

Memory connects these interfaces across time. Alongside its form, function, and dynamics [145], a retained fact carries physical conditions that determine when it remains applicable to the current robot and scene. Section 3.5 examines access and update mechanisms, and Section 5 examines how execution validates or revises retained evidence.

Intent adaptation changes the desired relation; *physical adaptation* changes its realization under new dynamics or embodiment. *Joint adaptation* requires both. Navigation additionally needs spatial knowledge even when the goal and dynamics remain unchanged. Section 6 organizes applications by these information needs and their implications for long procedures and task changeover. Any of the four computational families can serve several demands.

Table 2 connects each intermediate to its transfer assumption and shared mechanisms. A handle match can condition action inference or define a tracked motion reference. The execution interface determines which assumption must hold for the final motion.

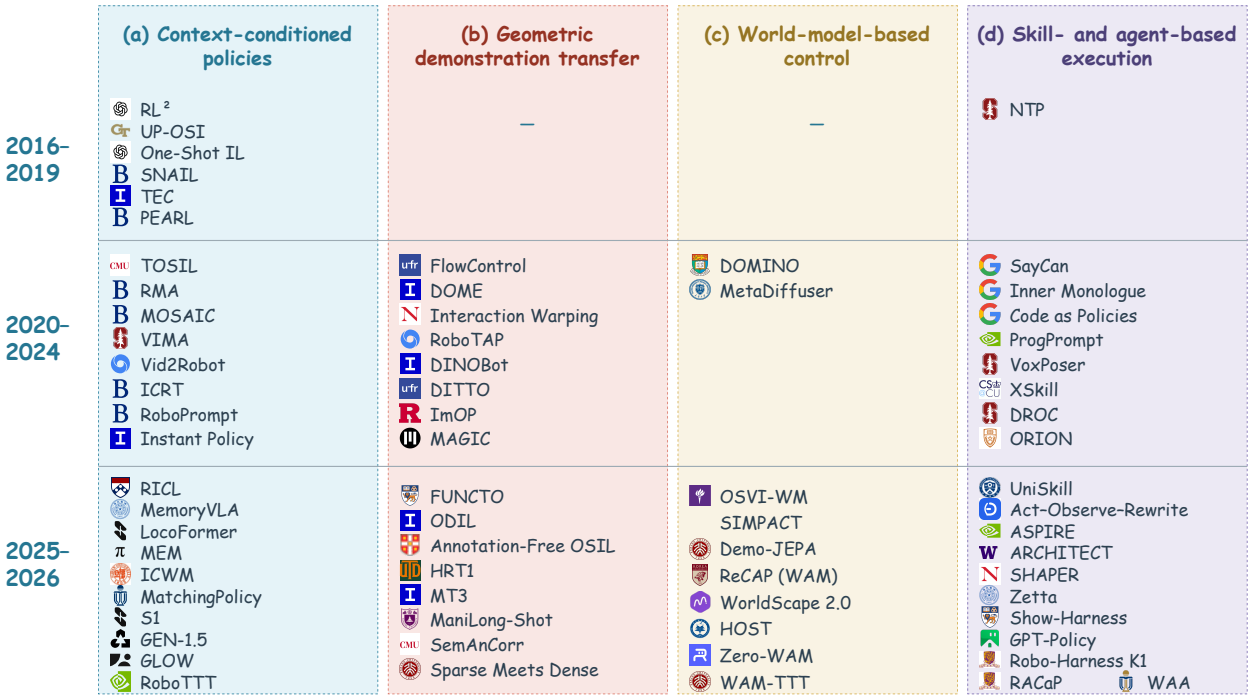

Figure 8 Evolution of the four method families by first public release period. NTP: Neural Task Programming.

2.4 Historical development

The historical account follows an expansion in what must be inferred from context. Task-family adaptation learned reusable responses to bounded variation; sequence interfaces made more of the preceding experience accessible; broader priors increased the behaviors that new teaching could direct. These developments overlap across the four families. The final publication census locates their recent growth within the literature reviewed here.

Task-family adaptation. Early contextual policies amortized adaptation across a task distribution: training learned how to interpret evidence so that deployment could change behavior without relearning the entire controller. In one-shot imitation, paired executions taught how one instance specifies actions in another [1]. In meta-RL, recurrent state carried adaptation across episodes [2]; probabilistic task inference separated evidence interpretation from control [4] and made task uncertainty available to action selection [333]. The resulting adaptation rule inherits the distinctions made available during training. Extending it to human demonstrations requires correspondence across bodies [8]; extending it to changed robot dynamics requires evidence about action response [39]. These extensions concern different unknowns, despite sharing rapid adaptation as their objective.

The four branches in Figure 8 respond to different obstacles in reusing experience. Context-conditioned policies learn which actions an example implies; geometric transfer identifies a relation under which a recorded interaction remains applicable. Predictive control makes consequences available before motion, while skill and program interfaces separate a new procedure from its motor realization. Their histories overlap because the obstacles coexist. Broader priors expand the available behaviors, but also make evidence selection and retention more consequential: a capable controller still needs to follow the current teaching rather than a familiar default.

Online system identification provided an explicit early version of the physical-adaptation idea: recent state–action history identified dynamics parameters for a universal policy [146]. Later methods could learn a latent instead of naming every parameter, but the central problem remained inferring how the present body would respond. The demonstration and interaction strands were already connected by Watch–Try–Learn: one policy attempted a demonstrated task and another used the resulting feedback [147]. A demonstration narrows the task hypothesis, while interaction can resolve information it omits.

Reusable sequence interfaces. A sequence interface addresses a tension between compressing experience and revisiting a decisive event. SNAIL combined temporal convolutions, which aggregate the preceding sequence, with attention, which retrieves relevant earlier information [3]. This made adaptation depend on both accumulated state and selective access. Transformers provide query-dependent access through attention [231]. Retaining more history increases the available evidence; attention and downstream computation determine which evidence informs the current decision.

Broader teaching made a fixed task summary less sufficient: two demonstrations can have the same goal yet differ in order, contact, or tool use. Multimodal task specifications placed goals and procedures in the input [100, 148], while trajectory models retained earlier observations and actions as evidence for later decisions [5, 6]. The benefit of a sequence interface was access to distinctions that a single task label could erase. Its cost was an alignment problem: the robot had to recover which part of that evidence mattered at its own execution stage. Temporal video correspondence and causal sensorimotor modeling addressed this pressure through learned representations [13, 14]; spatial representations imposed geometric correspondence before action prediction [16]. These complementary approaches preserve temporal evidence and geometric correspondence at different stages of action generation.

Adaptation over broad priors. Broad pretraining changes what adaptation must accomplish. When perception and a useful range of motion are already available, a new example can specialize an action model [17] or constrain an anticipated robot-domain future [24], rather than supply every ingredient of a new skill. Interaction histories similarly let reusable controllers account for a deployed body's response [40, 88]. Broader pretraining therefore expands the behaviors available to teaching, but also strengthens familiar continuations that may conflict with a new example. Evaluation must establish whether the example can redirect this competence.

Geometric reuse and learned contextual inference allocate generalization differently. Visual nearest-neighbor methods rely on a representation in which nearby observations admit compatible actions [149]. Explicit alignment instead restores the conditions under which a recorded robot motion or human-demonstrated object trajectory remains usable [34, 35]. When those conditions are insufficient, a video can initialize further interaction learning, as in WHIRL [150]. The same one-example interface can therefore support replay, transformed reference execution, or new policy learning. What distinguishes these routes is the assumption that makes the demonstrated behavior executable in the new setting.

World-model-based control exposes the consequences of behavior before executing it. Contextual dynamics models first made recent transitions useful for evaluating candidate action sequences [19, 20]. Demonstration-conditioned models then added a second role for context: specifying which future should be realized [21, 24]. This development joins physical prediction with task interpretation. The controller must preserve the procedure inferred from the example while predicting how it can unfold in the robot's current scene.

Task structure offers a different way to reuse competence: preserve the procedure while delegating its physical realization. Demonstration-derived hierarchies express a new task through existing routines [26]. Language-model programming expands the procedure to include perception calls, arithmetic, and feedback logic [27]; video-derived outlines let visual teaching specify that procedure for an existing planner and executor [31]. The gain is compositional reuse of capabilities. Its condition is that the intermediate preserve the task's binding details, because the executor cannot recover a contact or ordering requirement that was omitted during interpretation. Section 3 develops this trade-off across the four interfaces.

These developments connect teaching, interaction, and procedural reuse. General-Level distinguishes breadth from synergy in multimodal generalists [151]; for robots, the corresponding learning question is whether combining these capabilities improves subsequent acquisition. LMPC studies responsiveness to later teaching [117], while reward and learning-program search change how new skills are acquired [89, 115]. Section 8.5 develops this question for individual and shared learning.

Review scope and source selection. This review uses a mechanism-oriented narrative synthesis with targeted updates through 12 October 2026. Searches and citation chaining span visual imitation, contextual meta-RL, navigation from video and experience, system identification, example-conditioned programming, interaction memory, and robot foundation models. Sources include arXiv, official proceedings and journal

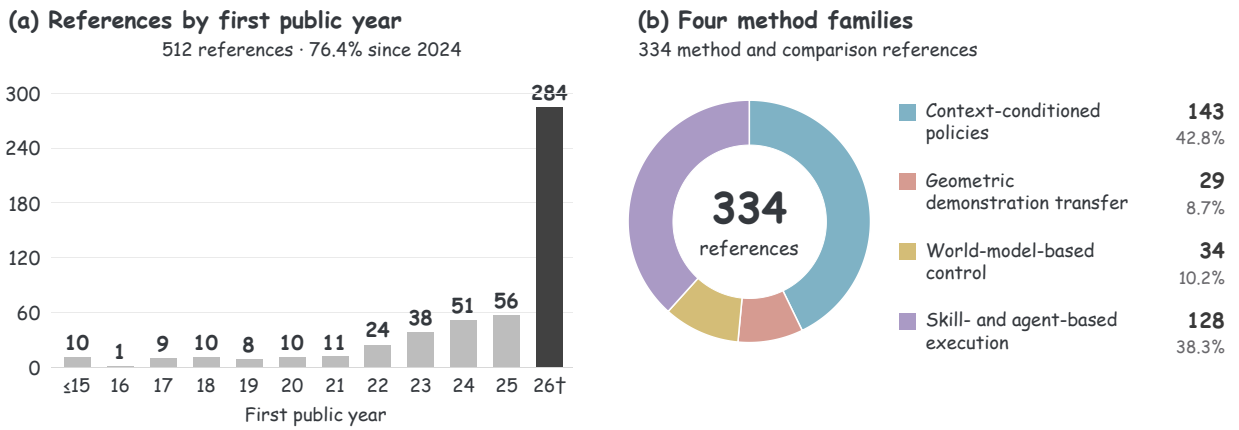

Figure 9 Composition of the reviewed literature. (a) First-publication years for all 512 references. (b) The four-family distribution of 334 method and comparison references. Percentages in (b) use this subset, which includes supporting mechanisms and parameter-adaptation comparisons. The first bin covers publications through 2015; † marks the partial year ending 12 October 2026.

articles, and author or organization releases. Technical claims are traced to these primary sources; release reports are identified as such where their evidence differs from a research paper.

Included works explain a route from new evidence to behavior or supply relevant data, evaluation, training, or theory. Fixed-weight contextual methods are central; parameter adaptation and prior competence provide comparisons. Versions share one entry, preferably the official publication; separately cited papers and dataset releases remain distinct. Hybrid families follow the analyzed deployment branch, independently of modality, storage, and parameter updates.

The reference inventory in Appendix A records this coding and the first public year, using arXiv v1 or an official release. Supplement S1 provides the item-level records and counting rules. The resulting counts describe the composition and publication history of this reviewed corpus.

Publication patterns. The 512 references include 391 first released since 2024 and 284 in the partial 2026 window. The concentration coincides with the recent expansion of generalist policies, multimodal teaching, and embodied agents. Figure 9 separates this complete reference set from the 334 method and comparison references; the latter includes supporting control mechanisms and parameter-adaptation comparisons.

Within the method subset, 143 references concern context-conditioned policies, 29 geometric transfer, 34 world-model-based control, and 128 skill- and agent-based execution. Branch-specific coding places ReCAP's Cosmos Policy branch under predictive world-action modeling [152], ENPIRE under program revision [89], and GPT-Policy under contextual tool use [94].

The recent concentration of publications and the distribution across control interfaces in Figure 9 motivate a closer comparison of how methods use context. The following chapter examines their execution intermediates and the assumptions that support transfer.

Takeaways

① Context resolves what the current observation and prior competence leave undetermined: the required task, relevant history, or physical response. Its value depends on whether that information changes a physically executable decision.

② Demonstration following and physical adaptation have distinct historical roots. Their convergence joins inference of a new task with inference of an unfamiliar body's response.

3 Methods: How Does Context Change Action?

Context reaches action through an action distribution, motion reference, predicted consequence, or skill/tool specification. We compare what each preserves, why existing competence can execute it, and which changes

invalidate that assumption (Table 2). Correspondence binds evidence, memory retains it, and external revision carries verified lessons into later attempts.

The factorizations use C_t, h_t, A_t, and fixed parameters θ from Section 2, exposing the intermediate consumed by stochastic or deterministic execution. Table 1 collects the notation; additional symbols are defined locally.

A shared primitive vocabulary. Tables 3, 4, and 5 use one primitive taxonomy. *Source* codes demonstrations or examples (D), retained history (H), instructions or goals (I), explicit feedback (F), and scene or map evidence (E). *Form* codes visual observations or features (V), state/action records (S), geometry (G), and linguistic or symbolic records (L). *Context operation* separates nine operations: *Retrieve* selects stored records; *Attend* makes a query-dependent neural read; *Align* establishes spatial or temporal correspondence; *Encode* builds a learned representation; *Estimate* infers hidden physical response; *Parse* constructs symbolic bindings or procedures; *Verify* checks candidates against feedback or constraints; *Simulate* propagates candidate actions through a scene model; and *Fit* solves query-specific coefficients or geometry with network weights fixed. The tables code task-relevant contextual evidence and the principal operations that use it during deployment. Commas list evidence types; slashes combine operations or carriers without specifying their order.

The resulting *carrier* is an action output, motion reference, predicted future, learned skill, program/procedure, or tool/controller call. A carrier identifies what passes between components; the family identifies its role in the complete control path. *Execute* then distinguishes six operations: *Replay* returns recorded controls; *Combine* blends retrieved action continuations; *Decode* generates actions with a learned model; *Track* follows an explicit reference; *Plan* searches for an executable realization; and *Invoke* runs a program or calls a tool.

Evidence and execution roles. Language contributes according to the information it carries, independently of the control interface. In direct action inference, numerical states and action examples can be serialized for a language model, as in RoboPrompt [153]; ICRT learns its sensorimotor interface without linguistic data [13]. For a geometric archive, language can narrow the intended task before geometric matching resolves the motion, as in MT3 [36]. Command–program examples instead teach an executable mapping [27], while textual event memory preserves information missing from recent visual history [44]. These uses resolve task selection, action interpretation, and retention at different stages. The relevant comparison is what the text contributes to the decision, including whether it supplies a goal, a new convention, or a remembered outcome.

The separation between information and execution also applies to geometry. Matching can condition a learned action generator or supply a retargeted reference to a separate motion executor. Table B1 in Appendix B (p. 97) compares these roles. Geometry can instead supervise further learning: using hand–object motion to guide residual reinforcement learning trains a new policy [188]. The role of geometry becomes clear by identifying what it changes: a policy input, an executable reference, or the policy parameters.

Semantic abstraction preserves roles, prerequisites, and skill order while allowing grasps or paths to change [31, 165]. Execution must receive each binding constraint: plate-first teaching specifies order; a handle grasp additionally restricts contact. A generic command can erase either constraint; the analysis therefore follows how teaching reaches execution.

3.1 Context-conditioned policies

Context-conditioned policies map task evidence and robot history to actions. They either reuse recorded actions selected by correspondence or generate actions from an interpreted context representation. We compare these two forms before examining three complementary demands on the representation: execution phase, spatial correspondence, and physical response. The final comparison considers how pretrained competence affects the evidence each form needs.

For a context-conditioned policy, write r_t for the representation of the evidence used at the current decision:

$$r_t = E^{\mathrm{c}}_\theta(h_t, C_t), \qquad \pi_\theta(A_t \mid h_t, C_t) = q^{\mathrm{a}}_\theta(A_t \mid h_t, r_t). \tag{4}$$

Here E^{c}_{θ} interprets context and q^{a}_{θ} generates actions from its representation r_t; both may belong to one network. The factorization requires r_t to retain the context information used by action generation. Autoregressive, diffusion, and flow-based generators can implement this interface. If an example teaches red-then-blue packing, the representation must preserve that order while the generator adapts the reach-and-place motion to the current scene.

Reusing actions through correspondence. Let the nonempty set J_t index the demonstration records selected for the current query, and let $\bar{A}_j$ be the action block stored in record j. A nonparametric action-reuse policy has the form

$$\pi_\theta(\cdot \mid h_t, C_t) = \sum_{j \in J_t} \beta_{tj}\, \delta_{\bar{A}_j}, \qquad \beta_{tj} \geq 0, \quad \sum_{j \in J_t} \beta_{tj} = 1. \tag{5}$$

The query-dependent weights β_{tj} mix point masses $\delta_{\bar{A}_j}$ at stored blocks. Nearest-neighbor replay gives one record weight one. Returning their mean, $A_t = \sum_j \beta_{tj} \bar{A}_j$, instead combines continuous controls deterministically, requiring compatible coordinates and separate treatment of discrete commands. All blocks must match the receiving robot's interface. A learned generator in Equation (4) transforms retrieved evidence; replay directly returns its recorded actions.

Action reuse assumes that proximity in a representation implies compatible control. The workflow encodes the current scene, retrieves comparable demonstration states, and returns their recorded actions [149]. Its generalization rests on representation quality and archive coverage. A retrieved demonstration can instead condition a learned generator [17], which must infer how the example applies rather than return its action directly. This introduces a second source of generalization: transforming available experience beyond local replay.

Archive coverage and interpretation can fail separately. Under unseen MuJoCo embodiments, REGENT's Retrieve-and-Play exceeds its frozen contextual transformer, while fine-tuning improves the learned agent [154]. A RICL retrieval study further distinguishes similarity from closed-loop utility [481]: on 25 tasks selected for prior RICL success, visual and VLA-based retrieval outperform random context, but their mutual differences are not statistically resolved. A useful retrieval metric must select evidence that remains actionable along the policy's own rollout, beyond matching expert-trajectory states.

Local continuation bridges replay and learned generation. Behavior Predictive Control (BPC) reconstructs the live observation–action history from retrieved trajectory windows, then applies the same coefficients to their action continuations [389]. Unlike the nonnegative mixture in Equation (5), its regularized coefficients may be signed, permitting local extrapolation. The retrieval metric and residual correction are fitted offline; deployment solves the continuation coefficients from the current history while keeping those fitted components fixed. Reliable continuation depends on whether nearby records jointly explain the query with stable coefficients, connecting retrieval quality to local control compatibility.

The use of correspondence determines which component carries motion generalization. A retargeted human motion can serve as a proposal that a pretrained diffusion policy refines, as in DemoDiffusion [155]. The learned prior then contributes to the final motion. Alternatively, registration can directly transfer the next demonstrated end-effector pose, as in IMOP [156]. In that case, geometric applicability carries more of the transfer burden. Section 3.2 develops this second interface, where the matched evidence becomes an explicit reference for execution.

Figure 10a separates relevance, execution phase, and physical response. Panel b adds task specification: language identifies an operation and visual prompts bind its spatial target. These operations construct the evidence that enters the four control interfaces in Figure 12.

From task identity to execution correspondence. When retrieved actions do not transfer directly, the learner must interpret what the example specifies. Early demonstration-conditioned policies compressed examples into a task representation. Task-Embedded Control Networks jointly learned a demonstration metric and an embedding-conditioned policy [157]. Such a summary makes few-shot conditioning economical, but the policy must recover the relevant execution phase from its current observation. As variation extends from task identity to ordering, viewpoint, and object placement, identifying the task is no longer enough: the robot must relate what it sees now to a particular part of the example.

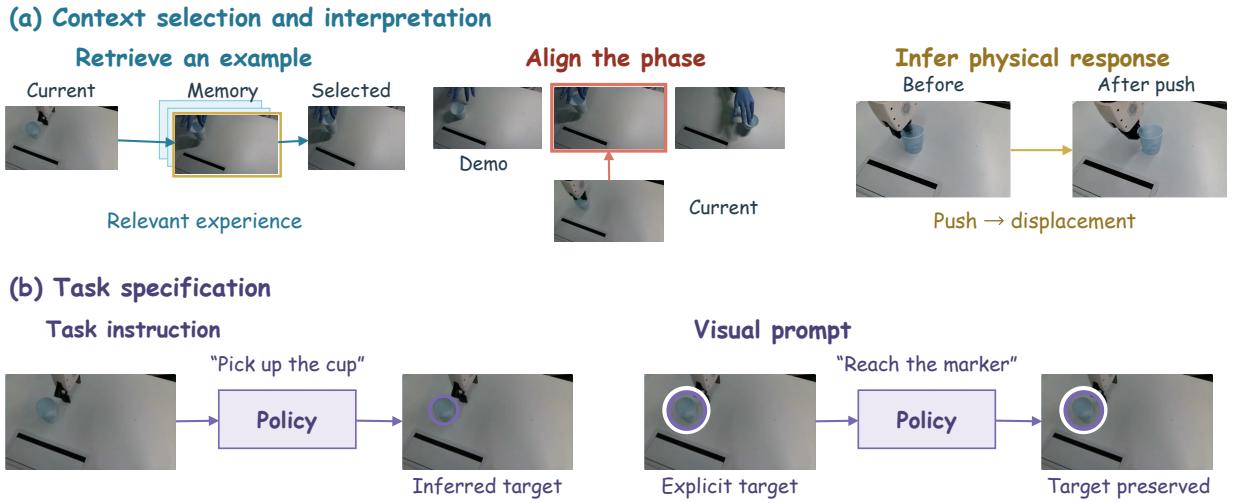

Figure 10 Constructing and interpreting task context. (a) Retrieval selects relevant experience, temporal alignment identifies the current phase, and observed action effects inform physical response. (b) Language specifies an operation, while a visual prompt makes its spatial target explicit. These operations supply task evidence to the execution interfaces compared in Figure 12.

Table 3 traces context operations into action rules. Retrieval can return actions or condition a generator; alignment can condition inference or construct a tracked reference.

An attention-based interpreter implements a query-dependent read over demonstration elements. Let $D^{\mathrm{s}} = (d_i^{\mathrm{s}})_{i=1}^{L}$ be a support demonstration with L elements. Its ith element d_i^{s} may contain an image or a synchronized observation–action record; k is a summation index over the same elements. One representative alignment is

$$\alpha_{ti} = \frac{\exp s_\theta(h_t, d_i^{\mathrm{s}})}{\sum_{k=1}^{L} \exp s_\theta(h_t, d_k^{\mathrm{s}})}, \qquad r_t = \sum_{i=1}^{L} \alpha_{ti} e_\theta(d_i^{\mathrm{s}}). \tag{6}$$

The score s_θ matches the robot history to an encoded demonstration element $e_\theta(d_i^{\mathrm{s}})$. Normalized weights α_{ti} select evidence relevant at decision t, allowing the read to change with execution progress. This inference-time read supports image-only and action-bearing demonstrations alike; its training objective is a separate choice considered in Section 4.2. Richer attention or geometric representations can preserve several relationships without imposing a monotone frame match.

Execution-dependent attention addresses the loss of phase information in a single task summary: each action can consult the relevant spatial and temporal evidence [158]. Cross-domain contrastive supervision makes that read more reliable when demonstration and query differ visually [12]. Video cross-attention and causal sensorimotor sequences expose different evidence to the read, respectively visual progression and observations with recorded actions [13, 14]. Execution mismatch creates a further problem when frame similarity is misleading. RHyME constructs semantically compatible training videos through clip retrieval and composition, then uses learned alignment to condition its policy [80]. These designs address complementary parts of correspondence: when to read, what to compare, and how training makes the comparison meaningful.

Representation determines which taught distinctions action generation can recover. RGB leaves motion implicit; sensorimotor records expose actions, and keypoints or spatial graphs establish correspondence [15, 16]. Cross-domain latents couple human and robot movement, while hierarchical tokens separate persistent demonstration structure from finer decisions [159, 160]. Flow matching generates actions from encoded observations and commands [55]. TP-Flow uses support-demonstration prototypes to condition both its initial prior and velocity field [384]; deployment recomputes them with fixed weights. Richer generation helps only when its context preserves the required distinction.

Table 3 Context-conditioned policies: retrieval, demonstration interpretation, and physical adaptation (Section 3.1). Neural components remain fixed during the listed deployment branches. Codes: D examples, H history, F feedback; V visual, S state/action, G geometry, L symbolic. Slashes combine operations. †: technical report; –: unspecified. Operations are defined in Section 3.

Method	Source	Form	Context operation	Carrier	Execute
Retrieval and trace guidance					
VINN [149]	D	V,S	Retrieve	Actions	Replay
BPC [389]	D	V,S	Retrieve / Fit	Actions	Combine
RICL [17]	D	V,S	Retrieve / Encode	Actions	Decode
TraceFlow [390]	H,F	V,S	Retrieve / Align	Actions	Decode
Demonstration interpretation					
TOSIL [21, 158]	D	V	Attend	Actions	Decode
MOSAIC [12]	D	V	Attend	Actions	Decode
RoboPrompt [153]	D	G,S	Parse	Actions	Decode
BiCICLe [92]	D	S	Parse	Actions	Decode
Vid2Robot [14]	D	V	Attend	Actions	Decode
ICRT [13]	D	V,S	Attend	Actions	Decode
KAT [15]	D	G,S	Encode	Actions	Decode
Instant Policy [16]	D	G,S	Encode	Actions	Decode
MatchingPolicy [161]	D	G,S	Align / Encode	Actions	Decode
StellaVLA [163]	D	L	Encode	Actions	Decode
SynthICL [77]	D	V	Encode	Actions	Decode
BPP [54]	D	S	Encode	Actions	Decode
Physical-response inference					
RMA [39]	H	S	Estimate	Actions	Decode
Self-Adaptive VLA [391]	H	V,S	Estimate	Actions	Decode
Generalist demonstration interfaces					
GEN-1.5† [98]	D	S	–	Actions	Decode
S1† [99]	D	V	–	Actions	Decode
GLOW† [387]	D	V	Encode	Actions	Decode

Temporal selection addresses a different limit from action encoding. A complete trajectory may contain many locally irrelevant stages, so ICI-VLA retrieves phase-aligned micro-demonstrations using a retriever trained with dynamic-time-warping supervision [18]. The deployed policy reads examples selected from observable query inputs; target-action masking during training reduces reliance on exact action-prefix continuation. Phase selection determines which evidence reaches the generator, whereas token prediction or flow matching determines how that evidence becomes an action block.

Spatial correspondence as conditioning. Temporal phase identifies when an example applies; spatial correspondence identifies where its action should be realized. MatchingPolicy makes this second requirement explicit [161]. Foundation-model features establish demonstration–query point matches; tracking propagates them through time; RGB-D lifts them into 3D. A graph diffusion policy then generates gripper motion from these relationships. This workflow assumes that matched task geometry remains informative about the required action, without fixing a trajectory for the robot to replay. Section 3.2 considers the complementary case in which correspondence transforms an explicit executable reference.

Navigation extends this correspondence from object interactions to places. A policy can read successive visual subgoals [354] or attend to an action-bearing path memory [355] to follow new teaching. A scene preview supplies a different spatial reference: NOLO [350] combines a video with optical-flow-derived action labels, the current view, and a goal image to predict actions. Its offline reinforcement learning trains the interpreter; deployment reads a new scene video with weights fixed. The shared computational interface is contextual action inference, while the context can specify a route or reveal an environment.

Identifying physical context from interaction. Temporal and spatial correspondence can identify a desired motion while leaving its physical effect uncertain. Interaction history supplies the complementary evidence when the goal is known but the current body's response is not. System identification makes this unknown explicit through physical parameters, whereas a learned adaptation latent can retain the response information needed by a policy [39, 146]. ICWM extends physical-context inference to changes in observation and embodiment [41]. These approaches require transitions that reveal how commands affect the present system; a task demonstration alone may leave that uncertainty unresolved. Their representation conditions an action policy directly. Predicting and evaluating candidate futures introduces the additional computational stage considered in Section 3.3.

Physical context needs both informative measurements and a transferable representation. Self-Adaptive VLA uses command–response discrepancies for actuation bias, while joint-encoder offsets require visual evidence because readings can agree with commands despite a wrong physical pose [391]. CAPABLE shares a temporal encoder across joints, combines their command–response histories with the current Jacobian, and conditions a residual action policy [492]. The encoder and residual policy are trained offline and fixed at deployment. History identifies current bodily capability; it does not specify a new task. More history helps when it distinguishes causes that require different corrections.

Connecting pretrained competence to new evidence. Broader robot and language priors change what must be learned for demonstration following. An existing action generator can be conditioned on retrieved executions [17]; grounded spatial descriptions can supply scene information missing from the current visual input [162]. These inputs resolve different unknowns. A demonstration may change the requested procedure, while a scene description may clarify how to execute an already known procedure. The contribution of each channel depends on the decision its information resolves.

Observation-only teaching leaves action correspondence implicit. Behavioral Cloning from Observation estimates actions through inverse dynamics before training a policy [167]. Fixed-weight teaching uses an existing action interface learned from demonstrations or constructed through pretrained reasoning and geometry.

Textual action interfaces move part of action inference outside a visuomotor backbone. RoboPrompt supplies object poses and keyframe end-effector examples to a frozen language model [153]. BiCICLe conditions one arm's keyposes on the other's prediction, preserving coordination within a smaller inference problem [92]. Its v3 TWIN comparison uses ten demonstrations per task and reports 70.5% mean success over 13 simulated tasks, versus 64.4% for the strongest ICL comparator. Ground-truth segmentation supplies simulated object centroids; three physical tasks provide a separate execution test. These interfaces rely on perception and coordinate conventions that make reasoning outputs physically meaningful.

Teaching budget can change the preferred adaptation mechanism. KAT compares a text-pretrained model reading keypoint–action examples with KeyAct-DP, a diffusion policy trained on similar representations [15]. Contextual inference is competitive with few examples; policy training becomes stronger with more. The crossover differs against an image-based diffusion policy, indicating that both representation and adaptation mechanism determine the useful demonstration budget.

Generalist systems expose different teaching representations. BPP and GEN-1.5 accept sensorimotor prompts [54, 98]; S1 accepts video prompts [99]. The GLOW technical report [387] describes a Skill Demo Encoder that converts one human video into reusable skill context. A fixed-weight autoregressive model combines it with current observations, language, robot state, and history to generate end-effector actions. Its teaching pathway belongs to context-conditioned policies (Figures 5 and 7); Harness manages tools and feedback, while KnowinWorld supplies candidate rollouts. The primary family follows the teaching-to-action pathway, with prediction and tool management providing complementary support.

Retrieved actions need compatible coordinates and archive coverage; learned generation additionally depends on the interpreter transferring the taught distinction, as REGENT illustrates [154]. Geometric transfer exposes this dependency through an explicit reference that execution tracks.

3.2 Geometric demonstration transfer

Geometric demonstration transfer preserves a motion or contact reference and adapts its realization to the query scene. Perception establishes correspondence; alignment or retargeting transforms the reference; tracking or replay executes it. This explicit reference is useful when contact geometry matters, such as preserving a handle grasp. We follow the mechanism from reference construction through spatial alignment and shape changes to reuse across a task repertoire.

IMOP makes the reference local to the next demonstrated action: learned invariant-region correspondences support analytic pose transfer, and a state router selects the relevant demonstration transition [156]. The reference can therefore be a target pose rather than a complete trajectory.

Reconstructing an executable reference. Let D^{s} denote the supplied demonstration. A geometric transfer computation can be written as

$$\zeta_{1:L} = E_\theta^{\mathrm{m}}(D^{\mathrm{s}}), \qquad g = \mathrm{Retarget}(\zeta_{1:L}, o_t, \mathcal{B}), \qquad A_t = \kappa(g, o_t). \tag{7}$$

The extractor E_θ^{m} produces L descriptors ζ_i of demonstrated hand poses, object-relative poses, or contacts. Retarget maps them to scene o_t and robot kinematics/control constraints $\mathcal{B}$; executor κ consumes the resulting motion reference g. Object-centric and hand-based transfer preserve different descriptors: object motion versus grasp and hand motion [35, 168]. Fitting these poses or trajectories can leave the extractor's neural weights fixed; superscript m denotes the motion interface.

For rigid object-relative transfer, this relationship has a particularly simple form:

$${}^{W}T^{\mathrm{q}}_{E,b}(\sigma) = {}^{W}T^{\mathrm{q}}_{O_b}\left({}^{W}T^{\mathrm{s}}_{O_b}\right)^{-1}{}^{W}T^{\mathrm{s}}_{E,b}(\sigma). \tag{8}$$

Here ${}^{W}T_X \in SE(3)$ is frame X's rigid pose in world W; E is the end effector and O_b the reference object for segment b. Superscripts s, q denote source and query; $\sigma \in [0, 1]$ is segment progress, and -1 denotes inversion. The source factors express object-relative motion; the query factor relocates it. A fixed reference and applicable relative motion are assumed. Shape change, moving references, or embodiment differences require Equation (7)'s richer correspondence, making reference selection part of task interpretation.

Table 4 traces four ways to preserve an interaction: align its visual reference, transform object-relative motion, match functional parts, or switch references across stages. A richer correspondence is useful when it preserves a constraint that a global pose transform would lose.

From image alignment to object-relative transfer. Geometric reuse begins with an applicability assumption: restoring the relevant relative configuration makes the demonstrated interaction useful again. Image-based approaches enforce this through visual feedback. FlowControl aligns live RGB-D observations to successive masked reference frames [32]; DOME separates a learned visual servo to a bottleneck pose from subsequent velocity replay [33]. The difference is where feedback acts: throughout reference following or primarily before the interaction segment. Tracking active points and gripper events allows RoboTAP to change the visual reference as a task advances [169]. POIL closes the loop on the transferred object points themselves: functional-part correspondences establish a reference, and point-set feedback converts tracking errors into a rigid-body twist [403]. Using the same points for transfer and execution ties disturbance recovery to the demonstrated interaction.

An object-relative invariant makes applicability more explicit than image agreement. Pose-based transfer preserves a recorded end-effector trajectory [170]; category-level shape fitting extends correspondence to interaction keypoints [171]. DITTO reconstructs manipulated-object motion from an RGB-D human demonstration, transfers that trajectory to the detected objects in the robot scene, and realizes it through grasping and motion planning [35]. Hand-based retargeting instead reconstructs the demonstrator's grasp and movement before converting them into robot configurations [168]. The embodiment gap thus appears in different places: realizing an object effect in DITTO, or translating a demonstrated hand configuration. In either case, successful transfer requires both a preserved interaction and a feasible realization.

Table 4 Geometric demonstration transfer, grouped by the reference preserved (Section 3.2). R-NDF lists descriptor alignment. Codes: D demonstration; V visual, S state/action, G geometry. Slashes combine operations. Tracking, replay, and planning denote different stages of realizing the reference.

Method	Source	Form	Context operation	Carrier	Execute
Visual alignment					
FlowControl [32]	D	V,G	Align	Motion	Track
DOME [33]	D	V,S	Align	Motion	Track / Replay
DINOBot [34]	D	V,S	Retrieve / Align	Motion	Track / Replay
RAGrasp [472]	D	V,G	Retrieve / Align	Pose	Plan
Object-relative motion					
MT3 [36]	D	G,S	Retrieve / Align	Motion	Track / Replay
Pose-based OSIL [170]	D	G,S	Align	Motion	Track
IMOP [156]	D	G,S	Align	Motion	Track
DITTO [35]	D	V,G	Align	Motion	Plan / Track
HRT1 [168]	D	G	Align / Fit	Motion	Track
CosmoH2G [503]	D	V,G	Encode / Fit	Motion	Replay
Functional correspondence					
NDF / L-NDF [421, 422]	D	G	Align / Fit	Pose	Plan
IGA [420]	D	G	Align	Pose	Plan
R-NDF [345]	D	G	Align / Fit	Motion	Track
Interaction Warping [171]	D	G	Align / Fit	Motion	Track
Part decomposition [346]	D	G	Align	Motion	Track
SemAnCorr [38]	D	G	Align	Motion	Plan / Track
Staged and coordinated execution					
ODIL [37]	D	V,S	Align	Motion	Track / Replay
ManiLong-Shot [93]	D	G,S	Align	Motion	Track
Annotation-free OSIL [177]	D	V,S	Align	Motion	Track / Replay

Semantic anchors and functional geometry. Rigid pose transfer preserves a motion only when the source and target support the same local interaction. A different handle or articulated part requires correspondence within the object. SemAnCorr [38] selects semantically consistent surface anchors and propagates them through a functional map. Its manipulation pipeline transfers contact regions and local frames, selects a feasible grasp, and reuses motion relative to that grasp. Contact keyframes and relative motions form its execution specification. Semantic correspondence supplies the local geometry needed to extend the rigid transfer in Equation (8) across object shapes. Descriptor-based transfer makes the matched invariant explicit: NDF optimizes poses to reproduce demonstrated descriptors, L-NDF localizes the representation to transferable geometry, and IGA aligns object graphs conditioned on examples [420–422]. The optimized object is an alignment or pose, not the descriptor network's weights. A template library makes this correspondence explicit at retrieval time. RAGrasp combines semantic features with geometric filtering, then registers a selected RGB-D grasp template to the query and refines its contact pose [472]. Reuse avoids task-policy fitting but depends on calibrated depth and a transferable local grasp; physical tests cover calibrated planar overhead grasping.

Shape variation makes whole-object similarity too restrictive for some interactions. What must remain compatible may be a grasp role and a contact role, rather than the complete tool geometry [172]. A complementary approach starts from global shape and refines local curvature correspondences to preserve the contact used by a motion [173]. Surface interaction functions and screw interpolation connect such functional geometry to a transferred trajectory [174]. These routes differ in how they identify the invariant, but share a control assumption: preserving the relevant interaction structure makes the adapted motion useful. This shifts the transfer problem from matching objects as wholes to matching the parts of their geometry that the task actually uses.

When both objects change, correspondence must preserve their interaction. R-NDF aligns task-relevant local frames on a new object pair [345]; part decomposition separates local geometry from object arrangement to recover the demonstrated relationship [346]. Both reduce interference from irrelevant shape variation, extending transfer beyond relocating motion around one object (Section 3.5).

Deformable manipulation requires correspondence to remain meaningful as the object changes shape. Sparse Meets Dense extracts task-relevant contact pairs from a demonstration and uses an offline learned dense representation to track them during constrained execution [175]. Its final-state relative-pose annotation supplies interaction geometry; sparse constraints express the task, and dense correspondence maintains that geometry through deformation. This extends the reference from a rigid frame to a changing set of contact relations.

Affordance prediction can supply a related execution interface from a learned prior. VLAff predicts interaction heatmaps, grasp poses, and motion trajectories from an observation and language instruction [176]. Its human videos provide offline affordance supervision. These outputs can ground a contextual planner's specification, while a demonstration-conditioned system additionally uses the newly supplied example to determine which interaction should transfer.

Cross-embodiment geometry can be learned before a new demonstration is supplied. CosmoH2G learns hand-to-gripper transfer from paired recordings, predicts endpoint poses and orientations, and refines the resulting trajectory through contact alignment, smoothness, and inverse kinematics [503]. Deployment optimizes a motion reference while keeping the learned mapping fixed. Its open-loop execution tests transfer of demonstrated geometry; correcting deviations would additionally require observation-driven control.

From one demonstration to a reusable repertoire. A geometric repertoire must select an applicable interaction and restore its starting configuration. DINOBot combines retrieval with wrist-view alignment before replay [34]. MT3 adds language filtering, geometric retrieval, and pose alignment [36]; its retrieval-versus-cloning comparison separates alignment from interaction. Its 1,000 single-interaction tasks require less than 24 hours of human demonstration; multi-stage tasks use one demonstration per stage. Adding references avoids task-policy retraining, while execution still depends on restoring each interaction's preconditions.

Independent object motion and arm coordination expose the limit of one global alignment. A grasp and a later insertion must refer to different objects when the part and receptacle move independently; dual-arm interaction must additionally preserve relative coordination. Multi-stage visual servoing establishes a coordinated reference before ODIL's replay [37]. Interaction segmentation with invariant-region matching, and VLM-based keyframe selection with repeated alignment, extend the reference-switching principle to longer demonstrations [93, 177]. Table 4 compares how individual systems bind and execute these references, while Table B1 in Appendix B contrasts the computational roles of correspondence. The key decision is when a change of interaction requires a different reference.

Geometric transfer exposes whether a matched interaction remains applicable after alignment. Correspondence and reference switching address shape variation and independent object motion; feasible tracking addresses physical realization. Predictive control instead represents the consequences execution should produce.

3.3 World-model-based control

World-model-based control predicts consequences to specify desired evolution or assess candidate actions. Learned predictors and physics simulators both support this interface when their predictions influence deployed control. The comparisons distinguish demonstration conditioning from prediction used only in training.

A representative factorization is

$$\begin{aligned} v^{+} &\sim p_{\theta}^{\mathrm{v}}(\cdot \mid C_t, h_t), \\ A_t &\sim q_{\theta}^{\mathrm{v}}(\cdot \mid v^{+}, C_t, h_t), \end{aligned} \tag{9}$$

Here v^{+} is a predicted observation or latent future. p_{θ}^{v} predicts it and q_{θ}^{v} decodes future-conditioned actions, with fixed neural parameters θ in the ICL regime. Superscript v denotes this predictive interface. Joint video–action generation may share a backbone [178]. For packing, the future can specify red-before-blue

placement; actions must realize that sequence. The law of total probability marginalizes future-conditioned actions over predicted futures:

$$\pi_\theta(A_t \mid h_t, C_t) = \int q_\theta^{\mathrm{v}}(A_t \mid v^+, C_t, h_t)\, p_\theta^{\mathrm{v}}(v^+ \mid C_t, h_t)\, \mathrm{d}v^+. \tag{10}$$

Discrete futures replace the integral with a sum. Sampling a future and then actions, or selecting one approximate future, implements this dependence. The future's role in control defines the interface.

Generating a desired future or evaluating an action's future. World-model-based control has two complementary forms. In Equations (9)–(10), context specifies a desired evolution and an action decoder realizes it. In model-based selection, candidate actions specify possible evolutions, and context determines which consequence is preferred. One planning form is

$$A_t^\star \in \arg\max_{\widehat{A}_t \in \mathcal{A}_t} \mathbb{E}_{v^+ \sim p_\theta^{\mathrm{d}}(\cdot \mid h_t, \widehat{A}_t, C_t)} \big[\mathcal{R}(v^+, C_t)\big]. \tag{11}$$

The set $\mathcal{A}_t$ contains admissible candidate action blocks, $\widehat{A}_t$ denotes a candidate, and $A_t^\star$ is a selected maximizer. The action-conditioned dynamics distribution p_θ^{d} estimates its future v^+; d distinguishes forward dynamics from the context-conditioned future generator p_θ^{v}. The score $\mathcal{R}$ measures how that future satisfies the contextual task requirement. The expectation averages over predicted futures. This form makes the two roles explicit: the transition model predicts what an action may cause, whereas teaching determines which consequence is desirable. Optimizing $\widehat{A}_t$ leaves the neural parameters θ unchanged. For explicit simulation, the predictive distribution can be induced by $s_{k+1} = F_\phi(s_k, a_k)$ and $v_k = R(s_k)$, where s_k is physical state, F_ϕ a transition operator with scene parameters ϕ, and R an observation renderer. Estimating ϕ instantiates the scene without updating the planner's neural weights. SIMPACT constructs this simulator from RGB-D and returns rollout images and object states to a VLM for candidate revision [423]. Thus, simulated consequences become planning context; their value depends on whether reconstructed geometry and dynamics preserve the interactions relevant to the task.

Unknown dynamics changes which candidate trajectories are credible. One response is to infer dynamics context from recent transitions and evaluate actions with a context-conditioned world model, as in DOMINO [19]. Another is to condition trajectory generation on those transitions and use reward and dynamics guidance during sampling, as in MetaDiffuser [20]. Both specialize prediction through new evidence while leaving learned parameters fixed. Context can also specify how actions are interpreted: SyncWorld uses paired action–video calibration segments to infer the current camera- and embodiment-dependent mapping from numerical commands to visual motion [506]. Coordinate augmentation and contextual training establish this capability offline; deployment supplies the new mapping and ranks candidate actions through imagined rollouts. Thus context can identify a control convention as well as a desired outcome.

Prediction supports control when the target is grounded and candidate actions remain feasible. LeWAM retrieves demonstration-derived future goals and searches its frozen policy's noise space with a fixed latent dynamics model [512]. Demonstrated progression sets the objective; the learned decoder constrains proposals. Its small-data simulation study tests goal-guided execution within this trained repertoire.

Inference and fitting change different parts of prediction. DeepJEPA recurrently refines imagined transitions using an offline-trained continuation head, with world-model weights and outer CEM settings fixed [448]. MetaPusher instead updates a pushing model and repairs its planning tree [367]; Sandwich-Residuals fits neural corrections around a frozen predictor [366]. These operations refine computation or change the predictor; contextual prediction specializes an unchanged model through new evidence.

From goal-reaching to demonstration-conditioned futures. Goal-reaching and demonstration transfer place different demands on a predictor. Universal Planning Networks optimize actions toward a supplied goal with model weights unchanged [179]; ViGAR turns a goal image into subgoals for joint video–action execution [460]. The image specifies an outcome. PEWAM instead compiles a demonstration into object-keypoint trajectories, contact points, and a terminal state, then conditions a fixed world-action model on this effect program [459]. Removing the demonstrator's body makes the desired object evolution an editable

interface. Its four-task physical test reports 36/40 successes against 22/40 for goal-image conditioning after training on other real tasks. Yet terminal-state-only conditioning exceeds the full program on its small held-out LIBERO comparison (49/90 versus 40/90). Abstraction follows the taught constraint: placement may need only an endpoint, while ordered effects and contacts preserve additional requirements.

This inference can remain in a latent space. OSVI-WM predicts demonstration-conditioned latent states and decodes them into attributed waypoints [21]. Demo-JEPA instead passes a predicted latent goal to a cross-entropy-method planner, retaining that goal until an achievement threshold permits progression [22]. Both combine world-model-based control with an explicit waypoint or goal executor: the predicted future supplies targets for a separate execution stage, rather than only features inside an action head. In Demo-JEPA's real-robot comparison, the diffusion action head has higher mean success on familiar behavior grounding, while planning has higher mean success on unseen tasks. The familiar-task advantage does not persist in its simulation comparison. The relative value of search and learned decoding depends on both task novelty and the execution setting, even when the two share a future representation.

Prediction can also supply a route or a discrete transition model. TADreamer reconstructs navigation waypoints from generated video and registers them to measured geometry before planning [334]; metric calibration connects imagined motion to feasible execution. GAVEL instead rolls plans through an explicit graph of action preconditions, effects, and location beliefs, repairing model-detectable violations before invoking semantic replanning [335]. Their shared predictive interface does not require a common visual representation.

Human-video world-action systems broaden the intermediate toward robot-domain visual evolution. Two dependencies become central: current robot progress must determine which part of the demonstration remains relevant [23], and the demonstrated procedure must change the predicted future even when the current scene admits a familiar continuation [24]. These dependencies connect the older goal-reaching formulation to one-shot task transfer. The predicted evolution must preserve the procedure selected by the example.

Procedure and physical history have different roles. WorldScape Policy 2.0 combines persistent demonstration context, event memory, and short-term dynamics [25]. Long-WAM instead studies the robot's own observation history: causal video pretraining supplies a history-to-future representation, and partially denoised futures condition action generation [483]. Its context-window variants are trained separately. Their gains therefore concern learned history use, rather than adding teaching to one fixed checkpoint. Stable procedure tracking requires appropriate training and timely access to physical history.

The planning and control clocks can be separated explicitly. MaP-WAM uses sparse observed segment histories to generate a language subgoal and visual plan; its executor reuses the cached plan until progress estimates trigger another planning step [399]. Matching the current observation to nearby plan frames calibrates that progress. At transition, actual execution frames replace the imagined plan in episodic memory. This separation matters: a forecast can guide action, but recording it as an achieved event would let prediction errors become historical evidence. The executor's auxiliary future-video head can be omitted at inference, while the upstream visual plan remains in control. On RMBench, a multitask planner uses task-specific executors; constant executor latency excludes the separate planning call. Memory coverage, motor transfer, and response time consequently require separate measurements.

Retrieval can constrain prediction before detailed motion is generated. The retrieval-augmented policy ReCAP supplies trajectories to a frozen residual predictor [152]; its Cosmos Policy branch jointly generates future images and actions. The retrieved motion narrows the continuation, while prediction relates it to the evolving scene. Other ReCAP backbones route retrieved evidence through different action interfaces, so the predictive classification applies to its Cosmos Policy branch.

The predictive interface also localizes transfer. OpenWAM freezes a local-context inverse-dynamics decoder while training the target-task video producer with action supervision [496]. On four held-out LIBERO-90 tasks, both counterfactual dynamics coverage and restricting the decoder's inputs contribute to transfer. The reusable component is the action decoder, while task adaptation remains in the producer. Across pixel, latent, and action-conditioned predictors [180], the central question is which component receives the new task information and whether its output preserves an executable interaction.

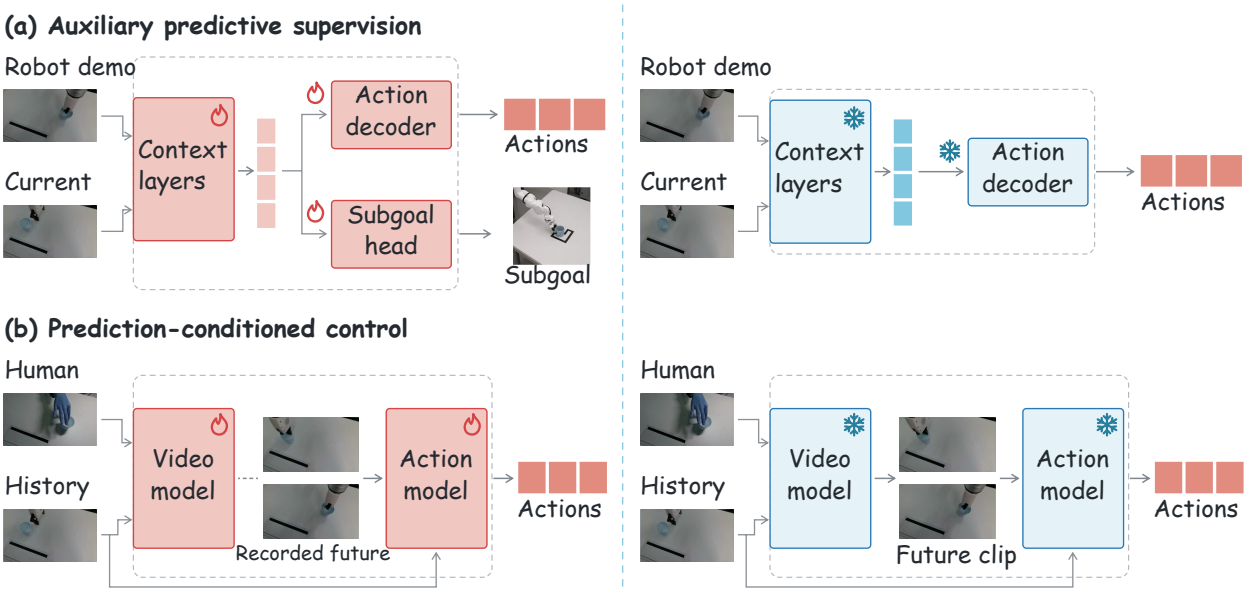

Figure 11 Two roles of visual prediction: training (left) and deployment (right). Flames mark trainable modules; snowflakes mark fixed modules. (a) SynthICL trains the context layers and both heads, then removes the subgoal head; its frozen DINOv3 backbone is omitted [77]. (b) Zero-WAM trains the video and action branches, then freezes both and replaces recorded future video with generated video [24].

Learning from futures without executing through them. Future supervision can improve an action representation without requiring a future to be generated online. Let $\mathcal{L}_a$ be an action-training loss and $\mathcal{L}_v$ an auxiliary future-prediction loss. The training and deployment interfaces can then be written as

$$\theta^\star \in \arg\min_\theta \big[\mathcal{L}_a(\theta) + \lambda \mathcal{L}_v(\theta)\big], \qquad A_t \sim \pi_{\theta^\star}(\cdot \mid h_t, C_t). \tag{12}$$

Here $\lambda \geq 0$ weights the auxiliary loss. Optimization acts on the designated trainable components, with any frozen backbone held fixed. The retained components of $\theta^\star$ initialize the deployed parameters θ_0; the auxiliary prediction head can be removed when action generation does not depend on its output. Equation (10), by comparison, retains a future-dependent action computation at deployment.

Predictive pretraining and contextual adaptation solve different problems. ZimaBlue aligns causal video priors with robot actions through video–action mid-training and target-robot post-training, then caches slow-model futures for fast execution [499]. Physical-demonstration prompting is a proposed next step in that work. Its trained generalization establishes a prior on which contextual teaching could operate.

Training intermediates need not remain execution dependencies. Gripper traces supervise action representations even when reasoning dropout or head removal permits execution without generating them [181, 182]. PonderPounce instead retains an asynchronous cognition token within action inference [183]. DeltaWAM uses future-delta supervision but deploys an action stream that reads cached observed changes [414]. Its streaming memory reduces scene-encoding cost; the runtime interface is history-conditioned action inference, not control through a generated future.

Figure 11 contrasts these roles. Auxiliary future targets shape a policy during training; prediction-conditioned control retains a generated future in action computation. Panel (a) shows the trainable context and state–context transformers after a frozen visual backbone. Panel (b) shows trainable prediction branches, not input frames. The latter couples control quality directly to the generated future.

Predictive control makes an anticipated consequence available for inspection or planning. Forecast accuracy and action decoding determine whether it reaches execution. Skill- and agent-based methods instead specify operations and delegate their realization to available skills or tools.

3.4 Skill- and agent-based execution

Skill- and agent-based execution uses context to specify the selection, order, and arguments of available skills or robot tools. A red-before-blue placement video can become an ordered procedure realized by an existing executor. We compare learned skill composition with VLM/LLM-generated programs and calls, then examine when feedback can revise the specification and when additional motor learning is needed.

The specification can take the form of code, spatial targets, or an ordered sequence of learned skills. Its defining role is to govern an execution stage that can track or advance it, as described in Section 2. At that action interface, the deterministic form is

$$g = f_\theta(C_t, h_t), \qquad A_t = \kappa(g, o_t), \tag{13}$$

where f_θ extracts the execution specification g from context C_t and robot history h_t, and κ realizes it as the next action block A_t using observation o_t. For stochastic specification generation and execution, the same interface becomes

$$\pi_\theta(A_t \mid h_t, C_t) = \int k_\theta(A_t \mid g, h_t)\, p^{\mathrm{g}}_\theta(g \mid C_t, h_t)\, \mathrm{d}g. \tag{14}$$

Here p^{g}_θ generates specifications and k_θ realizes them. Discrete specifications give a sum; deterministic components recover Equation (13). Operations, arguments, and composition reside in g, while the executor's neural parameters reside in θ. Code or argument edits can therefore change behavior with weights fixed. Programs can invoke geometric trackers, and predicted futures can supply specifications, composing the corresponding interfaces.

Skill composition and program execution instantiate this interface as

$$\begin{aligned} &\text{Skill composition:} && g = (z_1, \ldots, z_{N_g}), && A_t = \kappa(z_{\ell_t}, o_t), \\ &\text{Program synthesis:} && g = f_\theta(C_t, h_t), && A_t = \mathrm{Run}(g, h_t). \end{aligned} \tag{15}$$

Here z_ℓ specifies a skill, N_g counts skills, and ℓ_t selects the active one. Run executes program g until it yields an action block, including permitted perception and control calls. A tool agent can generate one request at a time: its name and arguments form g, the robot adapter implements κ, and returned observations inform the next decision.

Composing reusable skills. An early insight was that rearranging existing motor capabilities can express a new task. Neural Task Programming [26] made this explicit in 2017: a demonstration instantiated a hierarchy whose leaves invoked robot routines. The transferable knowledge was the decomposition and its arguments, while execution reused the available primitives. This separated generalization in task length, ordering, and object assignments from learning the physical skills themselves.

Human-video teaching adds recognition across bodies and timing. XSkill learns shared human–robot skill prototypes [29]; UniSkill learns from unaligned cross-embodiment video [30]. Both translate human examples into guidance for a robot-side skill-conditioned policy.

Waypoints preserve motion targets, object-relation graphs preserve dependencies, and staged procedures preserve interaction changes [93, 164, 165]. Their role depends on the executor: they may specify a separate controller or condition action inference internally, as semantic subtasks do in StellaVLA [163].

Correct skill order also requires compatible physical handoffs. SkillWeave scores a frozen predecessor policy's sampled chunks against start states retained from the successor's training demonstrations [511]. This steers execution toward compatible handoffs within the existing skill repertoire.

A composed procedure needs evidence for when to advance. StageGuard distills completion reasoning into a monitor that continues, advances, or skips skills [336]; Navi-Agent checks subgoals against visual-place topology [337]. RoboMonitor instead learns phase, failure, and completion estimates from instruction-conditioned visual history [413]. These monitors supply execution checks; recovery additionally depends on how the controller uses them.

Navigation illustrates two levels of example-conditioned execution. Instruction–landmark examples in LM-Nav [358] and instruction–subtask examples in A^{2}Nav [359] guide a language parser that specifies what pretrained navigators should execute. Select2Plan [351] retrieves visual decision examples for selecting marked movements or waypoints. The reasoning output is an executable choice, while motion realization belongs to the available navigator. Terrain feedback can further change the costs attached to those choices, as in VLM-GroNav [352]. Section 6.2 compares the information supplied by these contexts.

VLM and LLM agents: programs, tools, and feedback. Language-model pretraining expanded adaptation to program generation in 2022. Code as Policies [27] uses command–program examples to recombine perception, control, arithmetic, and feedback logic; program examples also specify available objects and interfaces [28]. Multimodal reasoning extends this route to demonstrated procedures. Show-Harness converts video frames into a textual outline for an unchanged planner and an embodiment-specific action interpreter [31]. The outline must retain the spatial and contact constraints needed by the interpreter. The representation–execution boundary is explicit in FoldingAgent: manually selected video keyframes guide a fixed VLM in reconstructing and revising a parametric folding program through simulator feedback [501]. Its Pureland study validates geometric procedure reconstruction; robot grasping, force control, and material response remain outside that execution interface.

Preserving contact structure requires a richer executable abstraction. RAPID derives trajectory-optimization primitives, relational constraints, and a verification environment from a human demonstration [388]. Predicates and simulated scene variants support program revision before freezing the primitives and strategy for transfer. Teaching thus constructs operations and their checks; reconstruction fidelity determines whether those checks reflect physical interactions. Its nonprehensile experiments use demonstrations; LIBERO-Pro instead constructs programs without them. Both incur reconstruction and program-search costs.

The command interface determines which decisions contextual inference must resolve. GPT-Policy uses goal images or demonstrations, available state and action records, and interaction feedback to request Cartesian targets or gripper commands [94]. RoboDawn instead grounds discrete translation, rotation, and gripper commands through in-context examples [364]. In both cases, fixed models request actions from existing motor competence and use returned feedback to revise subsequent requests. RoboICL makes demonstrations and interaction memory share an observation–action–receipt grammar, preserving early anchors alongside the latest outcome [438]. The common record binds taught actions to the feedback needed for subsequent correction.

Tools expand the evidence and behavior available to this inference. Robo-Harness K1 exposes calibrated depth, spatial measurements, persistent visual anchors, and grasp hypotheses to ground generic motion requests [377]. The agentic framework RACaP exposes typed Policy APIs: a ReAct agent combines working memory, retrieved experience, and visual feedback to choose calls, arguments, and recovery steps [378]. Both retain fixed neural weights during deployment; RACaP also freezes its generated source code. MOKA grounds motion through marked visual affordances, while RoCo coordinates waypoint plans through dialogue and collision feedback [424, 425]. Their prompting interfaces revise executable choices; MOKA's separate distillation branch instead trains a policy.

The memory–executor interface sets another limit. In 2AM, the agent retains task history but sends only subtask language and optional 2D grasp, place, and move hints to an episodically stateless action model [401]. These cues bind remembered intent to current-image instances without asking the agent to produce robot trajectories. Their validity differs: a grasp cue expires after acquisition, whereas a destination can persist during transport. The action model is trained to interpret noisy or incomplete hints. External task memory works through a learned steering interface. Its expressiveness determines which temporal and contact-dependent requirements can reach the action model.

Previewable actions add a further use of context: a candidate can be revised before it changes the scene. World Action Agent (WAA) selects interaction-centered views, renders proposed poses with kinematic planning feedback, and converts in-view corrections into robot motion [386]. Its main agent consults a separate Skill Agent for demonstration-derived procedures and visual checks. Tool requests form the execution specification, while geometric planning checks their feasibility before control.

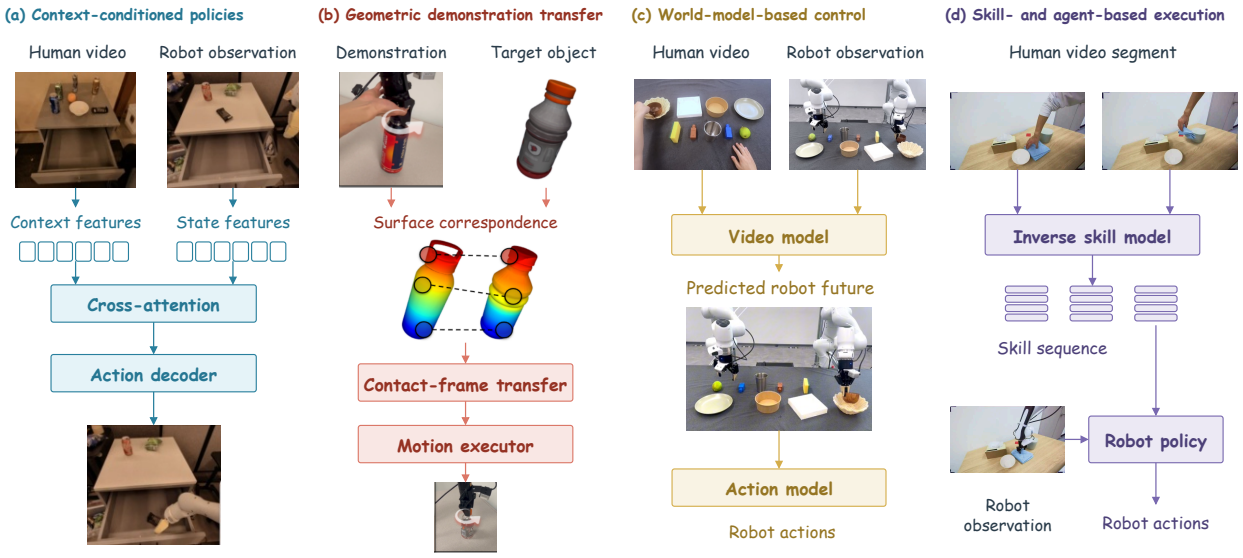

Figure 12 Four technical paradigms convert context into actions through (a) direct action inference, (b) geometric motion-reference transfer, (c) prediction-conditioned control, and (d) skill or tool execution. Representative implementations are Vid2Robot [14], SemAnCorr [38], Zero-WAM [24], and UniSkill [30], respectively.

Feedback must update the evidence on which a request depends. ETA observes after each world-changing call [348]; EmbodiedSkills checks typed skill inputs and artifact freshness before execution, verifies effects from new observations, and invalidates execution artifacts when a plan changes [502]. Its orchestration and action components are trained offline. Recovery can instead restore a learned policy's operating conditions: RoboHarness retrieves suitable entry states [349], while Causeway steers action-stream features toward demonstration-derived re-entry poses [412]. Verification determines whether the next attempt needs a revised request or restored operating conditions.

Spatial context connects these decisions to motion. VoxPoser constructs affordance and constraint maps for geometric planning [184]. HomeBody retains views and a reconstructed scene to ground skill, target, and hand selection; failures trigger retries or replanning [415]. Its digital twin supplies spatial context, while analytical manipulation and pretrained locomotion realize the requests. The division matters: context must resolve decisions left open by the supplied executor.

A plausible procedure can still request an action that the current scene cannot support. Skill-affordance estimates address this gap before execution by coupling linguistic relevance to physical feasibility [185]. Scene and execution feedback address it after a transition by revising what should happen next [186]. When feasibility depends on an unknown property, targeted measurements such as proprioceptive estimates of mass or stiffness can change the manipulation strategy [187]. These are complementary uses of evidence: predict applicability, observe the outcome, and acquire information missing from either. They connect program-level reasoning to the actual conditions under which a skill works.

Feedback-driven revision can change a procedure or its executor. ICPI adjusts primitive arguments [9]; CaP-X contrasts code revision with training the coding policy [166]. ENPIRE supports both program revision and neural policy learning [89], while VoxPoser's dynamics-learning extension updates a predictive component [184]. PP-ProPS searches controller parameters from encoded candidate–reward histories [411]: its language model stays fixed, but each proposal changes the robot-side controller. The adaptation claim therefore depends on which component changes, not whether the reasoning model is frozen.

Explicit structure trades *portability against physical detail*: a procedure can transfer across bodies yet omit a required grasp or compliance condition. Executors may recompute only details left open by teaching. DualManip separates update rates accordingly: geometric correspondence revises contacts, while failed geometric validation triggers semantic replanning [408].

Table 5 Predictive and skill/agent interfaces (Sections 3.3–3.4). Carriers identify intermediate representations; families follow the full control path. Listed deployment branches keep neural weights fixed; LeWAM and SkillWeave retain training demonstrations for goal and handoff selection. Codes: D examples, H history, I instruction/goal, F feedback, E scene/map; V visual, S state/action, G geometry, L symbolic. Slashes combine entries.

Method	Source	Form	Context operation	Carrier	Execute
I. World-model-based control					
OSVI-WM [21]	D	V	Encode	Future	Plan
SIMPACT [423]	I,E	V,S,G	Simulate / Verify	Future	Plan
SyncWorld [506]	D,H	V,S	Encode	Future	Plan
Demo-JEPA [22]	D	V	Encode	Future	Plan
LeWAM [512]	D,H	V	Retrieve / Simulate	Future	Plan
ViGAR [460]	I	V	Encode	Future / Actions	Decode
PEWAM [459]	D,H	G,S	Parse / Encode	Effects / Actions	Decode
HOST [23]	D	V	Encode	Future	Decode
Zero-WAM [24]	D	V	Encode	Future	Decode
WorldScape Policy 2.0 [25]	D,H	V,L	Encode	Future / Actions	Decode
ReCAP (WAM) [152]	D	V,S	Retrieve / Encode	Future / Actions	Decode
MaP-WAM [399]	H	V	Align / Encode	Future	Decode
FARE [400]	H	V,S	Verify	Future / Actions	Decode
FAVOR [467]	H,I,F	V,S	Verify / Parse	Future / Actions	Decode
II. Skill- and agent-based execution: learned skills					
XSkill [29]	D	V	Encode	Skill	Decode
AWDA [21, 164]	D	V	Encode	Motion	Track
ORION [165]	D	V,G	Parse / Align	Program	Invoke
UniSkill [30]	D	V	Encode	Skill	Decode
SkillWeave [511]	D	S	Verify	Skill	Decode
II. Skill- and agent-based execution: programs and tool calls					
Code as Policies [27]	D,I	L	Parse	Program	Invoke
Show-Harness [31]	D	V	Parse	Program	Invoke
GPT-Policy [94]	D,H,I	V,S,L	Parse	Calls	Invoke
WAA [386]	D	V,G	Parse / Verify	Calls	Invoke
RAPID [388]	D	V,G	Parse / Verify	Program	Invoke
Robo-Harness K1 [377]	I,E	V,G,L	Parse / Align	Calls	Invoke
RACaP [378]	H,I,F	L	Retrieve / Parse	Calls	Invoke
2AM [401]	H,I	V,L	Parse	Calls	Decode
Mimir [392]	H,I	L	Parse / Verify	Calls	Invoke
HomeBody [415]	H,I,F,E	V,G,L	Retrieve / Parse	Calls	Invoke
SparseNav [394]	I,E	G,L	Parse / Align	Motion	Track
RV-ICL [465]	D,H	V,S	Retrieve / Parse	Program	Invoke
MarvisNav [466]	H,E	V,G	Encode / Parse	Calls	Invoke
RobotUse [470]	H,I,F,E	V,G,L	Retrieve / Parse	Calls	Invoke
ProactiveVLA [474]	H,I,F,E	V,L	Retrieve / Parse	Calls	Invoke
TacZero [477]	H,I,F	V,S	Parse	Pose	Track
DROC [46]	H,F	L	Retrieve / Parse	Program	Invoke

Table 5 connects context operations to execution: a video can supply a future for planning or decoding, a tracked motion, or an invoked program. Retrieval and verification augment these routes; their carriers determine which taught constraints reach control.

Table 6 Object substitution illustrated by pouring. The demonstration uses a jug and a cup; all three settings preserve the goal of pouring into the receiver. Arrows indicate object replacement; the last column gives example motion adjustments.

What changes	Held vessel	Receiver	Motion adjustments
Held vessel	Jug → bottle	Same cup	Grasp; tilt angle
Receiver	Same jug	Cup → bowl	Pouring position; height
Both objects	Jug → bottle	Cup → bowl	Grasp; tilt; pouring position

Choosing an interface by the constraint it preserves. Interface choice follows the constraint most vulnerable to loss. Action inference relies on training relationships; geometric transfer exposes contact or motion; predictive control exposes consequences; skill interfaces expose order and preconditions. Hybrids can turn predicted futures into waypoints or invoke geometric trackers from programs. Classification follows the intermediate carrying the taught requirement into execution.

3.5 Shared mechanisms: correspondence and memory

Across all four interfaces, correspondence binds teaching to current objects, memory retains evidence, and the command interface determines which corrections can reach execution.

Figure 12 shows their control intermediates: demonstration features, contact frames, predicted futures, and skill sequences. Each depends on the evidence selected and interpreted in Figure 10.

Cross-object transfer of demonstrated tasks. A demonstration can specify a relation that remains meaningful after its objects are replaced. Consider a robot shown how to pour from a jug into a cup. It may then use a bottle with the same cup, keep the jug but pour into a bowl, or use both the bottle and the bowl. Table 6 contrasts these three settings using this illustrative task. The held vessel is the manipulated object; the cup or bowl is the receiving object, a role that can also be a support or contact surface in other tasks. Each role can change within a category or through a functionally compatible substitute. Novelty relative to the demonstration and novelty relative to training remain separate properties. We first examine which relations transfer between new instances, then consider functional substitution and the control needed to realize the transferred requirement.

The robot must still direct liquid into the receiver. A bottle changes grasp and tilt; a bowl changes pouring position and height. Replacing both couples the required geometric adjustments.

From object identity to task-relevant relations. An unchanged object in a new pose often permits relocation of a reference. A new instance can change the grasp, contact surface, or clearance even when its role stays fixed. R-NDF transfers relations such as placing a bowl on a mug or a bottle in a container to previously unseen object pairs [345]. Its original setting uses five to ten demonstrations with pretrained category representations and a marked 3D keypoint. Descriptor alignment recomputes placement with those representations fixed; the paper also studies a learned energy-based refinement, which requires training an additional neural component. A part-based representation supports one-demonstration transfer across larger shape variation [346]. Its real-robot tests include new mug–rack and bowl–mug pairs, using learned part models and segmentation assistance. These results connect the relation shown in a demonstration to execution on new objects while exposing the prior geometry and human input that make the transfer possible. Adapting by Analogy uses correspondence differently: expert feedback selects an in-distribution observation to substitute into an existing visuomotor policy [416]. The correspondence changes policy input rather than directly supplying a motion reference.

From new instances to functional substitutes. Category changes make the preserved role more important than whole-object similarity. FUNCTO transfers tool use from an actionless human RGB-D video and task description, distinguishing spatial, instance, and category generalization [172]. Its functional keypoints locate how a replacement tool should be grasped and used; the experiments emphasize tool variation, with less emphasis on changes to the receiving object. SemAnCorr transfers a pouring motion from a kettle to a toy

cup and a cutting motion from scissors to a nail clipper, using recorded contact and motion from kinesthetic teaching [38]. These different evidence interfaces support functional correspondence without requiring a common demonstration format. Cross-category matching can also be supplied: the part-decomposition study evaluates pre-pouring alignment for a watering-can substitution using provided part equivalencies [346]. This protocol isolates geometric alignment with the cross-object correspondence already specified.

Functional substitution must preserve the requirement actually taught. A general request to transfer contents can admit another vessel; a specified vessel or handle contact can remove that freedom. At a higher level of abstraction, preserving a purpose may require a different procedure rather than a transformed motion. The Imitator Game distinguishes category-preserving substitution from this affordance-adapted imitation [101]. The invariant is therefore a task-dependent choice: object relation, contact constraint, process, or intended effect. Section 7.1 tests whether the demonstration determines that choice.

Realizing a transferred requirement. Each interface gives the transferred requirement a different executable form: action features, a geometric reference, a desired future, or bound skill arguments. Geometry used during training [189, 190] differs from a deployment reference or demonstration-fitted motion system [191]. Spatial affordances and action representations supply priors [192, 193]; the new demonstration must select which requirement they realize.

Relational skill guidance provides an agent-based route to transfer. WAA's frozen LIBERO-derived skills transfer to robosuite lifting, stacking, and restacking, where objects, scenes, and viewpoints change; grounding resolves poses from the current scene [386].

When interpretation produces a desired pose or feature configuration, a separate controller can track it. For a stationary target, a local visual-servo rule is

$$\boldsymbol{e}_t = \boldsymbol{s}_t - \boldsymbol{s}_t^\star, \qquad \dot{\boldsymbol{q}}_t = -k_{\mathrm{vs}} J_{\mathrm{vis},t}^{\dagger} \boldsymbol{e}_t. \tag{16}$$

Here $\boldsymbol{s}_t$ and $\boldsymbol{s}_t^\star$ are current and desired visual features, $\boldsymbol{e}_t$ their error, $\boldsymbol{q}_t$ the joint configuration, and $J_{\mathrm{vis},t}$ the feature Jacobian. Its pseudoinverse and positive gain k_{vs} map feature error to joint velocity [194]. Context changes the reference while feedback corrects its realization. New contact or dynamics can still invalidate that realization, even with correct correspondence. The final part of this subsection examines this execution boundary. Transfer also requires the defining evidence to remain accessible when needed, motivating the memory mechanisms below.

Accessing and updating evidence across time. Alignment selects information within a demonstration, retrieval selects records from an archive, and memory preserves evidence across decisions. These operations can expose policy examples, motion references, predicted rollouts, or programs. The retrieved object's use in control determines its method family; the access mechanism itself does not.

Memory must identify what matters, how to retrieve it, and when it expires. Task requirements, response estimates, and completion records need different update rules (Table 7). Moving a part can invalidate its pose while preserving the taught order. The read–update decomposition connects selection and reset timing to control; Sections 5 and 8.4 develop validity across execution and reuse.

Carrying evidence between decisions. A recurrent state m_t and an external archive M_t provide complementary ways of retaining experience. Let $x_t = (o_t, a_t, o_{t+1})$ be an executed transition, augmented with feedback when available. A common read–update decomposition is

$$\begin{aligned} m_{t+1} &= u_\theta(m_t, x_t), \quad & M_{t+1} &= w_\theta(M_t, x_t), \\ R_t &= \rho_\theta(h_t, M_t), & C_t &= b_\theta(C_t^{\mathrm{task}}, m_t, R_t). \end{aligned} \tag{17}$$

The maps u_θ and w_θ update recurrent and external memory, ρ_θ retrieves records R_t, and b_θ constructs the policy input. Task evidence C_t^{task} includes instructions, goals, examples, and corrections; demonstration-only configurations use D^{s}. Retained state (m_t, M_t) differs from selected input C_t. Either memory component may be omitted, and the maps can combine analytic and fixed neural operations.

Retaining the information that changes control. Early latent task inference compressed transitions to identify reward or dynamics [196, 197]. Selective retention matters because a demonstrator's collection style can correlate with task identity in training logs without identifying the task reliably at deployment.

Physical adaptation depends on evidence age and validity. RMA uses recent transitions, ICWM collects task-agnostic probes, and LocoFormer carries interaction across trials [39–41]. RopeFormer retains action–response context across physical resets in dynamic rope tasks [363]. Zeva and Zeva-Ego encode action effects at two timescales; the latter combines recent traces with phase-matched memory across attempts on the same task instance [88, 362]. These fixed-weight interfaces trade acquisition cost against subsequent relevance.

Direct measurements can expose a changed response before it is inferred from action history. TeAR conditions bounded multiplicative and additive command corrections on robot state and motor telemetry, with a gate preserving nominal commands [458]. Both the base policy and adapter are fixed at deployment after offline training. Thus, the relevant context describes hardware operating conditions rather than a new task. Its controlled degradation models test response compensation; a physical Lift study with 20 trials per condition reports thermal gains of two or three successes. Direct telemetry and interaction history offer complementary evidence about whether the same command still produces the intended effect.

Event memory preserves facts no longer visible, such as whether a button was pressed. Perceptual retrieval and semantic records retain this evidence at different levels of detail [42, 43]; MEM combines recent views with a semantic history [44]. Recurrent updates [198, 199] and event-triggered keyframes [200] trade compression against direct access to earlier observations. MemBodied preserves an initial-scene anchor alongside associative interaction state [379], whereas ARMS records which arm acted and when [385]. HIRE's ordered wrench traces distinguish visually aliased contact states; shuffling them tests the contribution of temporal order [402].

Retention determines which evidence remains recoverable. Divide-and-Remember recursively selects K action-relevant visual tokens from each merged $2K$-token history block [444]. ECoMEM instead uses a fixed Writer for entity, state, and event records, followed by a learned Reader [445]. Selection preserves visual detail but discards other evidence; explicit records support targeted updates within the Writer's recognition vocabulary. Both use offline-trained action interfaces with fixed deployment weights. Their task-trained physical studies evaluate retained evidence under different protocols.

Evidence can occupy different computational carriers. ReCAT separates current and recurrent reads; DRAM learns associative state with a frozen backbone [417, 418]. RecastVLA updates shared fast weights between action calls [419]. TaskAnchor's CA-TTT instead reconstructs fast weights through a history-conditioned gradient-form update at each decision, uses them for current-step readout, and discards them [507]. The history buffer persists within the episode; learned slow weights remain fixed. This transient computation differs from both recurrent weight adaptation and direct fixed-weight attention.

Execution progress and historical detail resolve different ambiguities. TaskAnchor supplies a milestone-supervised language stage coordinate and a visual history residual; task-specific post-training teaches the VLA to use both [507]. Its human-cued matching study tests event-dependent target selection within a trained procedure. PMTRM instead re-encodes a bounded state–action history into a temporal sequence, with offline objectives separating distant phases while retaining action information [508]. Deployment changes the queue while keeping encoder and policy weights fixed. Explicit records expose updates directly: Mimir advances progress after verifying a postcondition [392]; OCC4M updates a tracked target while preserving its remembered location [393]. The latter uses calibrated support geometry with moving simulated views and a fixed physical camera. Each carrier must preserve the distinction that changes the next action.

SimpleMemVLA preserves timestamped video history and routes its evidence through subtask hidden states to an action head [45]. This defers evidence selection until the current decision; MemoryVLA++ additionally combines retrieved events with imagined states [209]. Both depend on a trained interface that makes retained information useful for control. μVLA's recurrence helps held-out tasks with familiar memory requirements but little when the required structure changes [199]. Recalling a past event and inferring an unfamiliar rule therefore impose different demands.

Table 7 Four roles of contextual evidence and when to recheck their applicability. One episode can supply several roles, but each can remain valid for a different duration. Representative mechanisms illustrate each role; Table B2 compares storage and access.

Information role	Evidence and mechanisms	Recheck after
Task specification	Plate-first order; corrections and relational programs [46, 388]	A new task or correction
Correspondence	Matched target or phase; object tracks and trace retrieval [390, 393]	Object or execution-phase changes
Physical response	Action effects; trial context and hardware-shift tokens [88, 391]	Tool, material, or contact changes
Execution state	Verified completion; task agendas and observed segment records [392, 399]	New action, observation, or scene reset

Workspace Models train latent workspace tokens using VLM-selected historical patches [395]. This amortizes online selection while inheriting the saliency target's information choices. Comparing retrieval with learned compression requires both memory cost and recovery of newly relevant evidence.

History can also reconstruct an unobserved part of the present, rather than a past event or future consequence. LIFD combines recurrent evidence with geometry-anchored scene-token completion before action prediction [338]. This distinction matters for classification: generating missing current state supports context-conditioned inference, whereas world-model-based control requires predicted consequences on the execution path.

Selecting, compressing, and using history. Compression must preserve decisive events. Selection and ordering matter in language ICL [124]; robotics adds brief physical transitions that change a trajectory's meaning. Robust Instant Policy's downsampling study identifies lost grasp transitions [201]; its full system also fits a policy at deployment.

The useful horizon follows the unknown. AMAGO's MetaWorld ML-1 analysis finds that short histories can identify hidden goals, while longer contexts can reduce sample efficiency [202]. ReLIC instead reuses experience across episodes to locate objects in an unfamiliar home, with deployed neural parameters fixed [203]. Section 6.2 relates this longer horizon to acquisition of spatial knowledge.

Retrieval must identify the event that resolves the current uncertainty. HALO learns relevance through question answering, while a calibrated history gate gives sufficient current observations precedence [105, 204]. Access resolution is another choice: RV-ICL organizes an annotated demonstration into task, phase, moment, and clip views that a frozen coding agent revisits as execution demands more detail [465]. This read-only hierarchy connects coarse planning to local interaction evidence without repeatedly loading the full video. Context repair addresses evidence missing from an initial answer: Find, Fix, Reason trains a student on targeted spatiotemporal patches supplied by a frozen teacher [453]. Selection and repair are assessed by whether the returned evidence changes the downstream decision.

For an archive $M_t = \{\xi_j\}_{j=1}^{N_t}$, ξ_j is its jth record and N_t is the number of records available at decision t. Let J be a candidate set of record indices and J_t the selected set. A budgeted retrieval rule is

$$J_t \in \arg \max_{J \subseteq \{1,\dots,N_t\},\, |J| \le K} \sum_{j \in J} \big[s_\theta(h_t, \xi_j) - \eta \big], \qquad R_t = (\xi_j)_{j \in J_t}. \tag{18}$$

Here $|J|$ is the selected-record count, K its budget, and η the relevance threshold. The score s_θ follows Equation (6), now querying an archive record. Fixed ordering and tie rules determine R_t. A nonempty read supplies the normalized action mixture in Equation (5); an empty read requires a fallback policy or evidence acquisition. This additive rule selects records independently; sequence-aware retrieval instead scores combinations when events depend on one another.

Motion-sensitive features preserve executed change and temporal order [205]. EvoMem-VLA encodes ordered pairs of observed frames as source-conditioned change tokens, retaining initial, recent, and selected historical states [469]. These tokens summarize what changed before the current decision; predicted futures instead specify or assess what should happen next. Offline training determines how the policy uses the memory, while episode-time updates change its contents with neural weights fixed.

Stage-dependent retrieval and action refinement. Relevant experience must match execution progress as well as object appearance. RTCF aligns history to successful trajectories and transfers bounded low-frequency action components, leaving the remaining motion and gripper decisions to a frozen policy [195]. Its fixed evaluation bank isolates trajectory use from archive growth. PACE instead writes realized action–observation transitions into an episode-local fast-weight MLP, whose readout biases attention over the demonstration [475]. Training supplies stage-alignment supervision; deployment uses online gradients without stage labels. Both use history to resolve repeated-looking stages. RTCF selects an action reference; PACE updates the neural state guiding retrieval.

Failure traces add evidence about continuations to avoid. TraceFlow steers a frozen flow-matching expert with a bounded field derived from progress-aligned successful and failed traces [390]. Its ordering gains coexist with counting and occlusion deficits: stage identification and action correction impose separate demands. SALT uses another temporal signal, the unexecuted tail of the previous plan [476]. A calibrated gate activates updates to bounded projection scales using that detached, time-aligned target; updates accumulate within the trial and reset each episode. The tail carries prior intent rather than observed effects. Its usefulness under visual disruption therefore depends on whether that intent still matches the physical state.

Persistence and update timing. Persistence depends on what survives between decisions or attempts. Recurrent state and verified subgoal records preserve evidence [206]; task memory can organize repeated calls to fixed primitives [207]. Fast-weight methods instead update policy parameters during a sequence [10] or before rollout [11]; an online critic changes another component while leaving the actor fixed [208]. HumanoidTTT places learning in memory maintenance: qualified motions are reused through a frozen tracker when entry conditions hold, while a Double-DQN scorer learns which motions to retain [449]. Freezing the executor therefore leaves two independent questions: whether stored evidence changes and whether the rule maintaining it learns. Resetting records and adaptive parameters separately isolates their contributions. TTT-RM combines bounded explicit history with fast weights that encode the history decoder's reconstruction residual [471]. Online gradients adapt those fast weights while slow weights remain fixed; resetting both carriers separates stored observations from learned residual memory.

Table 7 compares information roles and validity; Table B2 in Appendix B compares storage and updates. One memory design can serve several roles.

Capacity, access cost, and adaptation speed impose different constraints. KV caching saves prefix encoding while attention still grows with history; recurrence and fast weights compress it, and retrieval selects from an archive. Pre-rollout adaptation adds preparation time. Section 7 compares decisive-event retention and feedback-to-action delay.

Exposing changes the controller can execute. Memory makes evidence available; the command interface determines whether the inferred change can affect execution. ReSteer exposes the state dependence of this interface: policies that complete two instructions from their usual starting states can fail to switch between them midway [108]. Transition data and policy refinement improve switching. The implication for contextual control is that a new instruction or correction must be actionable from the state reached under the previous one.

Command granularity determines which correction can be expressed. Steerable Policies exposes subtask descriptions, motion instructions, and pixel targets [210]. The 2026 RoboPrompt system translates sparse points, trajectories, or directions into action drafts, then refines them through a frozen base policy [482]; its translator is trained beforehand. CIRRA instead revises an explicit skill queue through clarification, preemption, or fusion with the ongoing plan [486]. These interfaces modify an action proposal or its orchestration using available skills. The subsequent DAgger stage in RoboPrompt changes policy weights and constitutes a separate learning phase.

Table 8 External artifacts used with fixed neural weights. Plans can guide the current task; retained guidance and callable routines can support later tasks. Table B3 compares their update scopes.

Artifact	Execution content	Examples
Plans and programs	Procedure; controller	Code as Policies [27]; ProgPrompt [28]; Show-Harness [31]
Reusable guidance	Corrections; constraints; patterns	DROC [46]; ASPIRE [47]; ARCHITECT [48]; WAA [386]
Executable repertoire	Callable skills; composition	LRLL [50]; RATs [52]; RoboRSI [347]
Execution oversight	Context; verification; intervention	SHAPER [49]; Zetta [90]

Correction requires available evidence, an expressive command interface, and execution from the current state. These dependencies span all four families. Agentic adaptation additionally revises the records, programs, or context-construction procedures supporting later decisions.

3.6 External knowledge reuse and revision

External reuse preserves guidance, callable routines, or context and verification code beyond the originating attempt. Guidance changes later inference; routines extend the execution repertoire; revised code changes how evidence is used. Table B3 in Appendix B distinguishes these update objects. Separating acquisition, reuse, and neural consolidation connects immediate adaptation to improved learning on subsequent tasks.

To separate within-attempt inference from persistent adaptation, let B_n collect the external artifacts retained before attempt n: memory records, skill code, controller code and its numerical settings, or context-construction code. Let $\mathcal{T}_n$ be the executed trace and feedback from that attempt. A proposal-and-validation loop has the form

$$\begin{aligned} &\widetilde{B}_{n+1} \sim F_\theta(\cdot \mid B_n, \mathcal{T}_n), \\ &B_{n+1} = \begin{cases} \widetilde{B}_{n+1}, & \chi_n = 1, \\ B_n, & \chi_n = 0, \end{cases} \qquad \theta = \theta_0. \end{aligned} \tag{19}$$

The fixed model F_θ proposes $\widetilde{B}_{n+1}$; the indicator $\chi_n \in \{0, 1\}$ accepts or rejects it after validation, which may require further rollouts. The effective policy $\pi_\theta(A_t \mid h_t, C_t; B_n)$ exposes dependence on accepted artifacts for context construction or command execution.

Accepted code and settings can select executor κ_{B_n} or determine program g; its learned neural parameters remain in θ. Systems without an acceptance gate set $\chi_n = 1$. Attempt index n differs from decision index t: acquisition can span multiple model calls and transitions. Section 7.2 includes acquisition and validation trials before measuring reuse.

Table 8 groups artifacts by execution role; Appendix Table B3 gives their mechanisms. Plans specify steps, guidance retains applicable rules, and repertoires expose callable routines. Programs or their settings determine the task controller. Execution oversight constructs inputs or supplies a separate verifier; checks within a task program belong to that program. Guidance and callable routines can coexist.

Two uses of the retained artifacts explain how this update reaches control:

$$\begin{aligned} &\text{Guidance reuse:} \quad C_t = b_\theta(C_t^{\text{task}}, m_t, R_t; B_n), \quad A_t \sim \pi_\theta(\cdot \mid h_t, C_t), \\ &\text{Executable reuse:} \quad g = f_\theta(C_t, h_t; B_n), \qquad\qquad A_t = \kappa_{B_n}(g, o_t). \end{aligned} \tag{20}$$

The builder and memory inputs follow Equation (17). Guidance changes the inputs to otherwise unchanged control; executable reuse exposes library B_n to specification generation and executor κ_{B_n}. Equations (19)–(20) thus connect acquisition to later behavior with neural weights fixed.

Grounding an interface and learning through it. Show-Harness separates semantic motion commands from their calibrated, body-specific interpreter [31]. Video can supply a missing task order. Reading explicit command conventions is easier than discovering arbitrary meanings through probing: one supplies correspondence, the other infers it from action effects. Both share the same low-level interpreter.

High-level commands delegate grasp synthesis and feedback control; lower-level commands require more model-side geometry and timing. Guava compares workflows and action abstractions [211], while compositions of frozen motor skills, planning, and verification expose their system-level effect [212]. Context interventions isolate what teaching adds to this existing competence.

SkipVLA makes this allocation explicit within a hybrid policy: a classical planner handles free-space transit, while the instruction-conditioned VLA handles contact-rich motion [339]. Its learned target predictor connects the two. The routing design improves execution efficiency by assigning transit and contact to interfaces suited to their respective demands.

Outcome-conditioned revision extends open-loop synthesis. Feedback revises plans [186] or Python controllers [213]; executable checks handle some responses without another model call [28]. Constraint violations narrow revisions to code and sampling bounds [214], while primitive errors guide control-setting changes [9]. The interface defines the available edits; Section 5 asks which edit a failure justifies.

Retaining guidance and executable routines. Reuse requires identifying what remains valid after repair. Constraints preserve scope [46]; causal annotations explain effects [215]; distilled rules support synthesis [48]. Attribution separates failure to follow valid guidance from a defective skill requiring revision [216]. Recursive Harness Distillation revises intervention playbooks using recipient feedback [428]. Robot-body assets and synchronized interaction records instead ground controller calls [429]. Diagnosis determines what to retain.

Visual evidence makes retained guidance more explicit. WAA turns expert demonstrations and human corrections into procedures with applicability conditions, reference images, and outcome checks [386]. In its one-demonstration stove study, review replaces knob motion as the success criterion with an observed activation signal. Fixed agents then interpret the acquired guidance.

Executable reuse makes recurring behavior callable. Function synthesis supplies missing operations [27]; program clustering and human feedback refine reusable candidates [50, 51]. Video-grounded retrieval supplies implementation details such as contacts and target poses [217], while repeated successful code can enter a library [166]. Transfer then depends on scene-specific arguments and valid execution preconditions.

Autonomous trials acquire different reusable objects. Controller rewriting improves an implementation [218]; task-agnostic play accumulates callable code [52]; repair distillation retains failure conditions and strategies [47]. ProactiveVLA changes which experience is collected: after the initial task, remaining turns probe affordances, state changes, and compositions [474]. Outcome checks support rules consolidated across adaptation scenes, then frozen for evaluation alongside the models. Exploring beyond a successful solution can thus supply guidance for changed conditions without expanding the learned motor repertoire.

Reusable libraries need failure attribution and regression checks. RoboRSI's callable task–skill code yields 174/600 versus 129/600 successes without code across 120 tasks and five layouts [347]; its separate training branch consolidates corrections in weights. RPG diagnoses failures from simulation and videos, revises skills and prompts, and checks local and merged effects before frozen deployment [439]. DependEval tests dependencies, repository construction, and coordinated multi-file edits [454]. Physical regression and software consistency protect different aspects of reuse.

Transfer also determines when knowledge may change. Skill2Real validates local guidance and higher-level compositions in simulation, then freezes both memories for physical deployment [462]; code sketches guide synthesis through existing APIs. RoboBridge instead revises guidance from target-side outcomes, comparing initial transfer with ten update episodes [463]. Its shared interface links different pretrained VLA checkpoints; action-gradient refinement changes actions with weights fixed. Procedures, neural weights, and executors thus have distinct transfer conditions.

Semantic effects can remain reusable across different geometries. Teach and Grow's Skill Blocks specify effects, scope, executors, verification, and recovery [53]. Its simulation pilot expands a library with deterministic decomposition and a fixed executor; agent-based induction and Experience Memory updates remain proposed mechanisms for learning how to build that repertoire.

Reuse can encompass a complete closed-loop policy. Programs refined through demonstrations and simulation become examples for new policy generation [340]; generalized TAMP tests frozen programs on unseen instances [405]. RIVET composes perception and planning programs through object poses and a relation graph [406], while HuGo revises policy code over a frozen whole-body controller [407]. InterEvolve instead revises staged reward programs and tunes their constants with CMA-ES, retaining verified programs for later tasks [440]. Its fixed controller translates each active reward into motion. These interfaces expose different editable strategies; evaluation beyond the synthesis trials tests whether the acquired logic transfers.

Adapting how evidence reaches control. Retained guidance improves how available routines are invoked. Parameterized call sequences preserve useful orderings, while global heuristics describe the operating conditions and failures of fixed primitives [207]. Hi-VLA's tabletop study likewise finds greater benefits from prior-episode affordance summaries than longer within-episode history [219]. Such memory captures conditions of use, connecting agent reasoning to system identification of available capabilities.

Ledger separates persistent object records from short-term perceptual memory, and DynaMem updates spatial-semantic point clouds as objects move [426, 433]. Both make scene history usable by a planner; scene retention and acquisition of a new task remain distinct benefits. RobotUse separates local subagent histories from a persistent cross-attempt playbook, returning compact scene, outcome, and unresolved-constraint summaries to the main agent [470]. Guidance changes while models and the geometric execution backend remain fixed. MarvisNav exposes spatial memory through candidate routes: exploration history is rendered onto current views so a frozen VLM selects a route index for a classical planner [466]. Decision-aligned memory reduces the need to reconstruct spatial evidence from a separate history transcript.

Repeated selection or verification failures motivate revising the procedure that supplies context. SHAPER changes procedural guidance and context construction [49]; Zetta changes critics and recovery tools [90]. RACaP evolves Policy APIs, its ReAct harness, and memory before deployment [378], while RoboFoundry promotes validated context and skill-system changes [430]. PhysEvo extends revision to the meta-agent's diagnostic tools: a frozen model uses separate task and meta contexts to improve both execution artifacts and the means of inspecting failures [484]. Held-out evaluation then freezes the acquired artifacts. External revision changes what the controller can use; runtime context determines how those resources apply.

Compiling repairs moves computation from runtime to development. HarnessPAI retains observation-conditioned programs and revises them through stage-indexed repair memory; 15 of its 50 evaluation seeds also support program evolution [396]. Embodied Turing Machines develops explicit perception and control code, then runs fixed programs with changing state and no VLM calls [494]. Its simulation experiments rely on asset and calibration priors. Stateful execution demonstrates what the compiled controller retains, while fresh-task acquisition tests what its developer can learn. Development interactions and runtime computation belong to separate cost accounts.

Connecting contextual acquisition to neural consolidation. Neural consolidation changes the executor itself. Harness traces [211, 386], autonomous collection [220], and verified solutions [380] supply training data; CAPEX amortizes collection-time reasoning [432]. RoboCoach uses imagined failures to select demonstrations for LoRA skill updates while keeping its router, world model, judge, and backbone fixed [442]. Robo-COP additionally validates candidate policy updates and revises memory guidance after adoption or rollback [485]. A warning about an old policy's weakness can otherwise suppress a newly acquired capability. Retained context must therefore track the executor version it describes. Training on successful teaching exchanges, as in LMPC [117], targets the successor's responsiveness to later feedback.

Takeaways

① Context determines an action distribution, a geometric reference, predicted consequences, or an execution specification. Equations (4), (7), (9), and (13) identify these four interfaces.

② Context interpretation and motor competence limit transfer separately. Retrieval and memory support transfer by supplying usable evidence to a controller that responds to later corrections.

③ With neural weights fixed, context can revise task outlines, executable routines, and programs. Retaining these artifacts supports reuse, and successor comparisons measure their contribution to learning subsequent tasks.

4 Data and Training: How Is Context Use Learned?

An input channel makes context accessible; training determines whether the policy uses it. When the current scene already identifies every target action, a model can fit demonstrations without learning to consult earlier teaching. The central data question is therefore which relationships make earlier evidence informative about a later decision. Acquisition sources determine what is measured, training units establish these relationships, and construction at scale supplies their missing variations. This connects the transfer assumptions of Section 3 to the experience needed to support them.

Source taxonomies, such as the Data Pyramid survey [221], describe where embodied data originate and how closely they match robot execution. ICL for robots additionally asks what earlier teaching or interaction reveals about a later decision. A record's source, channels, and role in learning are therefore separate dimensions. Explicit pairs, continuous streams, and linked attempts organize these dimensions differently; pretrained interpretation and geometric construction can also connect evidence to execution without jointly training the entire path.

4.1 Input sources and context modalities

An acquisition source determines what is measured directly and what must be inferred. Teleoperation records robot commands, UMI estimates motion through a robot-like device, egocentric video records human activity, and simulation provides modeled transitions. The four sources are compared on this common basis.

Language can accompany every acquisition source. Instructions specify constraints, command–program examples establish conventions, corrections revise decisions, and summaries retain earlier events. Code as Policies, DROC, and MEM illustrate the latter three roles [27, 44, 46]. Records should preserve each statement's source, time, and referent: a teacher's requested change and a model's interpretation of an event carry different evidence.

Figure 13 contrasts the directly observed interaction: robot execution in teleoperation, instrumented hand motion in UMI, human activity in egocentric video, and modeled execution in simulation. These sources leave different correspondence problems for the learner, as the following parallel comparisons explain.

Teleoperation: measurements and commanded interventions. A real robot provides synchronized scene observations, measured robot state, and issued commands. DROID and BridgeData V2 supply such robot demonstrations [56, 58]; RH20T additionally records force–torque measurements and, in one hardware configuration, fingertip tactile signals [59]. These records can connect an attempted intervention to the body's response. A gripper-close command specifies an action, while measured aperture and contact help establish whether the object was grasped. Joint torque or motor current also requires interpretation through the robot's mechanics before it can describe external contact.

UMI: tracked motion without the robot arm. UMI reduces collection dependence on a robot arm by preserving a gripper-like interface in the demonstrator's hand. Images and inertial measurements support device-motion estimation, while visual tracking estimates gripper width [62]. The resulting trajectory can supply end-effector targets without measuring arm-joint motion. Sharing the end effector between collection and execution reduces a further source of correspondence error [66]; synchronized tracking improves the fidelity of the recovered motion [223]. Extensions address acquisition, whole-body realization, and richer interaction sensing [224–226]. Their common trade-off is deliberate: an instrumented interface restricts how a human demonstrates, but reduces the inference needed to turn that demonstration into robot-compatible evidence.

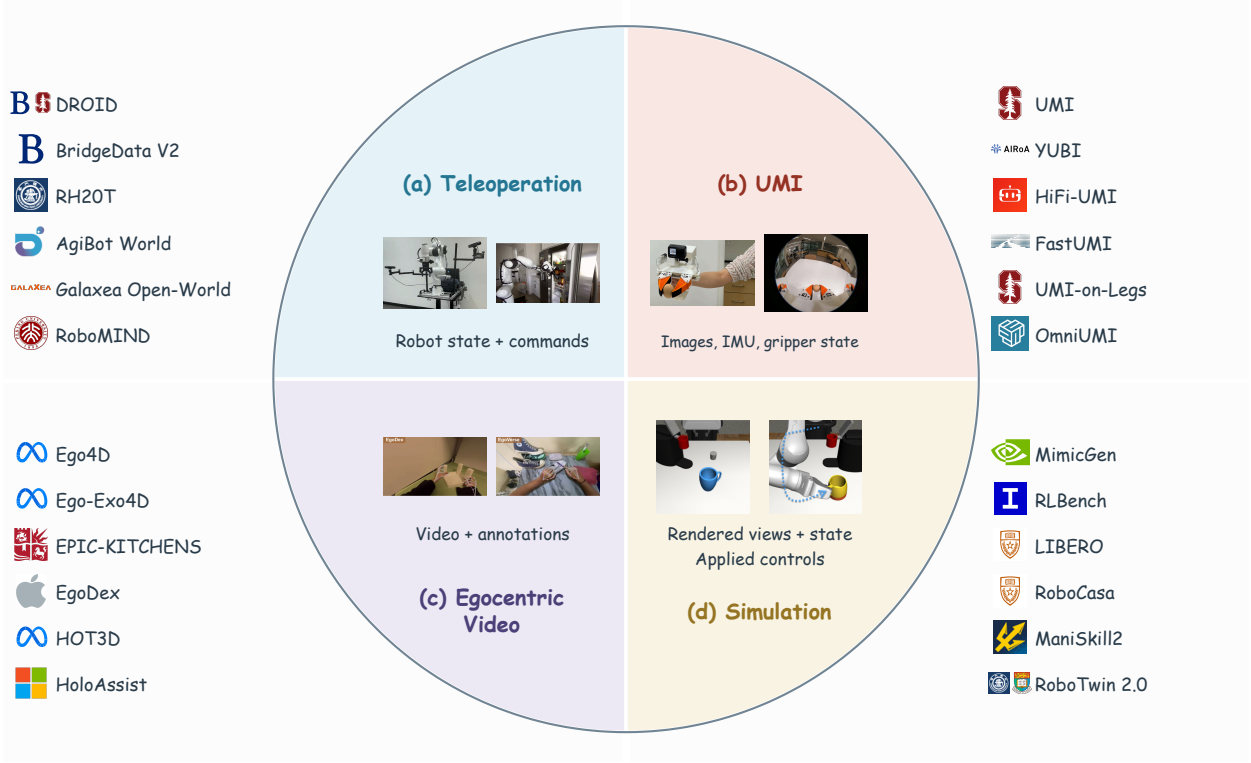

Figure 13 Four sources of contextual evidence for robots: (a) teleoperation, (b) UMI, (c) egocentric video, and (d) simulation. Representative resources include DROID [58], UMI [62], Ego4D [68], and MimicGen [74], respectively. Their recordings differ in visual, state, action, and contact channels; language can accompany each as instructions, examples, or feedback.

Table 9 Acquisition interfaces compared by state, actions, and contact evidence. UMI pose and width are estimated; extra sensing and simulated contact records depend on the setup. Table B4 lists resources.

Source	State	Actions	Contact
Teleoperation	Images; joints; gripper	Recorded commands	Force; tactile (optional)
UMI	Images; IMU; gripper pose	Mapped gripper targets	Visual; force (optional)
Egocentric video	Video; optional sensing	Inferred or paired actions	Visual interaction
Simulation	Rendered views; state	Simulator controls	Modeled contact

Egocentric video: visual procedure with optional human sensing. Egocentric recordings expose object interactions and their order without requiring a robot-compatible device. Ego4D and EPIC-KITCHENS-100 provide egocentric activity, and Ego-Exo4D adds synchronized external views [68–70]. Some collections also provide audio, gaze, inertial measurements, depth, or tracked hands. They measure human motion and the surrounding environment. Robot action targets must come from paired execution, reconstruction and retargeting, or a controller that grounds the observed procedure. An action-free video supplies visual observations without recorded robot commands.

Simulation: rendered observations and modeled responses. Simulation makes action consequences and privileged state available under a specified dynamics model. Demonstration environments such as RLBench, LIBERO, and ManiSkill2 support task execution with synchronized observations and controls [227–229]; trajectory or task-program construction produces additional executions [74, 230]. This control over the generating process is useful for contextual learning because a procedure can be held fixed while scene geometry changes, or varied while the scene remains comparable. Privileged object and contact states can verify

Table 10 Context units, retained evidence, and adaptation roles. Support actions may enter context; future query actions are training targets.

Context unit	Evidence	Role
Support example	Demonstrations; paired examples	Task; procedure; contact
Scene preview	Exploration views; spatial records	Layout; locations; routes
Query history	Observations; actions; dialogue	Progress; recent response
Transition sequence	Observation–action–outcome	Physical response
Cross-attempt record	Episodes; summaries; corrections	Reuse across resets

Table 11 Three training organizations compared by evidence, supervision, and record relationship. Support can omit robot actions; linked attempts can use predictive, corrective, or reward-based supervision.

Organization	Earlier evidence	Learning signal	Link between records
Paired episodes	Support demonstration	Query actions	Shared task or environment
Continuous stream	Observation/action history	Subsequent actions	Temporal continuation
Linked attempts	Trials; outcomes; corrections	Selected actions or feedback	Consequences across resets

those relationships during construction, even when the deployed policy receives cameras and proprioception. Section 4.5 examines how synthesis preserves the assigned teaching condition.

Channels and policy inputs. Table 9 distinguishes measured state, commands, and contact evidence. Collection channels need not enter the policy: UMI uses IMU data for pose estimation, while its policy observes images, end-effector pose, and gripper width [62]. Specify available fields separately for support, query, and supervision: human video may supply support while the robot queries joint and contact sensors; recorded support commands may condition inference, whereas future query commands supervise it. Measurements o_t and issued actions a_t form history h_t; selected records, language, and feedback enter context C_t.

Combining channels also requires preserving disagreement. TUNED combines feature and neighborhood evidence for multi-view classification [450]. For contextual control, the relevant question is whether conflicting observations remain distinguishable before fusion commits the controller to one interpretation: contact feedback can contradict an otherwise plausible visual estimate.

4.2 Training units and context dependence

Channels become useful training evidence through temporal and task relationships. Table 11 compares paired episodes, continuous streams, and linked attempts. We define the temporal units first, then use support–query supervision to distinguish input channels, action targets, and pairing criteria. Continuous experience and corrective records extend this dependency beyond explicit pairs.

Temporal units of contextual evidence. A contextual training unit must identify what was observed, what was commanded, and what the resulting feedback established. Language can specify the intended task or annotate the outcome. The sequence must preserve when each item became available, so that a later consequence can supervise learning without appearing as evidence for an earlier decision.

Temporal alignment preserves the relation between an intervention and its consequence through sensor timestamps, command application times, coordinate conventions, and episode boundaries. Measured motor state must remain distinguishable from its command or setpoint. Aligned windows or timestamp-aware encoding accommodate different sampling rates while retaining brief contact events. Missing channels and reconstructed motion must also remain identifiable, because they supply different evidence from direct measurements.

Table 10 groups records by what they contribute to learning. Support can specify behavior or reveal an environment; query history locates execution; transitions reveal action consequences. These units overlap:

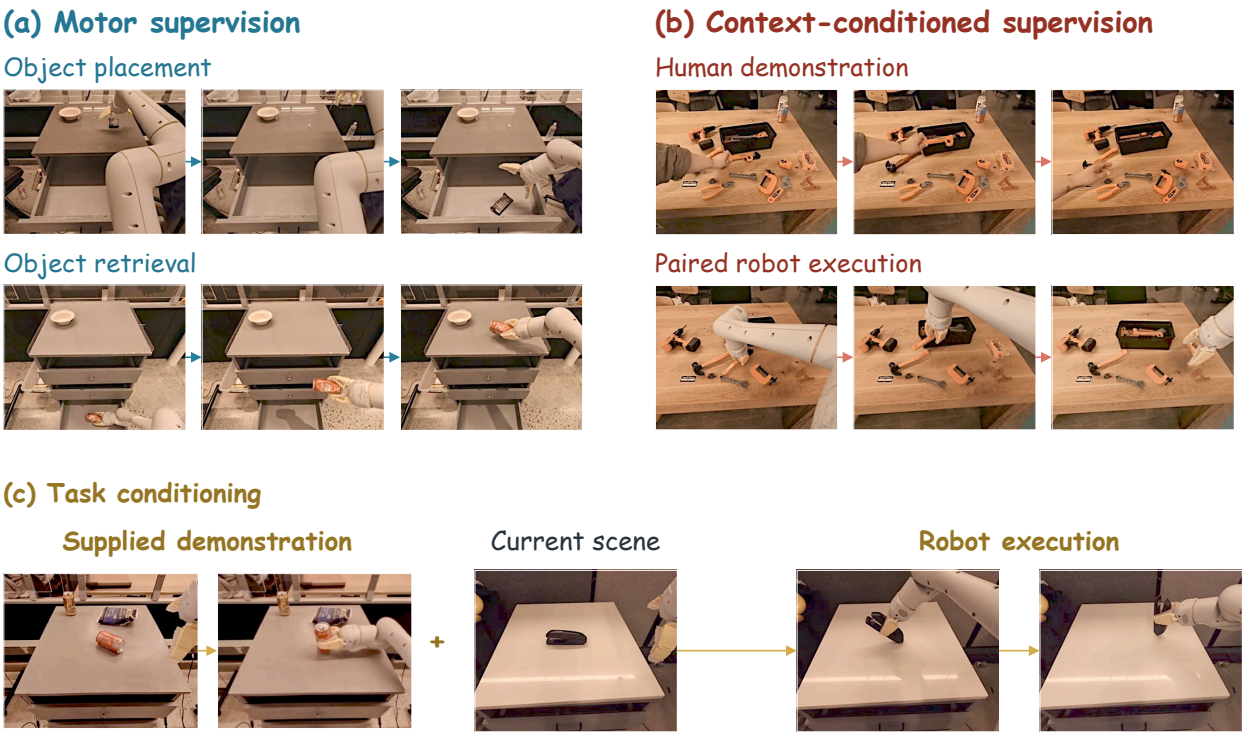

Figure 14 Demonstrations serve three roles: (a) supervise motor actions, (b) train context use through paired episodes, and (c) specify a new task to a fixed policy. Panel (c) illustrates a can-uprighting demonstration specifying stapler reorientation; it is a conceptual example of cross-object transfer.

history contains transitions, and retrieved segments can come from demonstrations or earlier attempts. Their role is independent of modality. Images or synchronized robot records can supply a demonstration, while state–action history can support physical adaptation without one, as in RMA [39].

Supervising a query from a support episode. Following the paired-execution principle of one-shot imitation [1], let support demonstration D^{s} and query observation–action trajectory τ^{q} share task specification ψ. The matched executions may differ in objects, timing, or embodiment. A common supervised objective predicts query actions from their history and support:

$$\mathcal{L}(\theta) = \mathbb{E}_{(D^{\mathrm{s}},\tau^{\mathrm{q}})\sim P_{\mathrm{pair}}}\left[\frac{1}{|I^{\mathrm{q}}|}\sum_{t\in I^{\mathrm{q}}}\ell_\theta(A_t^{\mathrm{q}}; h_t^{\mathrm{q}}, D^{\mathrm{s}})\right]. \tag{21}$$

Here P_{pair} samples executions matched by specification ψ. Query history h_t^{q} ends at the current observation, while $A_t^{\mathrm{q}} = a_{t:t+H-1}^{\mathrm{q}}$ is the recorded target block. The nonempty set I^{q} contains positions with complete targets; averaging over it normalizes each episode's contribution. The loss ℓ_θ may use likelihood, regression, denoising, or flow matching. These objectives differ in action modeling but share the same supervisory dependency: the support must explain which query behavior is appropriate.

Equation (21) specializes the general context interface to $C_t = D^{\mathrm{s}}$. In instruction-bearing episodes, the same query-action loss conditions on C_t containing both the demonstration and its associated text. Instructions specify goals and constraints; demonstrations can convey remaining choices of order or contact. A correction enters context when received, preserving what was available before each supervised action.

Figure 14 distinguishes motor supervision, training to use teaching, and deployment-time task specification. Uprighting a differently shaped object illustrates the third role: the operation must survive changed appearance. Paired supervision learns this dependency explicitly; continuous streams can expose it through the sequence of experience.

Support channels and action supervision. Equation (21) supervises query actions while leaving support channels open. Vid2Robot conditions on video without recorded support commands [14]; ICRT supplies support actions in a sensorimotor prefix [13]. Both supervise the query execution. Support content, the action loss, and the architecture that combines evidence are therefore separate choices.

Task identity and support–query pairing. The shared specification ψ determines the granularity of teaching. Pairing all executions labeled "put the object away" can preserve a destination while discarding grasp or route preferences. BPP makes this dependence explicit in its task-grouped demonstrations [54]. The pairing rule must retain the distinction the deployment example is expected to convey. For cross-object transfer (Section 3.5), support and query can change the manipulated object, the receiver, or both while preserving the same requirement. Conversely, retaining comparable objects while varying the required relation makes the support informative. Together, these pairings separate task meaning from object identity and provide examples of which substitutions preserve the intended function.

Pairing requires both semantic agreement and compatible action meaning. Additional demonstrations can distinguish task interpretations left ambiguous by one example, as Instant Policy illustrates with differently shaped boxes [16]. The resulting query target must still express the same intervention in consistent coordinates, timing, and gripper semantics. WALL-OSS-0.5 addresses this physical side through synchronized video and actions, camera repair, and end-effector calibration [233]. These requirements are complementary: semantic pairing determines which behavior is intended, while calibration determines what command realizes it. With shared action coordinates, each body retains its own reachability and contact mechanics.

A shared environment supplies a different pairing relation from a shared procedure. Route-following training pairs a demonstrated path with a perturbed execution, as in RPF [355]. Scene-conditioned training instead pairs a preview with navigation to goals within that environment, as in NOLO [350]; ReLIC [203] links episodes in one home while changing the target and start. The shared variable is therefore the route in the first case and the environment in the latter two. Scene identity and memory-reset boundaries determine which spatial knowledge remains available across examples or episodes.

From paired episodes to continuous experience. Explicit pairing makes the teaching–execution relationship part of the training unit. S1 supplies an in-context demonstration in its pretraining episodes [99]. GEN-1.5 instead reports next-action pretraining on randomly sampled continuous physical spans, without dedicated demonstration packing or an ICL-specific objective, and inserts sensorimotor examples as prompts at deployment [98]. A stream exposes earlier observations and actions without assigning them separate support and query roles. Both systems later accept task examples, acquired through different organizations of training experience. Action generation is a further choice: ContextFlow uses explicit robot support–query supervision with continuous flow matching [55]. The action loss and the organization of experience play different roles: one fits the action model, while the other links preceding evidence to subsequent behavior. Cross-Episodic Curriculum orders trajectories by learning progress or demonstrator quality [435]; controlled MiniHack and Procgen studies link task diversity and trajectory burstiness to new-task ICL [434]. These training results explain why history can become useful, without establishing the same transfer on physical robots.

Behavioral diversity and contextual identifiability. Access to earlier evidence becomes useful when it resolves uncertainty about the continuation. If the current scene already determines every training action, the loss can be minimized without reading a demonstration. Comparable scenes with different valid procedures instead give teaching predictive value; repetitions of one procedure across scenes teach which variations leave it unchanged. The practical consequence is to count distinguishable teaching problems as well as trajectories. The comparisons below examine this dependency, while Section 8.1 asks how far the resulting inference transfers.

Context can also support shortcuts: previous actions or a familiar scene may predict the training target without specifying the requested behavior [234]. CRAFT counters this dependence by varying the requested skill composition for the same observation and aligning separate skill and state representations [457]. Its counterfactual objectives train the VLA offline; deployment uses observation and instruction. Held-out compositions combine familiar constituent skills, making this evidence about compositional instruction use.

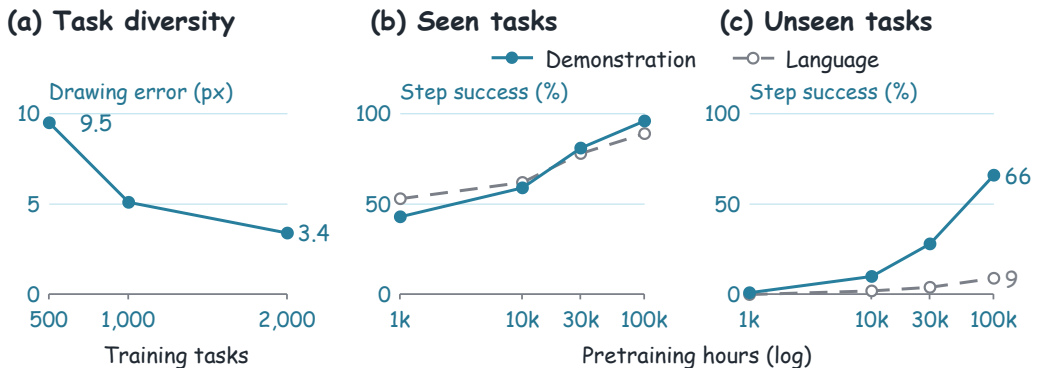

Figure 15 Task diversity and prompting affect transfer differently. (a) BPP drawing error with 10,000 demonstrations, averaged over three seeds [54]. (b,c) S1 compares demonstration and language prompting on seen and unseen tasks; its cumulative per-step success measure includes human recovery interventions [99].

For demonstration ICL, the corresponding test changes the taught relation while preserving the scene. Both designs ask whether task-defining evidence controls behavior beyond collection correlations.

The evidence distinguishes behavioral diversity from visual variety. ICRT's selected DROID-only condition fails its test tasks; its multi-task scenes instead require different actions from similar observations [13]. At a fixed 10,000-demonstration budget, BPP's mean drawing Chamfer error falls from 9.5 pixels for 500 tasks to 5.1 for 1,000 and 3.4 for 2,000 (Figure 15a) [54]. Its folding study can favor language conditioning. Its architectural comparison attributes much of BPP's advantage over ICRT to the diffusion action decoder, separating decoder quality from prompting. Demo-JEPA varies task classes and trajectories per task independently [22]. Useful diversity adds distinctions that teaching must resolve and the action decoder must realize during execution.

S1 separates scaling gains on seen and unseen tasks under matched data, compute, and architecture apart from the prompt embedding [99]. Across 1k–100k training hours, both prompting methods improve on seen tasks. On unseen tasks, demonstration prompting rises from 1% to 66%, while language prompting rises from 0% to 9% (Figure 15b,c). Thus similar seen-task performance can conceal different returns to new-task teaching. The cumulative per-step metric includes human recovery, mainly for the language baseline. The S1 comparison concerns combined data scaling and prompting, while BPP directly isolates task diversity at a fixed demonstration budget. Matching the procedural information conveyed by text and video would further isolate modality from teaching detail (Section 7.1).

Linked attempts and corrective supervision. Learning from a failed action or a diagnostic probe extends context dependence across attempts. The training unit must link what was tried to what happened next, preserving reset boundaries and any change in the task or physical condition. Algorithm Distillation models such learning histories [6]; LocoFormer retains experience over varied morphologies and dynamics [40]. Reward-based meta-training can value an informative action through its benefit to later attempts [2]. A short probe gains learning value when its outcome informs the next decision. Linked attempts expose this dependency during training.

In recovery-conditioned supervision, the failed attempt and its feedback enter context while a later continuation supplies corrective targets. RoboTTT's DAgger Distillation uses the full rollout as context but applies action loss only to human corrections, also improving its GDN comparator [10]. The failed actions record what was tried; the selected corrections specify what to imitate. This training organization can support different memory mechanisms.

Equation (21) extends to this case by replacing support-only conditioning with C_t. For block-level losses, I^{q} includes only complete blocks composed entirely of actions selected for imitation. An objective supporting action-wise masking can instead suppress unselected positions within a block. Other executed actions remain in history. The context mask controls available evidence, whereas the target mask controls desired behavior. Runtime context contains only evidence available before the decision; future outcomes may supply supervision. Attempt boundaries must separately record physical resets, memory resets, and changes of tool or dynamics.

Long-context control requires representations that preserve decisive events. Past-Token Prediction adds reconstruction of earlier actions to future-action prediction, first learning short-context visual features and then caching them for longer-context training [235]. Caching reduces repeated computation; the cached visual features determine which task details remain available to the controller.

The supervisory unit determines what evidence can explain a later action: pairs link teaching to realization, streams link earlier events to current control, and linked attempts connect outcomes to revised decisions. Their common requirement is a decision that earlier evidence helps disambiguate. The remaining analysis examines acquisition sources that preserve this dependency with usable action correspondence.

4.3 Data coverage and action correspondence

Data coverage has two distinct meanings: the range of behavior recorded and the range of teaching–execution relationships recoverable from it. Robot records, motion-bearing human demonstrations, and ordinary video trade direct action correspondence against collection breadth. The comparison below follows that trade-off, asking which annotations make each source usable for contextual training. Simulation supplies controlled counterparts in Section 4.5; Table B4 lists resource-level details.

From reusable experience to annotated procedures. Motor coverage supplies reusable behavior before teaching can specialize it. Autonomous multi-task experience and shared demonstration collections addressed this need [236, 237]; Bridge Data included 7,200 demonstrations across 71 tasks and ten environments. Later collections broadened scenes and embodiments [56–58]. Contextual training additionally needs relationships between these executions that preserve what a new example should teach.

Procedure annotations identify why a continuation changes. AgiBot World's March 2025 release spans 1,001,552 trajectories, 2,976.4 hours, 217 tasks, and 106 scenes, including subtask and recovery records [60]. Task intervals, step instructions, and later corrections connect demonstrations, autonomous attempts, and human interventions [238, 239]. Linking an attempt to its correction preserves the evidence that makes a different continuation appropriate.

Embodiment diversity separates task correspondence from action correspondence. Galaxea shares a platform and subtask timing [61]; RoboMIND 2.0 and RoboCOIN span more platforms [240–242]. Procedural metadata establishes task equivalence; kinematics describes the differing realizations. Pairing needs both.

Robot records supply executable supervision. Robot records become contextual supervision when a query execution is associated with an example of its intended procedure [56–58, 60]. Cross-scene retrieval can locate candidate executions [243], but matching must distinguish alternatives that begin in similar scenes.

Human–robot pairing exposes an asymmetry between teaching and supervision. A human video can specify the intended interaction, while robot measurements reveal the commands and contact response needed to realize it. RH20T provides corresponding human videos together with robot images, joints, gripper state, force–torque signals, audio, and tactile sensing in one configuration [59]. These richer query records can supervise a policy whose eventual human support contains fewer channels. Task-level human–robot organization helps define the transfer setting [101], but a shared label such as *fold* can still merge different procedures. Matching ordered events and terminal outcomes, as in WorldScape's robot and UMI pairs, preserves a finer teaching relation [25]. Pairing granularity therefore determines whether the model learns an activity category, a procedure, or an interaction detail.

Mixed capture can reduce pairing cost. XR-2 combines 531.7 hours of robot teleoperation with roughly 1,000 hours of dual-UMI demonstrations [244]. Robot views, state, and actions supply executable targets; egocentric views and gripper motion supply candidate support. Shared laundry tasks narrow the search, while procedure-level matching identifies compatible executions.

Motion-bearing demonstrations reduce correspondence uncertainty. Instrumented teaching trades restrictions on collection for less uncertainty in transfer. Handheld grippers make the demonstrator's observations and motion closer to a robot-compatible interface [62]. Preserving the end effector across collection and execution further reduces what must be inferred: YUBI reports 8,434 hours over 119 tasks

and transfers the same gripper between arms through per-robot inverse kinematics [66]. Sharing the gripper reduces hand-to-robot correspondence uncertainty, leaving each arm's reachability and control limits to be handled during execution.

Bimanual teaching makes relative motion fidelity more consequential than the visual plausibility of either hand alone. Stereo-inertial tracking, synchronized views, and inter-gripper reconstruction preserve this coordination [223]. HiFi-UMI's reported 2K release also includes subtask boundaries, and its policy studies use UMI post-training without target-task teleoperation. Calibration preserves the physical relation between hands; boundaries identify which procedure that relation instantiates. Modular tracking can simplify acquisition [224], while a simulation-trained whole-body controller can supply realization beyond the demonstrated gripper motion [225]. More accessible capture and more accurate coordination are consequently distinct improvements to the teaching interface.

Contact sensing extends what the teaching record can express. OmniUMI synchronizes vision and motion with tactile signals, internal grasping force, and external interaction wrench [226]. Such measurements can distinguish a visually similar motion performed with different contact conditions. Their transfer still depends on compatible sensing, calibration, and execution interfaces; a measured wrench and a commanded gripper setting carry different information.

Collection can be conditioned on the current policy's weaknesses. HIL-UMI compares handheld human trajectories with policy predictions on the same observation stream and targets additional data to disagreement and low-advantage segments [341]. The resulting demonstrations support further policy training. Thus policy-aware collection improves which experience is acquired, while remaining distinct from using that experience as fixed-weight deployment context.

Removing the collection device shifts this burden from measurement to inference. Tracked hands and segmented point clouds can be converted into gripper poses [16]; motion inferred from human play can supply pseudo-actions, retrieved context–query relations, and retargeted humanoid supervision [245]. These approaches broaden the usable recording pool, but the action evidence inherits reconstruction and retargeting error. The trade-off is most consequential near contact, where a small positional error can change whether a grasp or insertion is feasible. A common action format enables joint use of these records while their reconstruction reliability determines their supervisory value.

Mixed-source training can preserve this distinction instead of forcing all trajectories to carry the same supervisory weight. Aligning camera-space actions, embodiment, and chunk duration provides a common temporal interface, while reliability-weighted human pseudo-actions supplement a robot-supervised primary objective [246]. In ACE-Ego-0, the auxiliary objective emphasizes more dependable position channels. This weighting lets human records expand coverage while measured robot actions anchor execution. Carrying reconstruction confidence into contextual pairs likewise avoids treating uncertain hand tracking as an equally reliable robot target.

Zeva-Ego separates this acquisition of a physical prior from deployment adaptation [362]. Its Action-Centric Encoder converts egocentric visual transitions into continuous action-token supervision for mid-training. A later interaction-conditioned stage trains the action model to use observed action effects; deployment freezes the weights and updates memory. Human experience expands prior competence, while robot interaction supplies evidence about the current physical conditions.

Motion annotation reduces the uncertainty left by ordinary video while retaining a human-centered collection process. Hand–object geometry [63, 64] and contemporaneous hand and finger tracking, as in EgoDex's 829 hours [65], expose motion that otherwise must be reconstructed from RGB. Robot learning still requires conversion into feasible contacts and controls. EgoScale's 20,854 hours of action-labeled human data are connected to robot behavior through aligned mid-training and downstream post-training [67]; its one-demonstration experiment uses policy updates with one robot demonstration per task and additional human trajectories. These stages show why improved action evidence and fixed-weight task acquisition are separate contributions: the former can support either contextual inference or further policy fitting.

Figure 16 Temporal anchors for contextual teaching: opening the door enables loading, and closing it completes the procedure. These transitions align a teaching example with progress in a later execution.

The transition from human pretraining to robot deployment also changes the meaning of language supervision. Xiaomi-Robotics-1 combines over 100K hours of UMI and egocentric records with cross-embodiment post-training [247]. State-transition descriptions during pretraining and imperative instructions during post-training serve different prediction targets. For demonstration-conditioned learning, the corresponding design choice is which task details remain unspecified by the instruction and must be recovered from the example. This connects source breadth to the pairing criterion in Section 4.2.

Ordinary video broadens procedural coverage. Ordinary video reverses the trade-off of instrumented capture: activity coverage is broad, while robot-compatible action information is sparse. Egocentric activity labels help locate candidate procedures [68, 69]; paired viewpoints and short human–object changes offer cues about correspondence and outcomes [70, 71]. These sources can identify what happened without fully specifying how a different body should reproduce it. For contextual teaching, selection must preserve the choice that distinguishes this example from another clip of the same activity.

Annotation is useful when it removes uncertainty about the relation teaching must preserve. Common capture protocols reduce incompatible recording conventions [248]; atomic segments and reconstructed hand and camera motion expose where an interaction begins and ends [249]. Object-state and contact annotations can resolve physical events hidden in RGB [250]. These additions are not interchangeable: reliable boundaries help preserve order, whereas contact-sensitive transfer needs relative motion and object response. The appropriate annotation follows the task distinction the support example must convey.

Procedural and conversational datasets contribute another missing dimension. Assembly101 records natural variations, mistakes, and corrections during assembly and disassembly; HoloAssist pairs a performer's egocentric activity with an instructor's spoken guidance and intervention annotations [72, 73]. Their contribution is the relationship between a partial procedure and information that changes its continuation. Transferring that relationship to robot ICL requires robot-compatible continuation targets or an executor that can realize the corrected instruction. Preserving mistakes, instructions, and corrections as linked events retains this distinctive supervision.

Human-video priors and demonstration following use this evidence differently. Learning latent plans from human play can reduce the robot experience needed for control [251], even when human and robot executions are not paired. Following a newly supplied video additionally requires preserving the demonstrated choice in a compatible robot execution. WAM-TTT learns that connection through paired human–robot meta-training, while its deployment-time fast-weight update needs only action-free human video [11]. This is parameter adaptation under the distinction in Section 2. Robot supervision establishes the connection during training; the later teaching input can omit recorded robot commands.

Preserving a teaching relation requires more structure than a shared task name. A mobile-manipulation episode contains identifiable substeps and transitions that can anchor retrieval or a corrected continuation (Figure 16). Human support and robot execution can traverse those stages at different speeds, so task correspondence matters more than frame equality. Constructing this usable teaching signal incurs a cost beyond recording the raw activity, motivating the acquisition comparison below.

4.4 Acquisition cost and contextual capability

Data value depends on the cost of producing usable teaching, not recording alone. We first examine correspondence costs, then compare explicit contextual training with broad priors and robot grounding.

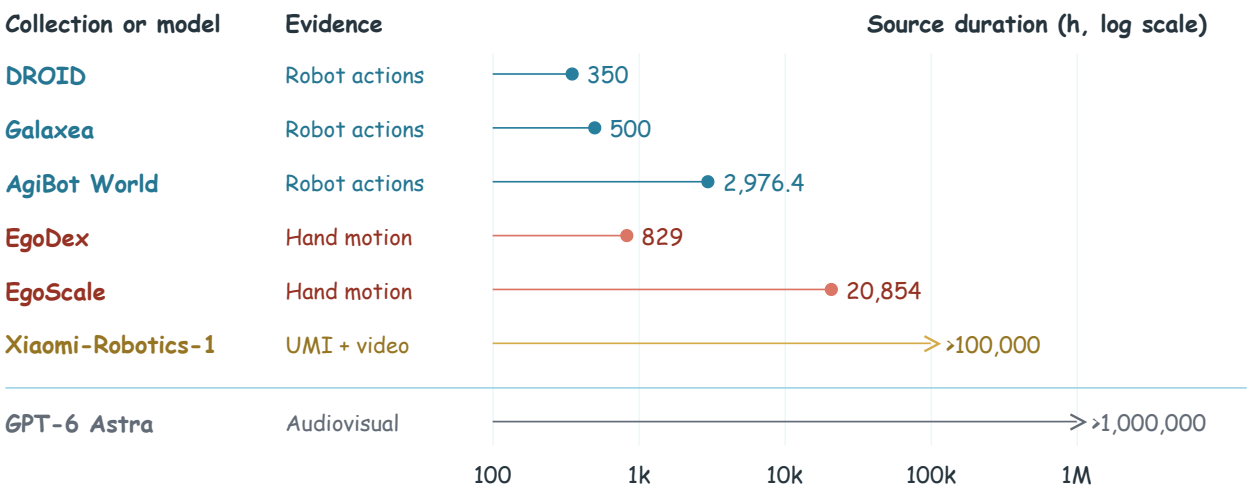

Figure 17 Reported source duration across embodied collections and general-model pretraining [58, 60, 61, 65, 67, 247, 252]. Action-aligned recordings support grounded teaching episodes; broader audiovisual corpora supply knowledge across activities and domains. The contrast motivates comparing explicit context-use training with broader pretraining. Arrowheads denote lower bounds.

Collection cost and correspondence. Filtering, synchronization, reconstruction, pairing, and validation contribute to acquisition cost. Egocentric video exposes hand–object interaction; UMI also measures device motion through a robot-like interface. The required conversion depends on the task: order-sensitive packing needs event boundaries and contrasting sequences, whereas tight insertion needs calibrated relative motion and contact evidence. Additional views or longer recordings improve coverage only if they preserve the distinctions needed for the task.

Figure 17 compares source duration and evidence type. Xiaomi-Robotics-1 combines UMI and egocentric recordings; general video spans broader activity with less robot-action correspondence. Hours measure exposure, while annotation effort and usable support–query pairs characterize the work needed to turn it into teaching.

GPT-Policy makes the contribution of correspondence concrete [94]. It compares robot video with the same video augmented by aligned end-effector poses, gripper states, and commands. Plug reinsertion succeeds in zero of three video-only trials and two of three augmented-reference trials. This is a combined state–action comparison. Goal-image and occluded-goal settings provide endpoint or self-interaction evidence (Section 6), showing how the content of the acquired evidence shapes its contribution.

Learning targets and acquisition routes. Pretraining can expand both executable behavior and the requirements inferred from teaching. Support–query episodes explicitly supervise the example–continuation relationship [1]; GPT-3 showed that broader pretraining can also support example-based specification at inference [121]. Both routes can yield fixed-parameter adaptation. Their transfer to unfamiliar rules is an empirical question distinct from the choice of training recipe.

S1 and GEN-1.5 illustrate explicit contextual pairing and continuous physical experience (Section 4.2). Their shared demonstration interface motivates comparing which training relationships produced context use, with tasks, information, motor interfaces, and training budgets matched.

Mechanisms of fixed-parameter context use. Fixed-parameter computations offer candidate explanations for context-dependent behavior. Bayesian accounts infer a latent rule shared within a sequence [318]; distributional studies identify repeated structure that favors contextual inference [111, 319]. Controlled models can bind items to labels and retrieve their association [232], or implement gradient-descent-like updates through linear self-attention [112]. Comparisons with LLaMA show that similarities between contextual and parameter learning depend on the setting [322]. These accounts distinguish what training makes inferable from the computation used at deployment. For robots, tracing a taught distinction into the representation that guides action connects the training relationship to its physical consequence.

General-model priors and robot grounding. Astra adds a different source of prior competence to this comparison. Its public training summary describes broad text, image, and audiovisual sources, including video sources above one million hours, represented partly through transcripts, captions, or metadata [252]. In GPT-Policy, a fixed general VLM interprets task references and interaction feedback, proposes robot-tool requests, and receives outcomes from a constrained controller [94]. This places contextual interpretation in a general model and physical realization in an explicit robot interface. The disclosure characterizes source exposure; the causal contributions of individual training components to the observed robot behavior remain unresolved.

Broad priors interpret procedures; grounded records connect them to robot commands. GPT-Policy's contact-sensitive action-reference comparison illustrates their complementarity. Compare task information and action correspondence under matched priors and executors.

From exposure to usable teaching. BPP varies diversity at fixed demonstration count, S1 varies scale and prompting, and general-model disclosures describe cross-domain exposure. Additional recordings can supply missing motions, distinguish procedures, or reveal repairs. Retrieval and synthesis should target the corresponding gaps in selection, composition, or rule inference (Section 8.1).

4.5 Scaling data construction

Data construction targets missing teaching relationships (Section 4.2): retrieval reuses evidence, spatial supervision and synthesis construct counterparts, and failure–correction collection supplies continuations absent from successful demonstrations.

Autonomous orchestration supplies executions [76]; reward and success signals select them for policy training [254]. Contextual training additionally needs each continuation's governing instruction, demonstration, or correction. Contrasting teaching–execution pairs reveal which behavior depends on that evidence. Acquisition aimed at motor competence selects successful motion; acquisition aimed at contextual responsiveness must also preserve the preceding uncertainty and its resolution.

Retrieval for training and contextual pairing. Training retrieval asks which external behavior can improve a robot's learned competence. RoboTok retrieves human clips through actor-centered hand-motion embeddings, then retargets trajectories into reward guidance for PPO training [255]. Semantic, hand-pose, and motion similarity can likewise select human samples compatible with robot demonstrations [256]. In both cases, relevance is defined by the supervision entering the policy's weights. This differs from contextual pairing, where the selected clip must explain a particular query procedure even if both behaviors are already within the motor repertoire. PHASE selects variable-length contact phases using tactile evidence, preserving insertion-stage compatibility in the retrieved training set [409]. This supplies a concrete counterpart to runtime retrieval: the selected evidence changes the policy learned offline.

Contextual pairing instead requires agreement with the intended procedure. RHyME composes human clips for a robot trajectory through sequence-level optimal transport [80]; shared constituent behaviors enable pairing, while clip boundaries can introduce errors. Runtime retrieval must also match current progress and motion compatibility. ReCAP supplies state–action chunks to a frozen residual predictor [152]. Expanding its archive changes deployment context, whereas training selection changes learned competence.

Constructing spatial supervision. Cross-scene point supervision teaches where analogous geometry occurs [161]; reconstructed contacts, hands, and motion teach plausible interaction geometry [176]. Contextual training links either prior to the example specifying which relation the robot should preserve. Spatial accuracy and responsiveness to teaching are complementary objectives.

One demonstration as a reference or a data seed. A demonstration can become a runtime reference, training data, or a learning objective. MT3 retrieves and aligns an executable reference [36]; bimanual and reconstructed-scene synthesis produce transformed training executions [257–259]. Dex-One2Many extracts a stage-wise relation graph from one human video and derives rewards, reset distributions, and state features for PPO [495]. Each embodiment then receives a newly trained policy; its zero-shot claim concerns sim-to-real

transfer. The same teaching budget can thus incur very different optimization costs. Contextual training additionally preserves the support–query relation so that the resulting learner can interpret another example.

Task-preserving synthesis. Synthesis must preserve the taught requirement while varying its realization. Human-arm renderings change appearance [260]; geometric pseudo-demonstrations and waypoints provide correspondence [16, 77]; simulated execution adds consequence and success checks [74]. Generated environments and task-derived costs broaden these variations [261, 262]. Keeping each counterpart tied to its governing task specification makes it useful for contextual supervision.

KnowDemo makes the preserved requirement explicit [368]. It extracts task conditions from human video, grounds them in the target scene, and screens alternative contacts and subtask orders before planning and simulation. The verified trajectories then train a robot policy. This connects behavioral diversity to task constraints: varying an incidental grasp can expand training coverage, while varying a required order changes what the example teaches.

Simulation provides controlled task and scene variation alongside executable action records. RoboCasa expanded household environments and compositional tasks in 2024; RoboCasa365 adds 365 tasks and 2,500 kitchen scenes, with 612 hours of human-operated demonstrations in simulation and 1,615 hours generated with MimicGen [75, 263]. These are simulated robot trajectories, including those collected by human operators. For ICL, the opportunity is to hold a procedure fixed across environments or vary the procedure within a matched environment, then verify that each generated execution preserves its assigned teaching condition.

Cross-embodiment synthesis must distinguish a changed appearance from a newly validated interaction. Generating human-video counterparts for robot tasks starts from a robot execution [24]; retargeting human motion and rendering robot arms into its scene starts from a human execution [222]. In the latter, visible object responses remain those of the human recording. Physical-consistency and forward–inverse checks can test whether a converted transition admits a compatible action [264, 265]. The teaching relation adds another requirement: the conversion must preserve order, goal, and binding contacts. A physically plausible transition can still be the wrong continuation for the supplied demonstration. Cross-shot generation exposes a complementary consistency requirement: VideoGen-of-Thought propagates identity across shots [451]. Preserving appearance can support visual augmentation, while motion and contact checks determine whether the resulting sequence preserves the demonstrated interaction.

Agent collection can obtain additional executions through shared interfaces. GUMI records pre-action observations and semantic commands for humans and agents [31]. An ICL collection could use this interface to execute contrasting valid orders in comparable scenes, retaining each task specification, recording, and outcome. Pairing each successful execution with its governing specification would preserve the distinction needed for contextual supervision.

Processing must preserve intentional variation. Demonstrator-specific speeds can assign different chunks to similar observations; reducing this ambiguity improves learning in an AgiBot study [266]. Normalize incidental style while retaining task-defining pauses, routes, and contact order. Insertion may require an alignment phase, and pouring may require a pause that uniform speed conversion would erase.

Failure–correction collection. Successful demonstrations leave recovery states underrepresented. DAgger queries an expert on learner-visited states [267], DART elicits corrections through perturbations [268], and selective intervention concentrates human effort [269]. Obtaining the correction also requires entering that state without destroying it. DITTO-X uses reverse actuation to align the operator's hand with autonomous motion before takeover, preserving ongoing contact [455]. Its diffusion policy is updated through DAgger with successful interventions; learning comparisons cover three tasks on Sharpa, while three-hand demonstrations establish interface feasibility. XR-2 likewise separates autonomous actions from takeovers during retraining [244]. These records expand recovery behavior in weights. Contextual recovery additionally retains the linked failure and feedback so later evidence can select an applicable repair.

Failure records must preserve the intervention and its physical consequence. Perturbations, predicted-rollout interventions, and planner exploration generate different continuations [270–272]; outcome-based learning need not imitate every action [273]. CureWM perturbs actions from shared starting states, executes them

to label failure or surviving success, and fine-tunes a predictor with the original objective [490]. Reduced optimism can coexist with more false alarms, so prediction sensitivity and successful control require separate measurements. For contextual repair, retaining the failed action, correction, and outcome, as ASPIRE illustrates [47], lets later inference distinguish applicable repairs.

A correction can teach the current controller and a future policy through different routes. In Bridging Frontier Reasoning and Robot Execution, five demonstrations and successful corrective segments condition a fixed frontier teacher; generated and human trajectories then fine-tune a local student [464]. Equal-size mixed and human-only training both yield 41/80 successes, while the language harness raises the mixed-policy result to 52/80. The comparison locates the added benefit in the surrounding execution interface rather than in synthetic data alone. Retaining linked failures, corrections, and outcomes supports later contextual teaching; using those trajectories as action targets consolidates behavior in neural parameters.

4.6 Connecting teaching data to a pretrained policy

Teaching data needs an interface through which its evidence can change behavior. An observation-conditioned policy such as ACT needs a context channel to distinguish procedures compatible with the same scene [274]. Language supplies goals and constraints [118, 119]; goal images supply endpoints [275]; demonstrations can specify order or contact preferences [13, 55]. Teaching-dependent targets connect the supplied evidence to actions.

The missing dependency determines the needed training records. Foundation policies supply perception and motor interfaces [118, 275–279]; contextual supervision connects task evidence to actions, geometric references, futures, or executable specifications. Subtask annotations and skill representations expose structure [30, 60, 61, 241], while phase-aligned examples connect it to a decision [18]. Joint training, pretrained interpretation, and geometric construction offer alternative routes to a teaching-sensitive intermediate.

Human pretraining and deployment teaching establish different dependencies. Robot pretraining can benefit subsequent human–robot fine-tuning [280]; ICL additionally requires responding to a new example with fixed parameters. The $\pi_{0.7}$ recipe supplies language, quality, speed, modality, and visual subgoals to distinguish executions [281]. Demonstrations can add procedural distinctions these inputs leave unspecified.

Action encoding must make the taught behavior expressible. Regularized ICRT tokens and hierarchical tokenization preserve different motion properties [282, 283]. LipVQ-VAE improves a low robot-success baseline while leaving execution difficult. Pairing, context encoding, and action representation must preserve the same task distinction. Deployment tests realization and repair under the current physical conditions.

Takeaways

① Data sources determine the available measurements; teaching–execution relationships determine what context can teach. Robot data supports motor competence; alternative taught behaviors within the same scene expose the additional dependence on demonstrations.

② Retrieval can select training data, construct a demonstration–execution pair, or supply examples during control. Only the last directly changes a frozen policy's input at deployment; the first two change the experience from which it learns.

③ Useful scaling adds missing motor competence, task-defining alternatives, or evidence that changes a later attempt. The value of added experience follows the new distinctions it makes available to the learner.

5 Grounding: How Does Context Guide Recovery?

Objects can move, grasps can fail, and resistance can change after a task has been correctly interpreted. Recovery aims to preserve the taught requirement while revising the state estimate and continuation. It establishes the current state, diagnoses the deviation, and selects a feasible repair. Retaining the repair with its applicable conditions makes that feedback useful beyond the current attempt.

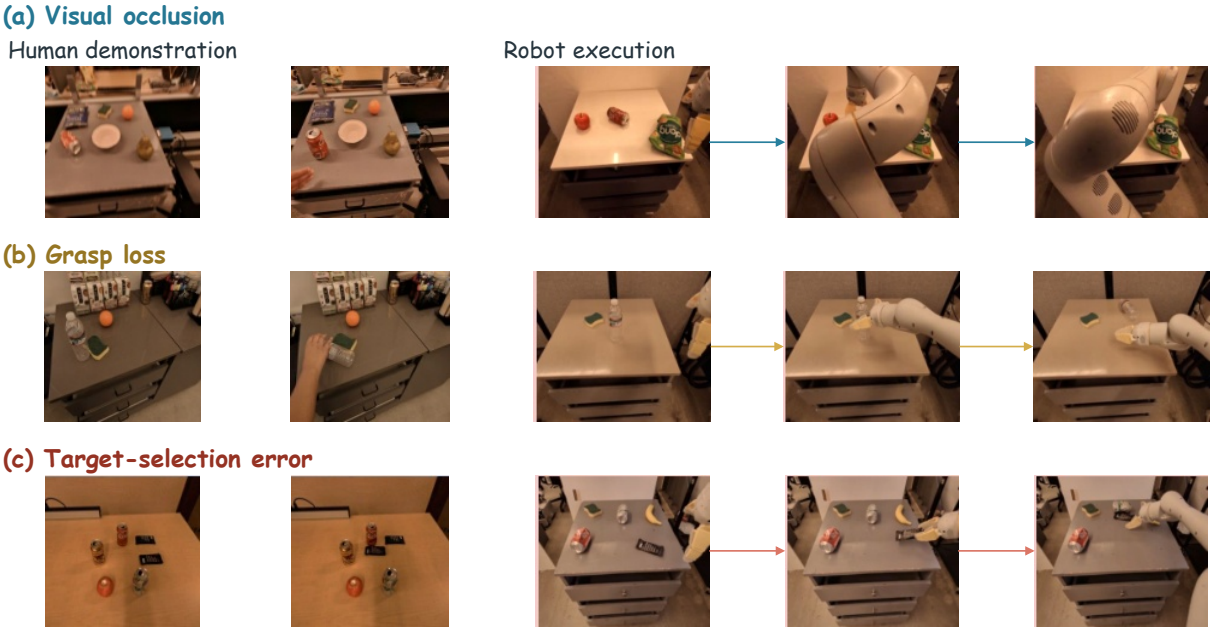

Figure 18 Three failures call for different recovery evidence: (a) visual occlusion, (b) grasp loss, and (c) incorrect target selection. Human demonstrations on the left establish the intended behavior; robot executions on the right reveal where realization departs from it.

5.1 Grounding the task in the current physical state

Grounding requires two complementary estimates: the task requirement that remains binding and the physical state that may have changed. An insertion can preserve its destination and prerequisite order despite a moved part. We establish these estimates before examining motion feasibility and the entry conditions of transferred segments.

Preserving the task reference during execution. A transferable task reference must distinguish progress toward a goal from adherence to the taught procedure. Cross-view progress representations can align executions despite appearance or embodiment changes [78, 79]; demonstration-conditioned skill inference selects which procedure that progress belongs to [29, 80]. Current observations must then establish whether its prerequisites occurred and its required contacts remain valid. Terminal appearance can conceal an incorrect order, so verification must use the relations supplied by teaching [284]. This connects correspondence in Section 3.5 to recovery: a deviation is meaningful only relative to the procedure the robot was actually asked to follow.

Figure 18 distinguishes three failures in video-conditioned manipulation. An occluded interaction calls for a view that restores the missing geometry; a lost grasp calls for checking contact before resuming transport; a wrong target calls for revisiting the instruction and demonstration correspondence. Each repair addresses a different missing fact. Contact-intensive tasks add another ambiguity: a stalled insertion can reflect misalignment, unfamiliar resistance, or excessive force. A bounded probe can distinguish physical responses, whereas an irreversible operation may require inspection or clarification before another attempt.

Maintaining a current, task-relevant scene. A repair must distinguish an intended change from its observed outcome. A placement command or predicted future specifies what should happen; completion evidence supports recording the step as finished. If the part subsequently moves, its remembered pose needs revision while the earlier placement instruction can remain binding. Memory and geometry-indexed retrieval recover unavailable observations [285, 286]; dynamic scene records replace spatial facts contradicted by new measurements [287–289]. When neither establishes applicability, another view or a reversible probe can distinguish the remaining explanations [290, 291]. A navigation failure can similarly invalidate a remembered route or its terrain assessment. VLM-GroNav [352] aligns visual terrain examples with sinkage or slip indicators before revising traversal estimates. This distinguishes physical feedback about a surface from geometric evidence that a passage is open.

Applicability includes both record validity and scene synchronization. CoreSense checks scope, provenance, time, and consistency [342]. SpatialHarness supplies a frozen controller with virtual views and object geometry from a reconstructed scene, correcting it differently during static, held, and contact-transition states [493]. These views expose relations obscured in the real cameras but inherit reconstruction and synchronization error. Their execution benefit concerns the quality of supplied spatial context. Scene construction, transition replay, and policy calls all contribute to its computational cost.

Connecting intended change to feasible motion. A corrected interpretation needs an executable continuation. Forward–inverse consistency checks agreement between proposed actions and modeled transitions; force-conditioned control adds measured contact feedback [265, 292]. These mechanisms make a context-selected correction executable. When the required motion lies outside the controller's repertoire, recovery also requires a new skill or a change in the physical setup.

Checking whether a transferred segment still applies. Geometric one-shot methods distinguish initial alignment from recovery during execution. DOME and ODIL first establish a relative starting configuration, whereas staged alignment can re-establish it at later subtask boundaries [33, 37, 177]. A transport segment also depends on the object remaining grasped: after a slip, geometric alignment can persist while the segment's entry condition fails. Its entry condition and intended completion therefore carry different information: the former establishes whether the recorded motion can begin, and the latter whether the next segment should follow. Learned subgoal transitions similarly govern progression within trained long-horizon policies [293].

Failure can invalidate a reference or expose missing control competence. Robustness via Retrying recomputes a continuation through image registration and predictive control [294]; ScrewMimic uses a demonstrated screw-motion representation to guide further bimanual learning [295]. The former changes the continuation through existing components; the latter improves the component that realizes it. This distinction directs recovery toward revising the current reference or expanding the motor capability needed to realize it.

5.2 From failure evidence to corrective execution

Recovery proceeds from assessment to diagnosis and correction. Assessment establishes a deviation from the taught procedure; diagnostic evidence distinguishes its plausible causes; the controller's correction interface determines which repair can be executed. The resulting continuation may be absent from the successful demonstration.

Task-conditioned progress assessment. Failure assessment requires a reference for what should have happened. Action consistency, visual progress, and calibrated rollout scores can detect departures from expected execution [81, 82, 296]. For ICL, the reference follows the current teaching, allowing an unfamiliar but valid order to define expected progress. GVL illustrates example-conditioned assessment by using observation–progress examples to improve a frozen VLM's estimates from shuffled frames, including human examples for robot trajectories [83]. The taught reference makes progress assessment responsive to the current task. Predictive control also supplies a visual intention reference. FAVOR compares the current observation and executed actions with a WAM's future-image anchor, interrupts a deviating chunk, obtains corrective language, and verifies again after repair [467]. Its verifier is trained offline around a frozen base WAM. The anchor makes recovery task-directed; rechecking establishes whether the correction restored the intended evolution.

Progress and physical acceptability can disagree. A connector can reach its apparent destination with an unsuitable contact force, just as a kit can reach the right inventory in the wrong order. Verification must therefore test both the context-defined requirement and the measurements needed to establish acceptable execution.

Diagnostic evidence acquisition. Diagnosis turns failure detection into a repair choice by identifying the likely cause. Multisensory diagnosis and learned failure knowledge organize clues to a stalled action [84, 85], but misalignment and an unseen obstruction can still require different interventions. Uncertainty about the intended task can be reduced by clarification and reasoning over alternative plans [86, 297, 298]. Uncertainty about resistance instead requires sensing or interaction, including tactile evidence absent from RGB [299]. The unresolved cause determines the information source: teaching clarifies intent, while sensed action effects reveal the resistance encountered by the robot.

Table 12 How teaching and feedback support assessment, correction, and reuse. Each row links supplied evidence to the decision or control element it changes. ENPIRE denotes its program-refinement branch.

Work	Available evidence	Affected output
Assessment and assistance		
GVL [83]	Observation–progress examples	Progress estimate
FAVOR [467]	Future-image anchor; executed actions	Verify; correct; recheck
IntroPlan [298]	Retrieved reasoning examples	Plan; clarification
Corrective execution		
ICPI [9]	Execution errors; correction examples	Motion-primitive parameters
RACER [87]	Current images; corrective language	Action continuation
CorrectVLA [142]	Human feedback	Sparse action bias
Experience reuse		
DROC [46]	Retrieved language corrections	Plan; skill calls
Zeva [88]	Action–effect history	Subsequent actions
ENPIRE [89]	Execution feedback	Executable program

Task-context exploration and question-based teaching formalize the choice to acquire evidence [300, 301]. The useful check is the one that discriminates causes relevant to the next action: a view for occlusion, contact feedback for grasp loss, or clarification for ambiguous intent. Affordance Agent Harness routes detection, segmentation, and other skills, verifies evidence, and retries before committing an actionable region [375]. Its episodic memory supplies priors for recurring objects. This illustrates how verification can govern evidence acquisition before a motion controller uses the result.

Correction granularity and feasibility. Correction granularity should match the cause of failure. A wrong target requires changing task selection; a displaced approach point requires a spatial update; insufficient displacement may require only a magnitude adjustment. Corrective-language training makes supervisory instructions actionable [87], while sparse adjustments around a frozen VLA can address magnitude errors when the target is already correct [142]. Subtask, motion, and pixel-target interfaces expose these changes at different resolutions [210]. The controller must also accept the correction from the state already reached, connecting diagnostic precision to the steerability analysis in Section 3.

Iterative in-context policy improvement links an observed error directly to a motion primitive's settings. A frozen language model uses primitive parameters, execution errors, and correction examples to propose an adjustment; its comparison favors relative errors over raw states and targets [9]. The error is useful because it identifies a change the primitive can make. Neural parameters stay fixed, while the generated control settings change within the primitive's available range.

When the frozen action policy cannot continue from the failure state, CoRe uses history and counterfactual continuations to choose a rejoin state, then realigns displaced objects before returning control [302]. The teaching still specifies the task; restoration makes an existing skill usable again.

Recovery can reconstruct a reference or revise retained history. RESETTLE triggers on persistent disagreement between independently sampled actions, retrieves a successful same-task demonstration, and combines state servoing with a guarded visual residual for one corrective control step [491]. Its modules are trained offline and frozen during recovery. FARE instead compares continuations from full, recovered, and reset causal caches under the same current observation [400]. Always searching or accepting an unverified prefix can underperform its base policy. Both approaches require a reference compatible with the present state: restoration of a known task differs from acquiring a new procedure. Task graphs and verified subgoal memory preserve completed steps and failed prerequisites [206, 303].

Feedback is useful when it identifies a decision the robot can revise: its estimate of progress, its next action, or a rule retained for later use (Table 12). A contextual repair must change that decision while preserving the task requirements outside the correction's scope. This connects immediate recovery to the longer-term validity of the lesson it produces.

5.3 Applicability and reuse of corrections

Reuse retains a conditional lesson: which intervention corrected which failure, and under what conditions. Guidance or executable artifacts preserve it through Equation (19); corrective training can incorporate such lessons into policy weights (Section 4).

Different lessons require different reset rules. Language-correction memory retains task constraints, whereas action–effect history retains physical response [46, 88]. A workcell reset clears completed-step memory; a changed connector or tool can invalidate a response estimate while leaving the assembly order intact. For a jammed connector, an adjustment justified by measured resistance should not be reused when misalignment is the cause. Retaining the observation, intervention, and outcome allows later attempts to reassess that condition instead of replaying the repair unconditionally.

Program-based reuse makes repairs executable. ENPIRE's program-revision branch uses fixed neural models; Zetta retains critics and recovery routines around a frozen policy [89, 90]. Interruptible execution exposes intervention points [304]. Later trials test the repair's benefit under applicable conditions. Task structure and irreversibility determine what can be repeated, linking recovery to the application demands below.

Takeaways

① Context specifies the recovery reference and helps interpret failure. Physical measurements establish quality; a responsive control interface makes the correction executable.

② Corrective training expands available repairs; contextual recovery selects or revises one from fresh evidence. Reuse requires retaining the correction with its outcome and applicable physical conditions.

6 Applications: What Must Context Resolve?

The value of context depends on the uncertainty an application makes consequential. Manipulation can leave the intended interaction unspecified; navigation can hide the route or local spatial structure; a long procedure can hide a completed prerequisite; an unfamiliar body can hide its response to action. These demands determine what teaching, observation, or interaction must supply. Repeated task changeover combines them with acquisition and reuse costs. Any of the four method families can serve several demands, provided its intermediate preserves the information the application requires.

6.1 Task specification in manipulation

In rearrangement, an instruction or demonstration can specify which object belongs in which container, how components should be ordered, or which tool serves a purpose. A moved destination requires a new path that preserves the demonstrated object–destination relation. The Imitator Game makes differences between demonstration and execution explicit through scene, object, and functional changes [101]. These changes test preservation of intent across altered object correspondence. We first consider the distinctions an example must convey, then how they survive replacement of the objects involved, and finally how geometry or contact response changes their realization.

When an example conveys more than a task label. The value of a demonstration depends on what the user needs to specify. A final drawing leaves stroke order, direction, and tool approach unresolved; BPP's DrawAnything study connects broader drawing-task diversity to better demonstration-conditioned generalization [54]. In its three-task laundry study, language is more reliable than behavior demonstrations, with some demonstration-conditioned failures executing the wrong fold. These contrasting settings identify an application condition: video is especially useful when it communicates a choice that a short task label or goal image leaves open. A goal image can specify the task when only a visible endpoint matters; a demonstration additionally conveys a required path or order.

A new workpiece can preserve an assembly relation while changing its grasp; a new fixture changes insertion geometry. Replacing both couples these adjustments (Section 3.5). The application defines allowed substitutions and binding identity, contact, order, and material requirements.

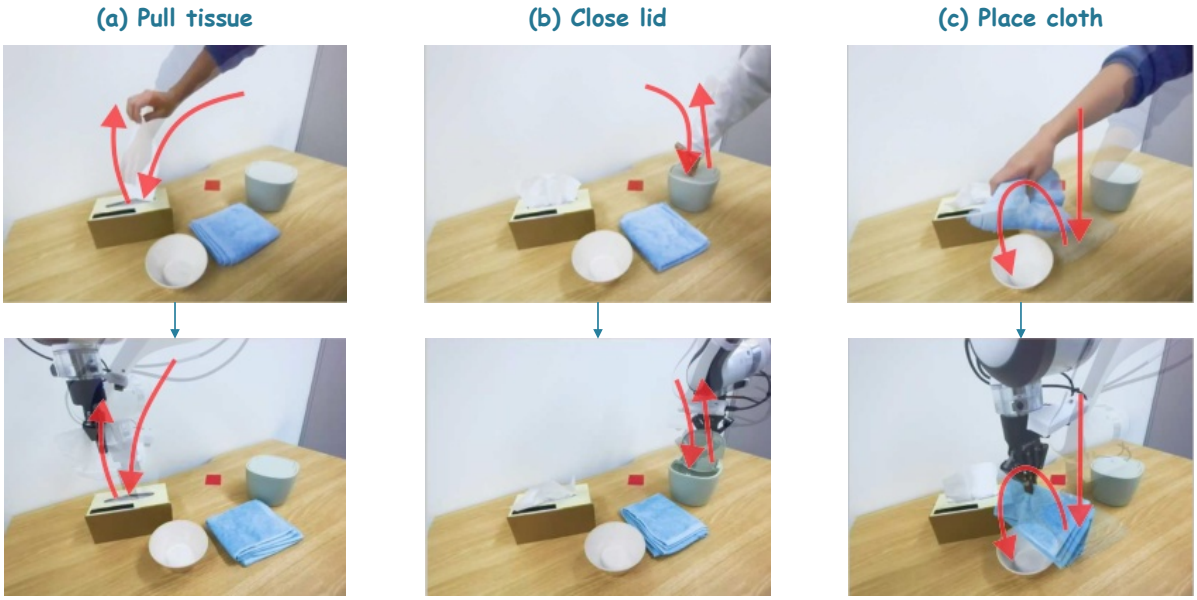

Figure 19 Cross-embodiment skill transfer in UniSkill [30]: (a) tissue pulling, (b) lid closing, and (c) cloth placement. Human demonstrations above and robot executions below preserve the interaction despite changes in embodiment and object appearance.

Personalized behavior adds persistent conventions to task specification. Language corrections can retain user preferences and guide subsequent plans [46, 305]. A request to place objects quietly should govern later placements, whereas a correction to one object's location need not. Such applications require both an acceptable endpoint and adherence to the user's continuing constraint.

Realizing these requirements across bodies adds a correspondence problem. In bimanual transfer, coordination must be preserved while each arm uses a feasible motion. Conditioning one predicted arm trajectory on the other addresses dependence within the action inference problem [92]. Sharing an end effector across arms reduces a different uncertainty, the physical mismatch between demonstration and execution [66]. Teaching specifies the intended cloth region or contact relation; action inference and embodiment design determine how coordinated motion realizes it.

Figure 19 shows the preserved interaction across human and robot executions: pulling tissue, closing a lid, or placing cloth. The relevant object part and destination remain identifiable while the robot changes its approach and contacts. A handle-specific instruction restricts that variation; an endpoint-only request leaves the grasp open.

Method selection follows the binding requirement. Geometric transfer exposes a contact or motion constraint; a skill interface exposes order and prerequisites; contextual policies and predictive control can encode either through their learned intermediates. The application supplies the criterion for judging these choices. Training pairs that equate executions solely by endpoint can erase a route or contact requirement that the user values (Section 4.2).

When intent and physical response change together. Deformable manipulation couples the procedure to a changing scene. A fold alters visible surfaces and accessible contacts, so the correct next action cannot be obtained by repeatedly matching the initial demonstration frame. Geometric and semantic information can jointly condition a folding policy [91]; preserving fold boundaries then provides a more useful correspondence than copying absolute hand trajectories. Instant-Fold's simulated mode variation and real garment variation examine different parts of this transfer. Its RGB-D and interaction interface, including estimated and potentially corrected human interaction points, also supplies richer information than ordinary RGB demonstration video.

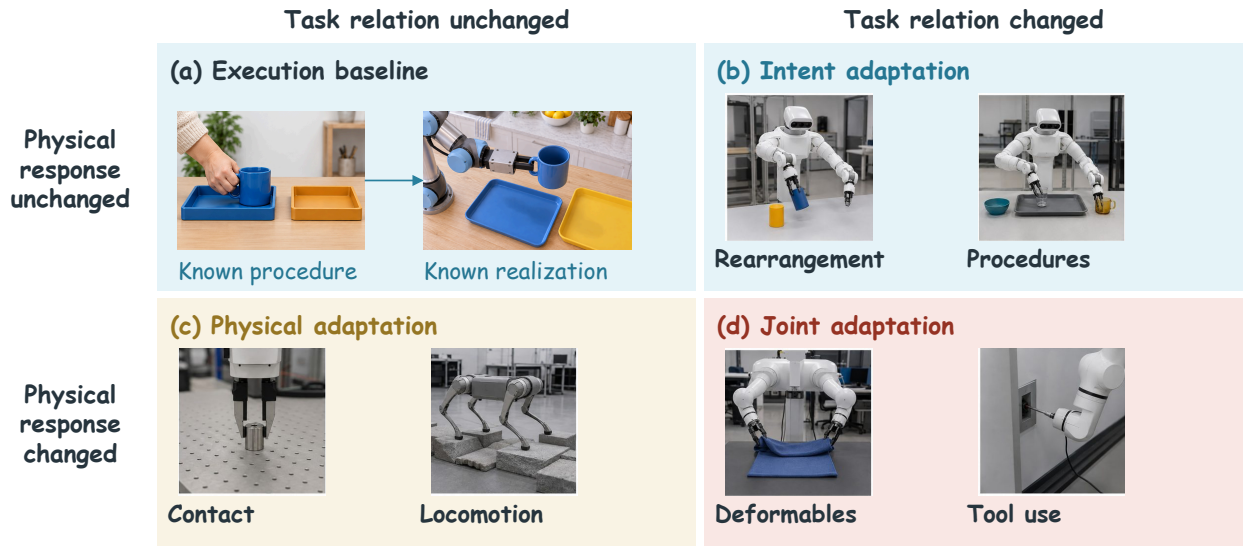

Figure 20 Task meaning and physical realization vary independently: (a) an unchanged baseline, (b) a new task relation, (c) a changed physical response, and (d) both changes together. Teaching specifies the intended relation; interaction reveals how the robot can realize it.

Material changes can preserve the procedure while altering the required force. Visuo-tactile measurements expose contact response [306]. TacZero places numerical touch, images, and within-trial action–result history in a frozen VLM's context to select Cartesian insertion targets [477]. Calibrated control executes them, with scripted grasp preparation and fixed orientation. PEARS separates slower revision from fast regulation: a frozen VLM revises force intervals from compressed cross-episode outcomes, while a force controller enforces them along manually corrected axes [478]. An online SAC actor and critics additionally learn diffusion-noise steering around the frozen policy. Contextual constraint revision and neural adaptation therefore complement each other. These interfaces connect contact evidence to different control timescales.

A new arrangement under familiar conditions primarily changes the requested relation. A familiar insertion under changed resistance primarily changes physical realization. A new fold on unfamiliar fabric couples both: the taught fold remains the reference, but each action alters the geometry and contact conditions for the next. This independent variation provides a basis for choosing teaching and feedback, even within one application.

Figure 20 crosses the two demands: a changed task relation, changed physical response, or both. This view identifies what evidence an application needs before comparing the method used to process it. Navigation extends this question from object interaction to spatial knowledge and traversability.

6.2 Navigation and spatial adaptation

Navigation context can specify a route, reveal an environment, teach an instruction interpretation, or correct a failed passage (Table 13). These information roles can coexist. They complement the method taxonomy in Section 3, which distinguishes how the supplied evidence enters control.

Route demonstrations specify where to go in sequence. A tour can teach a particular passage through an environment: pass the reception desk, traverse a corridor, and stop by the stairs. The ordered reference remains relevant even when execution deviates from the original camera poses. Zero-Shot Visual Imitation [354] treats demonstration images as successive goals for a pretrained goal-conditioned policy, including navigation with a TurtleBot in unseen offices. RPF [355] instead encodes views and recorded actions into a path memory that guides execution under motion noise and scene changes. Sparse visual memory [356] reduces the retained views while preserving the action sequence, examining how much visual evidence path following needs. These approaches differ in whether the reference specifies visual subgoals or an action-bearing route. Their shared mechanism aligns present progress with the demonstrated route.

Table 13 Navigation contexts grouped by the information they supply at deployment. Each type supports a distinct decision, and a system can combine several types. These information categories complement the four computational method families in Section 3.

Context type	Evidence and decision	Representative approaches
Route demonstrations	Ordered views or landmarks, sometimes with actions. Follow a demonstrated path or sequence of subgoals.	Zero-Shot Visual Imitation [354]; RPF [355]; sparse visual memory [356]
Environment observations	Scene previews or earlier visits. Locate objects and connecting passages for new goals.	SPTM [357]; NOLO [350]; ReLIC [203]; NavProbe [381]; SparseNav [394]
Instruction and decision examples	Instruction–landmark, instruction–subtask, or image–decision pairs. Interpret a new instruction or select a move.	LM-Nav [358]; A^2Nav [359]; Select2Plan [351]
Outcome feedback	Outcomes, route records, terrain response, or dialogue. Revise guidance, motion, or traversal costs.	VLM-GroNav [352]; CMMR-VLN [360]; HAM-VLN [361]; Talk2Escape [382]; NavGPT-3 [509]

Environment observations reveal where places and objects are. A preview or earlier visit can reveal the building without prescribing the next route. SPTM [357] converts an exploration recording into a graph of visual locations, then selects reachable waypoints for different goals using fixed retrieval and control networks. NOLO [350] conditions navigation actions on a scene video, while ReLIC [203] accumulates experience across episodes with changing starts and goals in one home. NavProbe uses a spatial index to find visual and geometric evidence relevant to the current subgoal [381]. Its executive consults these records when the available evidence is insufficient, then revises the subgoal before invoking navigation skills. The distinction is how local knowledge becomes available: a graph, a supplied sequence, policy memory, or actively retrieved evidence. A new goal can reuse these facts; a new building requires acquiring them again. CLUE additionally lets exploration revise what the task requires: hypotheses about relevant objects and functions are tested against an online language-embedded map [404]. Context acquisition then resolves an underspecified objective as well as an unknown route. HomeBody connects such spatial recall to manipulation: shared map coordinates bind remembered locations to navigation and object-directed skills [415]. This continuity matters when the next manipulation target lies outside the current view. NavHarness further carries maps, corrections, and structured recovery handovers across fresh reasoning sessions [427]. Transfer then requires reconciling retained records with new observations, not merely extending a single context window.

Spatial memory needs acquisition, interpretation, and expiry rules. SparseNav grounds instruction-relevant landmarks [394], NavProbe retrieves subgoal evidence [381], and ECROM conditions negative detections on viewing opportunity [447]. COOL retains people–object interactions and timestamps to infer ownership and choose search targets; active observation refreshes stale evidence [487]. LiSoNav instead moves small objects between episodes. Its IVAM-Nav method binds objects to support surfaces and updates obsolete records through revisits [488]. In its 245-subtask ablation, keeping memory without updating it lowers success below removing cross-task memory. Useful memory preserves both facts and their applicable conditions.

Spatial history can enter different control interfaces. LightNav-0 compresses time-marked views and generates pointing and action tokens through an offline-trained navigation model [497]. NavMCP instead connects fixed reasoning and navigation models through intent, observation, and memory channels, carrying negative evidence and unresolved goals between calls [498]. One exposes history through learned action inference; the other maintains evidence for explicit tool requests. In NavMCP, the full episodic-interface comparison changes all three channels, so its gain cannot be attributed entirely to memory removal.

Instruction and decision examples teach an interpretation. Example pairs can specify how a navigation request should be translated into executable choices. LM-Nav [358] supplies instruction–landmark examples to a language model; the extracted sequence is grounded in a topological graph and executed by a pretrained navigator. A^2Nav [359] extends this interpretation to action demands: passing a doorway and entering it require different subtasks even when the landmark is identical. Its prompted parser selects among navigators

trained for these demands. Select2Plan [351] moves example conditioning closer to the decision itself by retrieving image–answer pairs for movement or waypoint selection. Thus the example can teach a symbolic decomposition or a visual choice. Unlike a route recording tied to particular places, such a pairing can explain how to interpret a new query using an existing execution interface.

Outcome feedback revises what should be attempted next. Execution reveals wrong turns, premature stops, and slipping terrain. CMMR-VLN retains successful routes and first-error notes for later retrieval [360]; HAM-VLN attaches failure notes to visited places within an episode [361]. VLM-GroNav uses physical interaction to revise waypoint selection and traversal costs [352]. When retained evidence is insufficient, Talk2Escape's detected looping or trajectory divergence triggers a landmark-grounded question to a human or oracle [382]. Remembered outcomes, measurements, and clarification thus revise different decisions. Their lessons remain useful while the associated instruction, place, and physical conditions apply.

Evidence must reach the component able to change motion. NavGPT-3 retains route keyframes and visited places for a planner delegating execution to a VLA [509]. Reasoning, acting, and monitoring have separate contexts and permissions. Interruptions revoke motion authority; fresh state and renewed authorization permit resumption. Retention and timely intervention jointly determine whether feedback changes navigation.

The four navigation contexts retain different things: a prescribed route, facts about an environment, an interpretation rule, or a conditional lesson. Their validity therefore changes at different boundaries. A new goal can reuse a map but replace the route; a new building invalidates local locations while leaving an interpretation convention useful; changed terrain can invalidate an earlier traversal estimate. Section 7.1 translates these distinctions into controls. Combining navigation with manipulation adds procedural state: after reaching a pickup site, the robot must establish whether the object was acquired. The next subsection examines these dependencies across a complete procedure.

6.3 Long procedures and irreversible operations

Long procedures require tracking dependencies. Interaction phases and sequential end-effector goals expose them in RGB-D teaching interfaces [93]. An unfinished step, a completed step, and a failed prerequisite need different continuations even when their images resemble one another. Context must align the demonstrated sequence with actual progress; irreversible steps further constrain valid retries.

Irreversible procedures add a constraint on how progress and failure can be established. In laboratory workflows, intermediate products must remain valid while the robot decides how to continue. Benchmarks spanning atomic skills, compositions, and extended workflows expose the interaction of procedural inference and dexterity [95]; household planning with low-level embodiment abstracted away isolates dependencies among steps [307]. Varying the supplied procedure within these settings would test whether contextual teaching preserves dependencies through the available executor.

Long procedures also couple memories with different validity periods. Progress memory records which steps have been completed; physical memory estimates how to perform the remaining operation. Zeva's two-timescale analysis tests reuse of action effects across reset attempts [88]. In a laboratory workflow, however, a physical reset may be impossible: repeating an insertion and repeating a dose have different consequences. The informative next interaction must therefore respect both the taught dependency structure and the state of the intermediate product.

Reversibility determines how the recovery mechanisms in Section 5 can be used. A placement may permit local correction; dispensing or fastening can require inspection before a retry. The application therefore needs evidence of completed operations, alongside the demonstrated dependency structure, to decide whether a failed step can be resumed.

The distinctions in Figure 21 concern what information is missing from the next decision. In (a), *task specification* identifies the required object–destination relation among alternatives already present in the scene. In (b), *interaction guidance* conveys the motion relation that should survive the change from a human hand to a robot gripper. In (c), *execution-state tracking* establishes that the plate has been uncovered, making lemon placement the next feasible step [94]. These examples instantiate the information roles in Section 2:

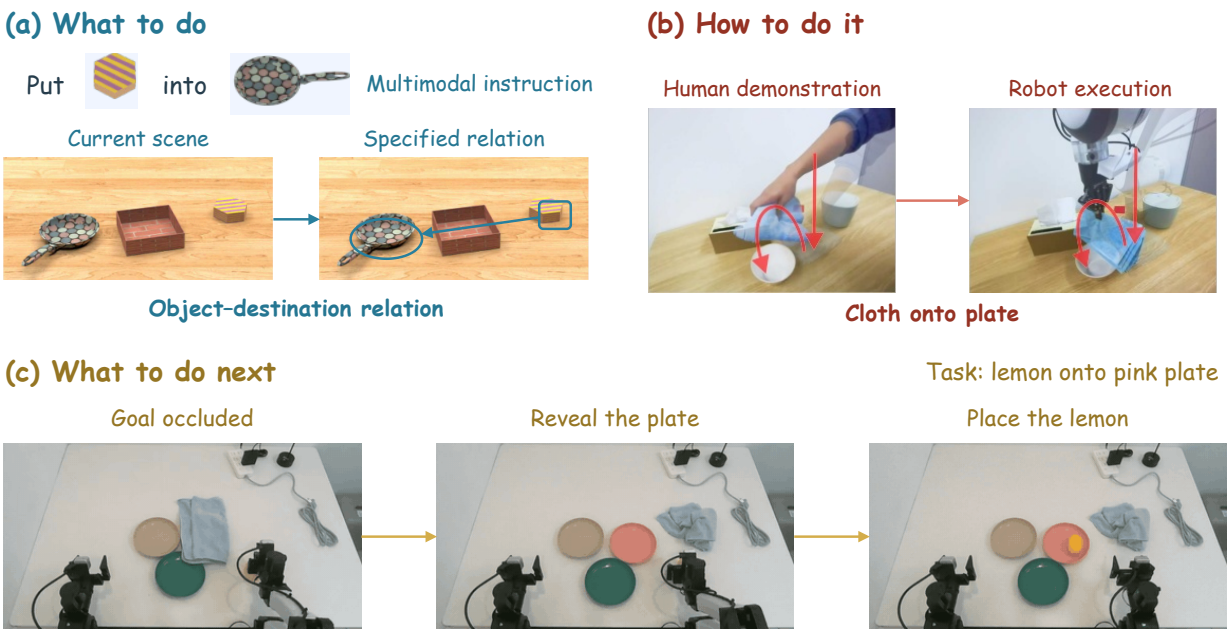

Figure 21 Three decision roles of context: (a) task specification selects the object and destination; (b) interaction guidance conveys how the cloth should move onto the plate; (c) execution-state tracking uses the result of removing the towel to advance from uncovering the target to placing the lemon. Representative implementations are VIMA [100], UniSkill [30], and GPT-Policy [94], respectively.

panel (b) combines a specified interaction with cross-embodiment correspondence, while panel (c) updates execution state under an unchanged goal. They are complementary information requirements within a task, and can be supplied together through language, demonstrations, and interaction history.

A new procedure can require either a new arrangement of available operations or a motion the current policy has not acquired. Assembly adaptation through fast weights [10] and GEN-1.5's brush-to-dustpan example after five minutes of fine-tuning [98] place new evidence into parameters. They provide a useful comparison with fixed-weight teaching: context can select, order, and parameterize available operations, including new motions within the executor's capabilities; further policy learning can expand those capabilities. This separation localizes long-procedure failures to task interpretation or motor competence.

Long-procedure evaluation must separate adherence to a new ordering from adaptation to changed physical response. Body and dynamics adaptation isolates the second demand: how a familiar requirement is realized when the physical system changes.

6.4 Adaptation to body and dynamics

Body adaptation isolates physical response while keeping the requested task specified. Locomotion makes this separation explicit; manipulation introduces analogous uncertainty through payloads and contact. Their combination in mobile manipulation then requires procedure and progress memory alongside the physical estimate. A direction or speed command can already specify the goal; recent action consequences reveal how the current body and terrain respond. Learned adaptation modules address this physical uncertainty [39], while longer-context experience broadens the setting toward different morphologies, actuator changes, and evidence retained across attempts [40]. A broader range of bodies expands the response variations that the policy must infer from its interaction history.

Locomotion also separates a change in the requested behavior from a change in the body that realizes it. Example-conditioned foot-contact generation can specify a gait-like procedure for a pretrained controller [96]. Active tracking across wheeled, quadrupedal, and aerial platforms instead uses a shared velocity interface, with body-specific realization delegated to lower-level control and visual history identifying the moving

Table 14 Teaching and reuse interfaces relevant to production. GEN-1.5, S1, and GLOW condition execution with fixed weights; physical feedback can instead revise programs or train policies.

System	Task evidence	Adaptation	Context or artifact
Demonstrations condition robot execution			
GEN-1.5 [98]	Sensors; trajectories	Physical prompting	Demonstration; history
S1 [99]	Video demonstration	Video conditioning	Demonstration
GLOW [387]	Single human video	Contextual action generation	Encoded skill context
HOST [23]	Human video; robot progress	Predict; decode	Demonstration
Physical feedback revises procedures and learning experiments			
ASPIRE [47]	Task; execution traces	Diagnose; repair	Repair guidance
ENPIRE [89]	Task; physical trial outcomes	Revise code; train policy	Programs; policies

target [97]. The shared interface supports cross-platform execution by assigning target tracking to visual history and body-specific motion to lower-level control. The contextual contribution depends on whether the evidence specifies a gait, tracks a target, or identifies an unfamiliar response.

Manipulation introduces analogous uncertainty through a new gripper, payload, or deformable object. RopeFormer retains action–response history across trials on unseen ropes, adapting rotation and whipping with fixed policy weights [363]. MetaPusher instead updates a dynamics model during pushing and revises the plan for the unfamiliar object [367]. The goal can remain fixed in both cases: interaction reveals how to realize it under changed physical conditions. Combining this evidence with demonstrations requires distinguishing a changed task requirement from a changed response.

Mobile manipulation combines procedural teaching, spatial memory, and response adaptation. In delivery, a demonstration specifies destination and handling order, lifting changes the payload, and memory records completed pickup. Evaluation must test procedural adherence and reliable transport under changing dynamics.

Procedural teaching and response adaptation become especially valuable when the same capabilities serve recurring tasks. Their practical benefit then depends on which acquired knowledge can be reused and which conditions must be checked again.

6.5 Task changeover, teaching, and reuse

Recurring kitting orders can share parts and grasps but differ in placement sequence. A demonstration specifies that difference when a count or destination alone is insufficient. Across batches, the sequence persists while progress resets; a new tool invalidates physical estimates without necessarily changing the order. Changeover combines specification, validation, and retention for later batches.

The changed requirement determines the teaching channel: language for counts, a sequential example for folds, and interaction for material response. Section 5 relates validation needs to precision and irreversibility.

Repeatable execution determines practical value. Retained procedures reduce later preparation, while changed tools or materials require revalidation. Section 7.4 compares teaching and operating costs.

Teaching and reuse interfaces. Teaching can specify a task for existing competence or revise the resources used in later execution (Table 14).

Sensorimotor and video prompts specify tasks without retraining [98, 99]; HOST instead predicts task evolution before decoding actions [23]. GLOW reports watering and box-storage examples that reuse encoded skill context across changed objects or positions [387]. The qualitative cases illustrate reuse of supplied teaching, while aggregate benchmark scores summarize broader capability. Context interventions isolate the demonstration's contribution (Section 7.1).

Outcome-driven reuse changes what is retained. ASPIRE preserves repair guidance [47]; ENPIRE can revise a control program or train a policy [89]. Prompt preparation, physical trials, and training incur different

changeover costs. Their value depends on whether the change requires another procedure, correction of a recurring failure, or a motion outside existing competence. The next chapter evaluates these contributions to task acquisition and reuse.

Takeaways

① Applications determine the missing information: an object relation, spatial knowledge, procedural state, or physical response. Each requires teaching or feedback that resolves the corresponding uncertainty.

② Unclear requirements call for teaching; uncertain execution conditions call for observation or interaction; missing motor competence calls for an expanded executor or additional skill learning. These deficits require different remedies.

③ Task changes create value for new teaching; repeated operations create value for retained procedures. Acquisition and verification costs determine that value, especially when actions are irreversible or time-critical.

7 Evaluation: When Does Context Help?

Evaluation traces context into physical success. Context interventions test whether changed teaching changes behavior; intermediate diagnostics locate losses during execution; retention tests measure benefits on later attempts. Matched scenes, executors, and retry budgets isolate the contribution of context.

7.1 Context use and transfer across tasks and environments

Context interventions test sensitivity to teaching; changes in tasks, objects, environments, or bodies locate its transfer boundary. Varying evidence quantity, retention distance, and distractors tests access to the required information.

Sensitivity to task-defining context. Demonstrations specifying different feasible behaviors in the same robot scene isolate the influence of teaching. In a kitting task, changing the demonstrated part count should change the selected count; changing the demonstrator while preserving that count should preserve the requirement. Together, meaning-changing and meaning-preserving interventions test what information the learner uses. Success without context provides the prior-only comparison. A fully specified target tests whether the required behavior is executable.

Instruction tests distinguish task sensitivity from scene robustness. LIBERO varies spatial, object, goal, and procedural factors, but task-index embeddings perform well when scenes reveal task identity [228, 309]. LIBERO-CF changes feasible instructions within familiar layouts; tested VLAs often retain the scene-associated task [107]. Encoded but Not in Control finds that an alternative target can remain decodable while actions follow that association [468]. A language-transfer study makes this distinction concrete: after offline action-adapter training, high success with correct Greek instructions remains close to success with incorrect instructions in single-goal scenes; a shared-scene, multiple-goal test exposes language dependence [505]. Probes reveal representation; controlled swaps and physical adherence reveal influence. Demonstration swaps similarly change order or relations, whereas robustness tests preserve them across views, appearances, and rephrasing [106, 310]. RoboStressBench separates material, viewpoint, lighting, and geometry stresses to diagnose recognition, reasoning, and planning [376].

Show-Harness's video-order comparison provides task information left unspecified by the no-video instruction [31]. An equally informative textual order is the relevant control for a claim about video's particular advantage. Comparing language-only, demonstration-only, and joint context with matched task information separates modality from information content. Holding the instruction fixed while changing the demonstrated convention tests learning from examples; holding the demonstration fixed while changing a correction tests use of feedback. Discovering an unfamiliar action convention proved harder: without supplied conventions, the system succeeds in one of 20 episodes and infers 23.3% of mappings correctly. Interpreting a demonstrated procedure and discovering an unfamiliar action interface thus require separate evidence.

Historical interventions can isolate the same dependency at a particular decision. SimpleMemVLA v2 holds weights, instruction, current observation, and robot state fixed while masking or replacing earlier video [45]. Removing a completed event changes the inferred count; replacing a color cue or demonstration prefix redirects

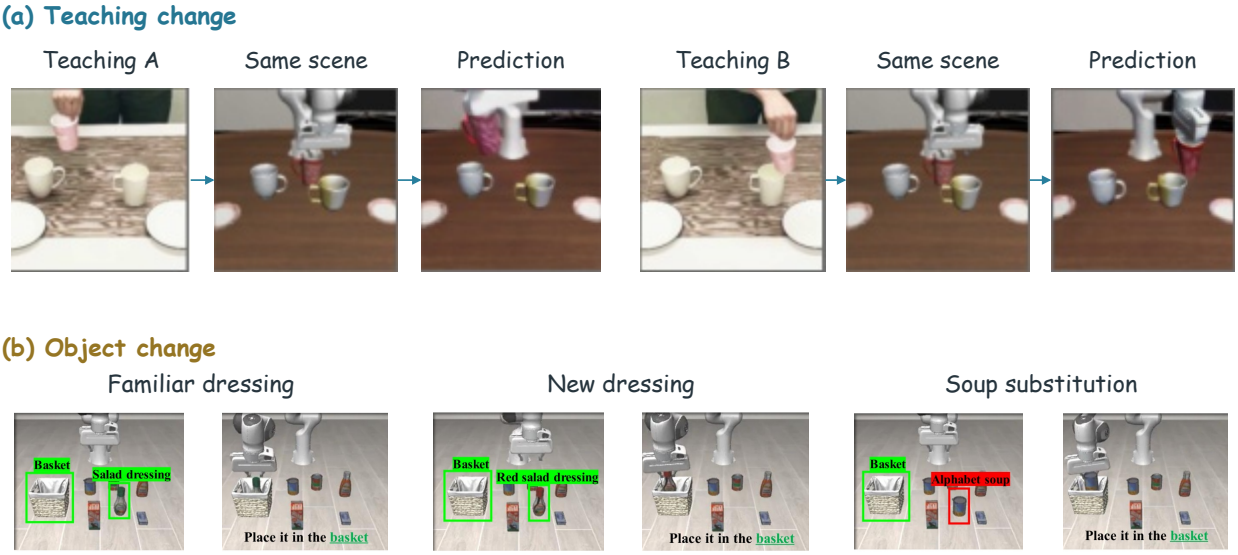

Figure 22 Two tests of contextual transfer. (a) Changing the demonstration while holding the robot scene fixed changes UniSkill's predicted continuation [30]. (b) LIBERO-PRO changes objects to test target selection [106]. Together, these axes motivate tests that vary teaching and object identity independently, measuring both the inferred requirement and its physical realization.

the output, while matched irrelevant edits preserve it. These tests identify which historical evidence controls the response. Task transfer additionally asks whether the inferred requirement remains useful through a complete execution in a new setting.

Figure 22 separates interpretation from realization: UniSkill's decoded futures expose inferred skills, while LIBERO-PRO's substitutions test target selection and execution. Both require controlled changes in supplied evidence to attribute the resulting behavior to contextual teaching.

Table 15 pairs evaluation questions with resources. RoboSPA varies spatial and procedural complexity and reports both success and progress [504]; its task-trained policies diagnose execution demands. Contextual tests additionally intervene on deployment-time teaching or retained evidence, whereas the human-data row measures transfer into trained weights, following Figure 14.

Object substitution and relation transfer. Task novelty has to be defined at the level of the relation being learned. Task suites make adaptation to held-out tasks measurable [227, 311], but withheld names can combine several kinds of change. New object combinations and task templates probe different generalization in VIMA [100]; the Imitator Game separates changes in demonstrated motion from changes in functional intent [101]. A familiar procedure on a new body tests realization, whereas a new ordering of available actions tests contextual composition. A useful transfer claim identifies both the unseen relation and the motor competence available to realize it.

Following Table 6, preserve the requirement while replacing the manipulated object, receiving object, or both. Within each condition, supply the correct demonstration, none, or another feasible procedure in the same scene, holding instruction information constant. Object changes measure substitution scope; demonstration changes measure dependence on teaching. Separate task interpretation from physically verified task completion.

FUNCTO separates instance and category changes [172]; object-pair transfer changes both ends of a relation [346]. Visually dissimilar functional substitutes and similar-looking incompatible distractors test whether matching follows the taught interaction. Report training exposure, demonstration count, correspondence assistance, updates, and retries. Training novelty additionally requires known pretraining exposure.

Withholding a relation while preserving familiar objects and executable actions isolates relation novelty; substituting objects while retaining the relation tests its physical transfer. Cross-robot imitation and paired human-video interfaces then extend the change to embodiment and observation [12, 312]. Whether the

Table 15 Proposed evaluation controls and resources that illustrate the corresponding capabilities. Comparisons hold other task conditions fixed; cross-attempt tests separate physical resets from memory resets.

Capability	Intervention	Readout	Example resources
Context use	Change teaching	Task-consistent actions	LIBERO-CF [107]; LMAct [102]
Relation transfer	Hold out relations	Relation adherence	VIMA [100]; MOSAIC [12]; Imitator Game [101]
Temporal grounding	Swap event history	Target adherence	GiT [473]
Evidence retention	Retain or clear history	Recall; memory updates	RoboMME [103]; MEMOBench [104]
Hidden-cue retention	Vary cue delay; reactive controls	Memory-dependent success	MIKASA-Robo-VLA [446]
Cross-attempt reuse	Separate reset types	Gain; interference	RopeFormer [363]; Zeva-Ego [362]
Physical transfer	Change dynamics	Adaptation; goal fidelity	MemMimic [105]; PACE-Bench [308]
Human-data transfer	Vary human data; fix robot data	Trained-policy success	H2RBench [365]
Scene robustness	Change scene	Target; outcome	LIBERO [228]; LIBERO-PRO [106]
Execution steering	Change command	Redirection; success	ReSteer [108]
Physical execution	Supply target or prerequisites	Target selection; success	RoboSemanticBench [109]; Behavior-Skill [110]
Route teaching	Change route; retain goal	Path; landmark-order adherence	RPF [355]; sparse visual memory [356]
Spatial adaptation	Change preview; reset memory	Success; route efficiency	NOLO [350]; ReLIC [203]
Example interpretation	Change examples; retain query	Parsed subtasks; selected moves	A^2Nav [359]; Select2Plan [351]
Outcome reuse	Remove notes; retain retries	Repeated errors; path cost	CMMR-VLN [360]; HAM-VLN [361]
Traversability	Withhold terrain feedback	Success; physical disturbance	VLM-GroNav [352]

demonstration remains available is a further axis: executing with a supplied reference and retaining its lesson after that reference is removed are evaluated separately in Section 7.2.

RoboReel links 2,000 human recordings from four views to simulated trajectories in ten environments, distinguishing clean/distracted capture, evaluation-only distraction, and pose-defined tasks [312]. These tests vary the observation interface while preserving the demonstrated requirement. The accompanying update rule identifies whether transfer uses context or task-specific fitting.

H2RBench reconstructs four real manipulation tasks in simulation and varies human-video quantity at a fixed robot-demonstration budget [365]. Its shared training protocol compares human-data transfer across simulation and reality; fixed-policy context interventions instead test the influence of new teaching.

Navigation across goals and environments. The four context types in Table 13 require different interventions. Changing a route demonstration while keeping its destination fixed tests adherence to the taught path or landmark order. Replacing a scene preview and clearing memory between goals test spatial evidence and within-environment reuse, respectively; NOLO's context-removal experiment [350] and ReLIC's cross-episode attention comparison [203] examine these contributions. Changing instruction–decision examples while retaining the current query tests the demonstrated interpretation. Removing failure notes while retaining backtracking tests whether a remembered outcome changes later decisions, as in HAM-VLN's reflection ablation [361]. A new building then tests acquisition after environment-specific locations cease to apply.

Completion judgments are another source of variation. DroneCATS evaluates image- and action-history-conditioned MLLM calls in AirSim, separating target approach from the agent's explicit declaration of completion [500]. Separate attainment, valid calls, and stopping errors when comparing policies.

Success weighted by path length (SPL) combines reaching the goal with route efficiency [353]; a prescribed route additionally requires an adherence measure, since a shortcut can reach the same destination. Success across successive goals measures environment reuse, while repeated wrong turns and stopping errors expose the contribution of feedback. Goal visibility, stopping distance, action discretization, and access to maps or localization define the comparison. Terrain adaptation adds physical cost: VLM-GroNav [352] reports trajectory length and IMU energy density alongside success. Exploration and retries spent acquiring context must be included when relating these gains to total interaction.

Navigation evidence also depends on the execution setting. NOLO tests unseen layouts in simulation; its real maze uses training and context recordings from that maze. Select2Plan combines simulated object navigation with waypoint assessment on recorded rover scenes [351]; LM-Nav executes routes on a physical robot [358]. HAM-VLN uses 100-episode unseen simulation subsets. These settings support different claims about physical execution.

RoboFollow supplies several feasible behaviors in shared scenes and separates stage-level intent from execution scores [397]. Layout and semantic recombination tests distinguish instruction grounding from physical completion. Unfamiliar-rule tests must instead withhold rule families and supply distinguishing examples.

Evidence quantity, retention, and distractors. Teaching quantity, retention distance, and distractor load test different dependencies. LMAct's expert-episode and replay comparisons show that more examples often provide little benefit to the tested models [102]; comparing ambiguous and fully specified tasks helps separate weak context use from inadequate control. MIKASA-Robo and RoboMME instead organize the earlier evidence that must remain available [103, 313]. MIKASA-Robo-VLA extends this evaluation to 90 language-conditioned tasks in ten memory categories, with 80 hidden-cue tasks and ten reactive controls [446]. Its $\pi_{0.5}$ reference experiment covers 14 tasks. The relevant delay is the interval between observing a cue and needing it, which can differ from total episode length. These tests measure retention under specified tasks; novel task acquisition additionally varies the rule conveyed by teaching. AMAGO's MetaWorld setting, where short history identifies the task, illustrates why a long episode need not demand long memory [202].

Checkpoint-based evaluation localizes memory failures. MEMOBench's 30 tasks distinguish Storage, Update, and Compression [104]. Supplying a missing record directly in the policy's expected format separates unavailable evidence from failure to use it. WorldLines pairs Memory QA with planning probes of current state and preconditions [374]; EvoEmpirBench tests dynamic spatial reasoning, with Agent-ExpVer revising experience across tasks [373]. These targeted tests complement rollouts by separating state recall and revision from physical realization (Section 7.3). STAR adds a structured spatial-reasoning comparison: turn-point supervision and segment-level preference optimization target cascading route errors [452]. Local accuracy diagnoses reasoning errors; navigation rollouts then test their physical consequences.

7.2 Sources of improvement and comparable evidence

Fourteen reported comparisons (Table 16) connect context use, physical transfer, and later learning to observable outcomes. Their findings motivate controls that attribute gains to supplied evidence, parameter updates, or retained artifacts.

What reported comparisons establish. The first group separates evidence quantity from usefulness. ICRT links context use to within-scene task alternatives; LMAct finds little benefit from expert-episode quantity alone [13, 102]. BPP favors trajectory-defining drawings, Show-Harness exposes missing action conventions, and NOLO tests scene-preview removal [31, 54, 350]. LIBERO-Agent holds execution fixed while varying video and action context for three agents; gains depend on the agent, and trajectories do not consistently improve on video alone [443]. OpenRUA instead resets workspaces each episode, retaining API documentation and feedback but no demonstrations or prior-episode memory [461]. This tool-competence baseline supports testing the additional value of acquired guidance.

The second group connects transfer to the execution interface. MT3's low-demonstration advantage relies on alignment followed by interaction replay [36]; part-wise warping improves the tested transfers with supplied part correspondence [346]. Demo-JEPA's comparison with an action head shows that the preferred realization can change with task familiarity [22]. RAPID's scene-variant ablation adds a complementary question: whether the acquired program is verified beyond the demonstrated configuration [388]. Transfer therefore depends jointly on the represented relation, correspondence assistance, motor competence, and conditions used to validate the result.

The third group distinguishes using retained experience from acquiring a better learner. Zeva tests access to earlier interaction; RTCF and TraceFlow test action correction from successful or failed traces; FARE tests selective revision of the history conditioning prediction [88, 195, 390, 400]. Their scope differs: TraceFlow's ordering gains coexist with unresolved simulation deficits, while FARE's always-search variant underperforms its base policy. LMPC instead trains a successor between sessions and measures its response to new teaching [117]. Cumulative success over retries measures yet another quantity, as in Zeva's evolution

Table 16 Reported comparisons of context use and transfer. → denotes baseline-to-variant change; ↑/↓ denote increase/decrease; > denotes ordering. SR: success rate (%); pp: percentage points; fractions: successes/total; qual.: qualitative; BC: behavior cloning. Comparisons are within study; weights stay fixed during deployment (LMPC trains successors between sessions). Full protocols and source locations: Supplement S1.

Study	Target	Contrast	Readout	Setting
I. Training and task evidence				
ICRT [13]	Training data	DROID-only / multi-task	DROID-only: no progress	Shared test tasks
BPP [54]	Prompt	Goal image → demo	Fidelity ↑ (qual.)	Unseen drawings; matched start
LMAct [102]	Context size	0 → 512 episodes	Often little gain	Games; simulated control
Show-Harness [31]	API semantics	Arbitrary names; no conventions	1/20 successes	Fixed harness
NOLO [350]	Scene context	No video → video	SR: 33.58 → 43.65	Habitat; 3 seeds
II. Transfer and physical realization				
MT3 [36]	Action reuse	BC → retrieval; whole → phases	SR ↑; demos ↓	New instances; align + interact
Part-based transfer [346]	Warp granularity	Whole object → parts	11/27 → 23/27	Mug–rack placement; assisted parts
Demo-JEPA [22]	Action interface	Action head / planning	Known, real: head > plan; unseen: plan > head	Sim. + real; task-group means
RAPID [388]	Verification	Scene variants: off → on	SR: 53.2 → 75.9	8 tasks; 50 scenes/task; 3 runs
III. Retained experience and subsequent learning				
Zeva [88]	Persistent memory	Off → on	SR: +10–20 pp	5 real tasks; brief trace retained
RTCF [195]	Memory correction	Off → on	SR: 86.4 → 88.4	Fixed bank; 2,000 episodes/variant
TraceFlow [390]	Trace guidance	Base → guided	Ordered: 21/50 → 39/50; sim. aggregate: unchanged	1 real packing task; simulation suite
FARE [400]	History revision	Base → selective; always-search ablation	SR: 91.5 → 93.2; always-search: 91.0	RoboTwin Easy; fixed checkpoint
LMPC [117]	Successor training	Base → successor	SR: 39.4 → 66.3	Held-out teaching tasks

Table 17 Attribution controls expressed as target–intervention–readout. ↔ denotes paired test conditions; × crosses checkpoints with context conditions. Match tasks, physical resets, base execution resources, and attempt budgets. Measure both behavioral change and task completion.

Target	Intervention	Readout
Training checkpoint	Checkpoint × context	Learned context use
Task teaching	Original ↔ replacement	Requirement adherence
Retained history	Retain ↔ clear	Use of prior evidence
Control intermediate	Original ↔ corrected	Intended action; success
Neural parameters	Adapt ↔ restore	Adaptation gain
Reusable guidance	Retain ↔ withhold	Lesson transfer
Skill library	Expanded ↔ base	Use of new skills
Active observation	Enabled ↔ disabled	Acquisition benefit

milestones. These comparisons locate gains in a specific use of experience; the interventions below test whether that use explains the outcome.

Controlled attribution. Table 17 separates interventions on the training checkpoint, deployment evidence, adaptive parameters, and retained artifacts. Evidence-acquisition controls additionally test how informative observations are obtained. CodeActionBench standardizes robot APIs, resource budgets, and hidden physical-outcome verification for code-as-policy agents [437]. Such controls distinguish successful tool execution from successful task completion.

The first control compares checkpoints under the same context interventions; the remaining controls change what is available during execution. Repeated-task and transfer trials then establish whether the identified dependence improves subsequent behavior.

Acquiring the ability to use context. Contextual training should improve responsiveness to evidence beyond general execution competence. Paired examples, multimodal prompts, sensorimotor streams, and corrective histories train different dependencies [1, 10, 12, 13, 100]. Compare each checkpoint under informative, absent, and meaning-swapped context on matched tasks. Within-checkpoint differences measure evidence use; differences between checkpoints measure its acquisition. A fully specified task separately tests motor competence.

Using new teaching and retained observations. Changing an example under a fixed checkpoint, scene, and executor tests whether teaching changes behavior [14, 24]. Clearing history at a matched state instead tests earlier evidence. Dynamics-revealing transitions [39, 363] and remembered task events [42, 44] resolve different unknowns, as memory-focused benchmarks distinguish [103, 104]. Counterfactual Memory Audit compares histories yielding identical current inputs, controls policy-sampling randomness, and tests each selected action under both histories [383]. This separates memory-driven action changes from justified choices and physical benefits.

Adapting neural parameters. Parameter adaptation requires a matched-input comparison and a separate reset. Meta-imitation updates an initialization from demonstrations [7, 8]; fast-weight methods restrict the updated subset and timing [10, 11]. PACE's demonstration attention also depends on transition-trained fast weights [475]. Restoring those weights separately from removing history identifies their contribution under matched tasks and inputs. SALT's visual-corruption tests preserve task and embodiment while adapting projection scales [476]; their gains concern execution robustness rather than acquisition of a new task. Interaction and optimization costs complete the comparison.

Reusing guidance and executable skills. Artifact ablations separate retained guidance from expanded motor competence. Withholding lessons tests fresh inference [46, 47]; restoring the original library tests newly acquired routines [50, 52]. Resetting a geometric archive tests acquired references [36]. Hold task instances, base executor, and attempt budgets fixed to isolate reuse from additional chances to succeed (Table B3).

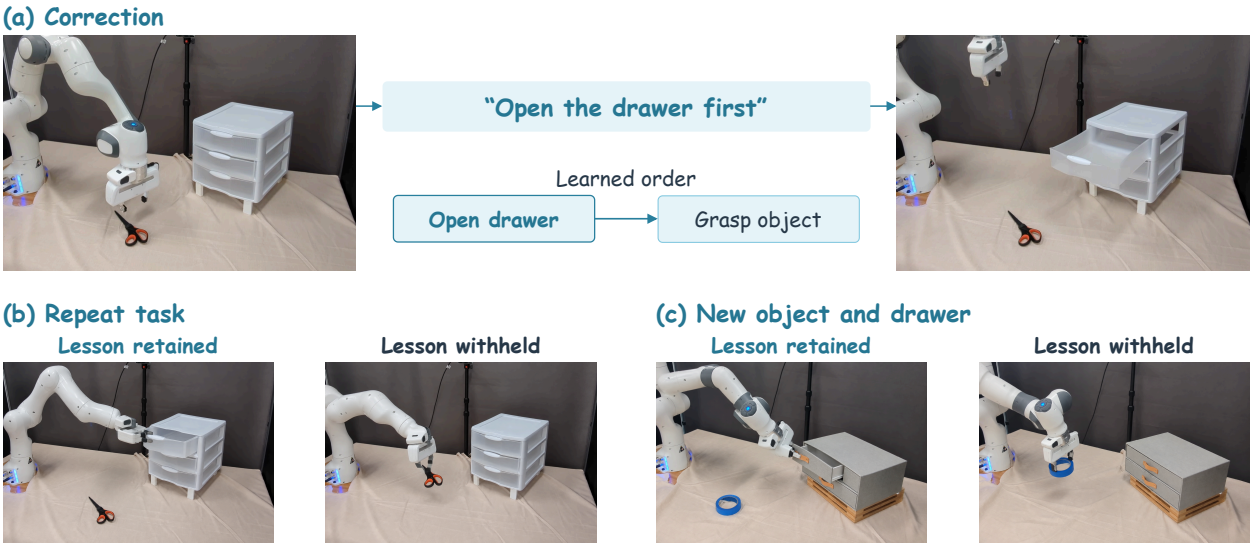

Figure 23 Testing the use of a learned ordering constraint. (a) Feedback establishes that the drawer must be opened before grasping the object. (b,c) Conceptual illustrations contrast following that order with premature grasping on repeated and new instances. Panel (a) follows the DROC example [46]; panels (b,c) propose candidate outcomes for retention tests. Success and further human corrections provide the readouts.

Acquiring new evidence. Active acquisition tests how useful context is obtained. VA-Bench varies camera control with a shared executor [343]; RoboQuest requires search, inspection, and physical tests before the agent presses a SUBMIT button [489]. Some probes irreversibly change the scene, coupling information gain to execution feasibility. Its scripted-oracle demonstrations have access to hidden task specifications, whereas evaluated agents must obtain that information. Comparing supplied evidence with agent-acquired evidence under matched action budgets separates interpretation from acquisition; final-state success tests whether either survives physical execution.

GiT supplies a complementary temporal test: identical current observations and instructions can refer to different targets after different event histories [473]. Its reported physical reference policy receives no history, establishing a reactive baseline. Comparing valid history swaps with a history-conditioned policy would test whether event order changes target selection. The input, its representation, and its effect on actions are separate stages of that test.

Matched retention and reuse tests. In Figure 23, feedback teaches a robot to open a drawer before reaching for scissors [46]. Panel (a) records that constraint; (b,c) propose retaining or removing it on repeated and substituted object–drawer instances. Opening first supplies an adherence readout alongside success and further corrections. MemTransfer varies initial pose, route availability, and history relevance under a shared frozen policy [436], testing when retained knowledge remains applicable.

Retention tests vary access to a lesson while holding instructions, initialization, base competence, and resources fixed. Separate scene resets remove physical progress; equal retry allowances control additional chances. A selective reset tests dependence conditional on other components, which may retain redundant evidence. Figure 6 locates the carriers; Table 17 specifies their resets. Undisclosed pretraining still leaves task novelty uncertain.

Acquisition consumes experience before reuse can save it. Separating play from downstream testing, as in RATs [52], and synthesis feedback from fresh validation [90, 218] distinguishes a candidate that worked once from knowledge that helps later. Comparing reuse gains with acquisition trials and corrections establishes whether total learning effort falls.

Teaching adherence and physical success at a common attempt budget characterize reuse; unsuccessful trials and variation across repetitions belong in the comparison. Physical diagnostics locate the cause of improvement.

7.3 Physical realization and learning across attempts

Channel and intermediate interventions trace contextual information into physical outcomes. Independent physical and memory resets then test whether retained evidence improves later attempts and recovery.

Removing language, vision, proprioception, or contact measurements under the same task, checkpoint, and history identifies the evidence supporting adaptation. Inputs must remain valid, with missing measurements distinguished from measured zeros. Goal completion and adherence can differ: transporting a cup does not establish that a taught handle grasp was preserved.

Intermediate substitutions localize failure by supplying a relevant demonstration segment, corrected correspondence, a task-consistent future, or a correct executable specification. Preserve the interface format and measure whether actions change in the intended direction before testing completion. TRUST makes this distinction concrete: an offline-trained value estimator reweights reasoning-token candidates in a frozen VLA [456]. On 1,509 paired DeepThinkVLA LIBERO-Plus rollouts, improved reasoning accompanies essentially unchanged success (78.86% without steering versus 78.26% with it). Reasoning substitution diagnoses whether an accurate statement reaches the action decoder. Target-selection and restored-precondition tests provide related controls [109, 110]. Such interventions locate where task interpretation diverges from physical execution.

Physical adaptation requires evidence about the changed condition. MemMimic separates event recall from unknown-response adaptation; PACE-Bench varies physics under a stable goal [105, 308]. The video-generation benchmark H2R-Bench tests goal, event, contact, and embodiment consistency before execution [314]. Probe-then-commit audits distinguish action-relevant belief coverage from failure prediction [410]. Watch, Infer, Coordinate instead infers a partner's physical constraints from joint demonstrations for CEM/MPC on new tasks [441]. Inferred, oracle, and capability-blind inputs to the same planner isolate constraint inference; behavioral cloning tests whether copying motion suffices.

Separate physical and memory resets distinguish continuing an unfinished task from reusing acquired evidence. Retaining either independently also controls gains from repeated chances to succeed.

For trial-memory adaptation, $\Delta_n^{\text{reuse}} = S_n^{\text{keep}} - S_n^{\text{reset}}$ compares the same success or adherence rate at attempt n with memory retained or cleared. Task distribution, physical initialization, executor, other artifacts, previous attempts, and execution budget remain fixed. Per-attempt rates isolate retention benefit; cumulative success also benefits from independent retries. A subsequent tool or dynamics change tests revision of obsolete physical knowledge. RobotUse's playbook comparison evaluates improvement on the same development tasks [470]; held-out acquisition tests would measure transfer to newly taught procedures.

Recovery evaluation separates added knowledge from additional chances. A frozen-VLA study evaluates learned correction against a plain rerun and a scripted retreat across 3,888 paired RoboTwin episodes [480]. The learned pipeline gains 13.5 percentage points on one of four tasks, with no detectable gain on three; 43.2% of failed base trials on the responsive task succeed on a plain rerun. A retreat comparison with no detected difference does not establish equivalence. Matched intervention timing, continuation budgets, and seed-cluster uncertainty are therefore needed to attribute repair gains. End-to-end success includes all tasks, while conditional repair success includes only detected eligible failures.

RoboRecover reconstructs deviations by replaying action prefixes before evaluating continuation under the original goal [398]. Its 2,000 RoboTwin and LIBERO scenarios distinguish resumption stage from physical deviation. Comparing retained and cleared history at that same reconstructed state, with equal continuation budgets, would extend this design from recovery competence to contextual recovery.

7.4 Teaching effort, response delay, and reuse cost

Contextual adaptation incurs acquisition, response, and reuse costs. Acquisition includes instruction preparation, demonstrations, exploration, and validation. Its *changeover time* runs from receiving a specification to satisfying a shared operating criterion; labor, interaction, and resets identify the resources consumed [89]. SAIL's reconstruction and simulated trials illustrate costs beyond inference [317]. Varying context independently of search budget separates information gains from additional computation [315, 316].

Separating preparation from execution makes these costs comparable. BiGym 2.0 coding agents use one three-view demonstration and bounded interaction to produce frozen programs without runtime LLM calls [479]. Its main nine-task comparison shares controller and observation/action interfaces with policies trained on 60 demonstrations per task. iAm.md's simulated TIAGo study instead uses robot descriptions, scene records, and live interfaces for code generation [510]. Reuse reduces interface rediscovery, yet the second task requires 49 edits and 94.5 minutes to verified completion: saved discovery does not eliminate debugging.

Response delay runs from informative feedback to the affected command. SimpleMemVLA reduces inference latency from 1.02 to 0.68 seconds without changing outputs [45]; ReSteer identifies where feedback can alter control [108]. Episode costs include 125.3 seconds of median LLM calls for BiCICLe's LiftBall [92]. TacZero's numerical-touch insertion averages 239 seconds waiting for APIs, 349 with motion, and 446 including overhead [477]. NavGPT-3 varies execution overlap and monitor priority to separate inference cadence from trigger-to-halt latency [509]. These intervals locate the bottleneck between evidence and intervention.

Reuse gains depend on recurrence and continued applicability. HumanoidTTT's 171.4× accepted-hit preparation gain becomes 16.4× end-to-end on a recurrence-rich 200-request trace [449]. Misses, qualification, and retention updates belong in total cost; varying request mix and entry conditions tests whether savings persist. The next chapter asks how retained experience improves subsequent learning.

Takeaways

① Context interventions establish dependence on task-defining evidence beyond familiar-scene recognition. Physical execution and held-out relation tests then establish how far that requirement transfers.

② Common task outcomes expose the contributions of training, supplied context, adaptive neural state, and retained knowledge. Acquisition and reuse carry different resource costs.

③ Teaching effort, feedback-to-action delay, and cost per accepted execution connect contextual learning to deployment value across repeated attempts.

8 Future Directions: What Limits Transfer and Improvement?

Broader contextual learning must infer unfamiliar rules, preserve binding physical constraints, revise outdated corrections, and improve subsequent learning. We connect these objectives to training, representation, and experience acquisition through behavioral tests.

8.1 From context dependence to a reusable learning rule

Section 2.1 separates familiar-task selection, composition, and unfamiliar-rule inference. Section 4.4 motivates training relationships that could generalize beyond familiar tasks.

Table 18 Proposed tests of behavioral transfer in a placement task. Keep available objects and motor actions matched; vary what teaching must supply. Novelty is relative to the controlled training exposure.

Learning claim	What is already familiar	What the example must establish
Familiar-task selection	Object-to-location assignments	Choose a known left- or right-bin routine
Compositional transfer	Placement relations; order constraints	Combine familiar color and order rules
Rule inference	Objects; executable placements	Infer an unfamiliar size-to-slot rule

Table 18 distinguishes selecting a familiar task, composing known relations, and inferring a new rule. An example can identify a routine without teaching a new relation: demonstration benefits under incorrect labels in some NLP settings expose the role of task-format recognition [320]. Conversely, function-learning studies show inference of unseen functions, and controlled regression connects task diversity to more general learning behavior [253, 321]. The illustrative robotic tests keep available objects and motor skills fixed while varying what teaching must establish: a familiar bin choice, a new composition of known constraints, or a size-to-slot mapping from a family excluded during training. Rule inference requires enough examples to distinguish

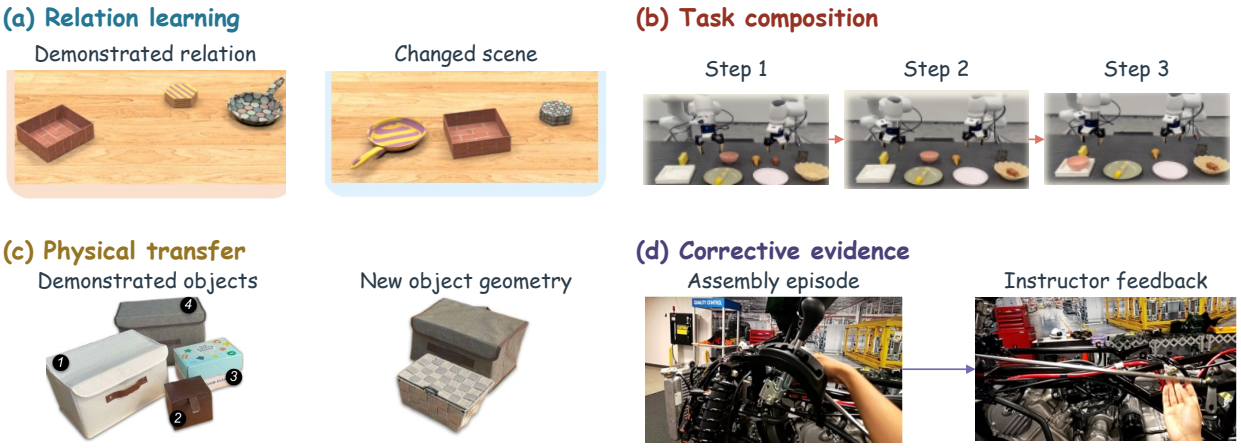

Figure 24 Four research directions: (a) infer taught relations, (b) compose task steps, (c) preserve function across physical changes, and (d) use corrective feedback. Together they extend contextual learning toward interpreting, transferring, and retaining task requirements.

the candidate mappings. BPP's continuous drawing tasks and the Imitator Game's functional substitutions provide concrete task variations from which to design these tests [54, 101].

The acquisition routes in Section 4.4 suggest that task-relation coverage and motor coverage contribute differently to transfer. Varying broad pretraining and grounded pairing independently, with executor and task information fixed, would test this hypothesis. Relation diversity, interaction experience, and model size are further factors; agentic systems add library size and acquisition cost [53]. Relation-level outcomes and continuous errors complement all-steps success, whose thresholds can obscure gradual gains [323].

Cross-modal transfer supplies a complementary test. GPT-Policy provides a concrete interface in which a general VLM receives teaching and acts through fixed robot tools [94]. Expressing the same unfamiliar rule through text, demonstrations, or both would test whether its inferred meaning transfers across formats. Matching task information and motor capabilities isolates rule inference from access to additional details, such as a procedural step visible in one format but absent from another.

Figure 24 makes these questions observable: (a) a taught relation determines actions in a new scene; (b) dependent steps test composition; (c) changed contact geometry tests preservation of function; and (d) corrective feedback revises an action. The following sections develop teaching, transfer, and retention requirements; Section 8.5 asks whether experience improves subsequent learning.

8.2 Instruction tuning for more complex contextual learning

MetaICL and Flan provide language-model precedents for example-based learning and instruction tuning [113, 324]. For ICL for robots, the training problem is to make new teaching select among feasible continuations within a controlled motor repertoire. The three cases below progress from combining requirements to revising them and resolving missing information.

Figure 25 illustrates these cases through separate kitting episodes with comparable parts and motor skills. Paired, continuous, and corrective records (Section 4.2) provide possible training formats; the intermediate tray states reveal which requirements survive execution.

Requirement binding and composition. Panel (a) combines a demonstrated plate-first order with a request for two plates and three bushings. Reversing the demonstrated order should reverse intermediate tray states while preserving the final inventory. The example separates *compositional complexity*, which varies skill order and dependencies, from *constraint complexity*, which adds jointly binding counts, contacts, or dwell times. Measuring each requirement and their joint satisfaction tests whether a new composition was followed through the method's intermediate representation.

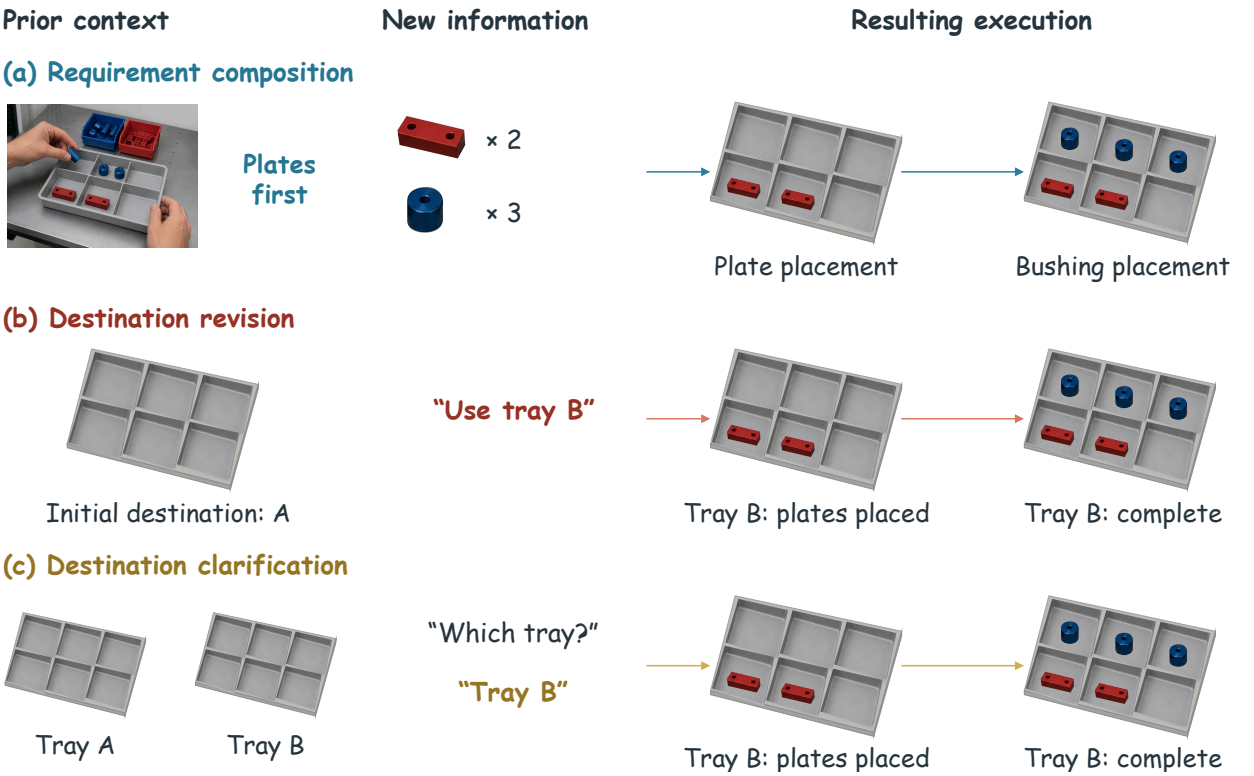

Figure 25 Illustrative kitting tasks for three contextual learning problems: (a) combine demonstrated order with requested quantities, (b) change the destination while retaining the order, and (c) resolve an ambiguous destination before acting. Intermediate tray states reveal adherence to teaching beyond the final inventory.

Selective revision of a procedure. Panel (b) changes the destination to tray B before placement, retaining the order and quantities. DROC's language guidance, RACER's corrective supervision, and RoboTTT's separation of failed context from corrective targets offer components for learning such revisions [10, 46, 87]. Training should reward the targeted change together with unaffected requirements. Varying correction timing, scope, and precedence then tests *revision complexity*: reaching tray B while losing the taught count is an overbroad revision. Intervening tasks test whether the revised procedure remains available later.

Decision-directed information acquisition. Panel (c) leaves both trays feasible without specifying which to use. Asking for the destination resolves this ambiguity before placement, building on question-based teaching and uncertainty-aware assistance [86, 301]. The acquired answer must preserve the already taught order and quantity. Other uncertainties call for different interventions: a view or probe can reveal execution conditions (Section 5), whereas a missing grasp skill requires an additional controller or motor learning. Behavioral Exploration offers a related action-based route: a policy uses interaction history to select expert-like behaviors that explore beyond previous attempts [332]. Clarification and exploration target different unknowns; training should connect each request or action to the information it supplies.

These cases motivate comparing action pretraining, demonstration-conditioned training, and mixed instruction–demonstration–correction training under a common backbone and experience budget. Withheld compositions and corrections test requirement adherence, physical completion, and later reuse. The next section examines which requirements should remain invariant when objects and execution conditions change.

8.3 Preserving taught relations across physical change

Cross-object transfer in Section 3.5 establishes useful relations when the corresponding parts and admissible substitutions can be identified. The remaining challenge is to infer which demonstrated details are binding and when no valid substitute exists. A handleless vessel may serve a pouring goal but cannot preserve an

explicitly required handle grasp. Similarly, a pause that lets adhesive set is part of the process, whereas an incidental pause may be shortened. These cases require selective invariance: retain the taught requirement while changing only details that its meaning permits. The part-decomposition study supplies cross-category part equivalencies [346]; learning to infer and validate such equivalencies would extend its transfer scope.

The crossed comparisons in Section 7.1 locate these limits by separating changed requirements from changed objects. Beyond successful substitution, a learner must recognize when the available objects cannot satisfy the demonstrated relation and acquire the missing information. Contrasting demonstrations or a clarification can establish whether a pause or contact is required; interaction can reveal whether a geometrically compatible replacement has unfamiliar resistance or dynamics [40, 88]. Correspondence and physical feedback therefore answer different questions: where an interaction should occur, and whether that interaction realizes the intended effect.

The available executor sets the boundary of this transfer. If the revised requirement can be expressed through existing actions or skills, contextual inference can select a new realization. If the necessary motion lies outside that repertoire, further motor learning contributes a separate stage, as in human-video-initialized WHIRL [150]. Such a hybrid process can preserve a context-inferred task relation while learning its physical realization.

Navigation introduces a related separation between reusable strategy and local knowledge. An exploration strategy can transfer across buildings, while a remembered object location is tied to one layout. ReLIC [203] illustrates the value of retaining such local experience across goals. Broader transfer requires identifying which spatial facts remain valid when the environment changes, alongside preserving the task relations considered above.

8.4 Retaining teaching while revising obsolete experience

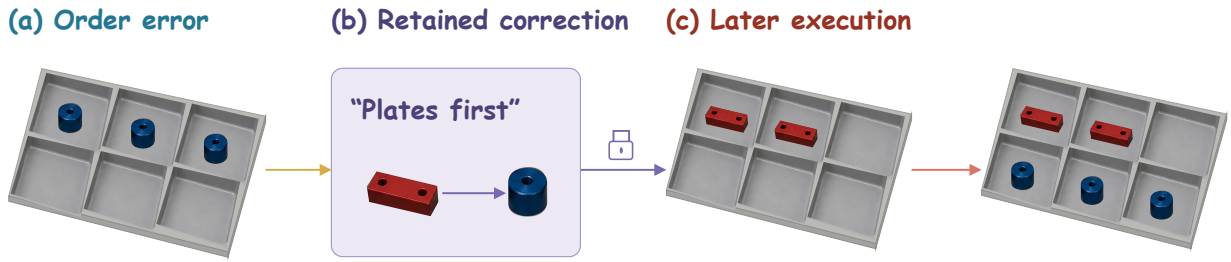

Figure 26 Illustrative reuse of corrective context: (a) an order error, (b) plate-first feedback, and (c) execution in a new layout with fixed policy weights. The intermediate tray makes retention of the taught order observable before the kit is complete.

Physical transfer preserves a requirement across changes in its realization; temporal reuse must also accommodate new teaching. Video outlines and lower-cost planning latents retain different details [31, 325]. A later correction can make an initially incidental contact or pause decisive, after a task-specific summary has discarded it. Delayed corrections can test what new teaching compact summaries and retrievable source segments support at matched storage and access costs. Preserving current task meaning and enabling future reinterpretation impose different requirements on retained evidence.

Consolidation into weights adds training after contextual acquisition. The $\pi_{0.7}$ coaching example and the RECAP training framework for $\pi^*_{0.6}$ connect interaction or correction to subsequent training [273, 281]. Immediate context-driven behavior and persistence after consolidation or reset characterize different stages of acquisition. Retained programs, records, or policy weights can help later tasks, but obsolete corrections can also interfere with new teaching.

A taught requirement can remain useful after the scene that elicited it has changed. In the kitting example, retained plate-first feedback supplies information absent from the new layout, illustrating how a fixed policy could combine an earlier correction with current geometry (Figure 26). Keeping, omitting, or replacing that

correction tests whether the later decision depends on the acquired lesson. Retention must also survive the representation changes used to make such lessons cheaper to store and apply.

Retained context must remain compatible with both the environment and the interpreter [127]. Robo-COP's memory revision after policy adoption or rollback illustrates why: earlier failure guidance can become obsolete when capabilities change [485]. A successor test should vary stored guidance and executor version independently, measuring whether valid teaching remains effective and outdated restrictions can be replaced. Section 7.4 adds the cost of this revalidation to the benefit of reuse.

8.5 Physical recursive self-improvement

Physical feedback can improve a solution or the process for acquiring future solutions. Retained context, program revision, and neural updates serve this second objective when they make subsequent teaching more effective or less costly. S5 names this objective; S6 adds knowledge exchange across robots.

Table 19 Feedback-driven improvement through update target, operation, and reusable output. Targets can overlap; successor comparisons measure their contribution to subsequent learning. ENPIRE and REVOLVE denote their program and policy branches, respectively.

Update target	Operation	Reusable output	Examples
Interaction context	Retain	Action–effect memory	RopeFormer [363]; Zeva-Ego [362]
Executable artifacts	Revise	Skills; programs	RoboRSI [347]; ENPIRE [89]; HarnessPAI [396]
Neural components	Fit; retrain	Dynamics; policy	MetaPusher [367]; REVOLVE [344]
Acquisition process	Refine	Teaching; training procedure	LMPC [117]; Eureka [115]; DrEureka [116]

Table 19 separates storage from learning benefit. RopeFormer and Zeva-Ego retain action–response history for fixed-policy adaptation [362, 363]; programs retain executable solutions; neural updates change predictive or motor competence. Their contribution to physical recursive self-improvement depends on whether subsequent acquisition or verification improves.

Program improvement can supply data for subsequent model improvement. HarnessPAI uses validated program rollouts to fine-tune the action model; its published sequence ends at policy refinement, before a further round of harness evolution around the improved checkpoint [396]. TraceFlow instead grows an external trace bank with weights fixed, and its reported stacking gains peak before the final round [390]. These examples motivate measuring the successor's acquisition rate on new tasks and the persistence of gains across rounds, beyond demonstrating a stronger current solver.

Learning to become more teachable. LMPC links accumulated teaching to improved subsequent acquisition [117]. Within a session, a fixed language model converts instructions and corrections into reward code for model-predictive control. Across sessions, successful interactions train it to predict teaching exchanges; the rollout variant searches those exchanges for a shorter route to success. Test-task success rises from 39.4% to 66.3%, while mean turns in successful sessions fall from 2.4 to 1.9. This connects experience to a successor's response to new teaching.

For action policies and VLM agents, the corresponding comparison holds new tasks, motor interfaces, and interaction budgets fixed. Evaluating each successor with and without informative context separates improved use of teaching from stronger prior execution. Familiar tasks and fully specified targets measure execution competence; withheld requirements and matched teaching measure acquisition. This extends the controls in Section 7.2 across successive versions of a learner.

Improving execution, supervision, and acquisition. Retained guidance and context-building code can improve selection or interpretation of future evidence [47, 49]. Autonomous practice can improve resetting, execution, or a successor policy [114, 141]. RoboRSI's successive skill revisions exemplify repertoire growth; its release-level cumulative coverage records tasks solved at least once [347]. REVOLVE updates two distinct

components: failure and correction records retrain the manipulation policy, while mismatch memory refines the supervising agent's judgments [344]. Selective resets and successor comparisons attribute gains to the retained solution, control, or supervision. A 123-round skill-discovery study exposes the converse: accepted local revisions need not produce end-to-end success when early failures starve later stages of evidence or verification measures the wrong outcome [431]. Improvement must therefore be demonstrated on the composed task, not counted from accepted edits.

Predictive components offer further update locations. Q-Planning trains a value function on deployment experience while retaining the action policy [208]; SILVR trains a visual planner on its executions [326]; joint refinement changes both policies and the models that assess or train them [327–329]. These loops improve the control and training signals available to contextual learning. Their contribution to teachability is measured by how the resulting successor responds to new evidence, alongside its execution success.

Changing the acquisition procedure makes the recursive objective explicit. Eureka revises reward code from policy-training feedback; DrEureka also configures domain randomization for transfer [115, 116]. ENPIRE extends agent-directed experimentation to physical environments, revising programs, data use, and learning algorithms through established interfaces [89]. Retaining effective learning choices together with their applicable conditions offers a route to improving future acquisition. Successor comparisons measure this benefit alongside repertoire growth. Evidence from simulated practice and physical experiments characterizes the operating conditions of this improvement.

Making the next experience informative. An improving learner can also change what it asks, observes, or tests. EnvGen, OMNI-EPIC, and SimWorld Studio use learner performance to shape subsequent simulated experience [261, 330, 331]. For ICL, teaching–execution pairs near interpretation failures could target conflicting examples, unfamiliar compositions, or corrections with a limited scope. Physical trials then test whether the inferred requirement survives contact and embodiment changes.

A failed insertion makes the distinction concrete. An ambiguous target calls for clarification; an uncertain contact response calls for a diagnostic motion. Retaining when each intervention resolved uncertainty could reduce unnecessary demonstrations and trials on later assemblies. The improvement loop then links requirement inference, evidence acquisition, physical verification, and reuse of an acquisition strategy.

Collective knowledge evolution. S6 extends the learning cycle to exchanges between robots (Figure 1). An arm can demonstrate a placement relation, a humanoid realize it under different kinematic constraints, and the receiver return a verified execution or correction. The exchanged lesson preserves task order and object relations while allowing grasps and balance control to change.

Shared progress and skill representations preserve task content across appearances [29, 30, 79]; multi-robot training supplies compatible competence [57, 132]; morphology interpolation and kinematic constraints support physical realization [131, 133]. Collective knowledge evolution adds a reciprocal loop: each execution helps revise knowledge another robot can use. Measuring transfer in both directions, with the teaching and practice required by each body, tests whether the exchange improves the group's subsequent learning.

Requirement adherence, correction efficiency, and transfer measure these learning gains; interaction, intervention, and training measure their cost.

Takeaways

① Diversity in what examples teach, motion coverage, and model scale address different limits. Learning unfamiliar combinations and relation families requires more than recognizing new instances.

② Composition, selective correction, and information acquisition expand what an episode can teach over an available motor repertoire.

③ Retained corrections should change later decisions through context and remain applicable when layouts change. Across longer learning cycles, the central outcome is a successor that learns more effectively from new teaching.

9 Conclusion

ICL for robots connects reusable competence to task-specific requirements through teaching and interaction, with deployed neural parameters fixed. Four interfaces carry this evidence into action: contextual action inference, geometric reference transfer, predicted consequences, and skill or program execution. Their benefits depend on compatible representations, transferable geometry, realizable predictions, and capable executors. Correspondence and memory bind the evidence across situations and time.

Useful context must preserve the distinction that changes execution: a task requirement, an earlier event, or a physical condition. Retention alone is insufficient if evidence cannot reach a supported action or arrive in time to revise it. Across manipulation and navigation, evaluation therefore tests whether teaching redirects behavior, its requirements survive physical change, and retained knowledge benefits later attempts. Acquisition cost determines the practical value of these gains.

The next challenge is to infer unfamiliar combinations of requirements, revise affected constraints selectively, and replace obsolete experience. Physical recursive self-improvement asks whether the successor learns new tasks more effectively; collective knowledge evolution extends this objective to verified exchanges between robots. Both compare subsequent learning with its teaching, interaction, and training costs.

Image attribution. Figures combine original vectors, AI-assisted illustrations, and cited task imagery. AI-assisted scenes appear in Figures 1, 10, 20, 23, 25, and 26; the S6 robot illustrations in Figures 1 and 3 are original vector drawings. Source motion and task annotations are preserved.

Method imagery comes from Vid2Robot, SemAnCorr, Zero-WAM, and UniSkill [14, 24, 30, 38]. Additional examples come from ICRT, MemoryVLA, WAM-TTT, DROC, MT3, and CorrectVLA [11, 13, 36, 42, 46, 142]; research examples use VIMA, Instant Policy, and HoloAssist [16, 73, 100]. Acquisition examples use DROID, UMI, Ego2Robot, and MimicGen [58, 62, 74, 222]. Other recordings come from Galaxea Open-World [61], LIBERO-PRO [106], GPT-Policy [94], and HumanAndRobot.

Vid2Robot imagery is adapted under CC BY-SA 4.0; SemAnCorr, Zero-WAM, UniSkill, LIBERO-PRO, VIMA, Instant Policy, HoloAssist, and DROID imagery under CC BY 4.0; and Galaxea imagery under CC BY-NC-SA 4.0. Editable sources include image provenance and institution-mark sources.

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A Complete Reference Index

Table A1 maps the bibliography to Figure 9 and the accompanying reference data.

Table A1 Reference index for Figure 9, grouped by first public year. Method families: C, context-conditioned policies; G, geometric transfer; W, world-model-based control; S, skill/agent execution. Other groups record data, evaluation, foundations, and training references.

Year	Family	Count	References
2016	C	1	[2]
2017	C	4	[1, 3, 7, 146]
	S	1	[26]
2018	C	4	[8, 157, 354, 355]
	W	2	[179, 294]
	S	1	[357]
2019	C	4	[4, 147, 260, 333]
2020	C	5	[158, 196, 297, 300, 356]
	G	1	[32]
2021	C	3	[12, 39, 149]
	G	1	[422]
2022	C	6	[5, 6, 100, 148, 197, 276]
	G	2	[33, 345]
	W	1	[19]
	S	5	[27, 28, 185, 186, 358]
2023	C	6	[57, 114, 202, 243, 274, 277]
	G	6	[34, 169–171, 420, 421]
	W	1	[20]
	S	10	[29, 46, 84, 86, 96, 164, 184, 251, 359, 425]

Table A1 Complete reference assignments (continued).

Year	Family	Count	References
2024	C	16	[13–16, 80, 118, 119, 132, 153, 154, 189, 203, 275, 278, 279, 350]
	G	4	[35, 156, 173, 295]
	S	13	[50, 76, 87, 117, 165, 213–215, 298, 351, 352, 424, 433]
2025	C	19	[17, 40, 42, 43, 61, 120, 155, 159, 190, 193, 201, 235, 245, 273, 280, 282, 315, 332, 416]
	G	8	[36, 37, 93, 168, 172, 174, 177, 191]
	W	3	[21, 326, 423]
	S	5	[9, 30, 51, 293, 305]
2026	C	75	[10, 18, 41, 44, 45, 54, 55, 77, 88, 91, 92, 97–99, 105, 160–163, 176, 181–183, 195, 198–200, 204, 205, 208, 233, 244, 247, 270, 281, 283, 292, 299, 317, 325, 327, 328, 338, 339, 341, 362, 363, 379, 384, 385, 387, 389–391, 395, 402, 412, 414, 417–419, 444, 445, 458, 469, 471, 475, 476, 478, 482, 491, 492, 497, 507, 508]
	G	7	[38, 133, 175, 346, 403, 472, 503]
	W	27	[11, 22–25, 152, 209, 302, 316, 329, 334, 335, 366, 367, 399, 400, 442, 448, 459, 460, 467, 483, 490, 496, 499, 506, 512]
	S	93	[31, 47–49, 52, 53, 89, 90, 94, 142, 166, 187, 206, 207, 210–212, 216–220, 262, 272, 290, 291, 303, 304, 307, 336, 337, 340, 342–344, 347–349, 360, 361, 364, 375, 377, 378, 380–382, 386, 388, 392–394, 396, 401, 404–408, 411, 413, 415, 426–431, 438–440, 447, 449, 456, 461–463, 465, 466, 470, 474, 477, 484–487, 493, 494, 498, 502, 509–511]
Data resources			
		46	[56, 58–60, 62–75, 221–226, 237–242, 246, 248–250, 255–259, 261, 263, 330, 331, 368, 451, 455]
Evaluation			
		54	[81–83, 85, 95, 101–104, 106–110, 151, 227–230, 284, 296, 306, 308–314, 353, 365, 373, 374, 376, 383, 397, 398, 410, 436, 437, 441, 443, 446, 454, 468, 473, 479–481, 488, 489, 500, 504, 505]
Foundations and surveys			
		41	[111–113, 121–130, 134–138, 143–145, 178, 180, 194, 231, 232, 234, 253, 301, 318–323, 369–372, 450, 501]
Training and representations			
		37	[78, 79, 115, 116, 131, 139–141, 150, 167, 188, 192, 236, 252, 254, 264–269, 271, 285–289, 324, 409, 432, 434, 435, 452, 453, 457, 464, 495]
Total		**512**	

B Method and Resource Comparisons

Table B1 Correspondence interfaces: matched evidence conditions actions or specifies an execution reference.

Method	Correspondence	Control interface
Correspondence conditions or corrects action generation		
MatchingPolicy [161]	Demonstration–query 3D points	Graph-conditioned gripper motion
RTCF [195]	Current and recorded progress	Bounded correction
Correspondence specifies an executable motion or constraint		
RoboTAP [169]	Stage-specific points	Visual-servo reference tracking
FUNCTO [172]	Grasp; function; center	Function-preserving tool motion
MAGIC [173]	Shape; contact geometry	Retarget; simulate
SemAnCorr [38]	Surfaces; local frames	Grasp; relative motion
Sparse Meets Dense [175]	Changing contact pairs	Constraint; pose updates

Table B2 Memory mechanisms by retained evidence, access, and update. Groups (a)–(d) use fixed neural weights; (e) separates transient fast weights, persistent adaptation, and learned maintenance. Fixed banks admit no new records.

Method	Retained evidence	Access and update
(a) Demonstration records		
KAT [15]	Keypoint–action examples	Load before rollout
ICRT [13]	Demonstrations; own trajectory	Read prefix; append execution
RICL [17]	Retrieved demonstrations	Query-dependent selection
ReCAP [152]	Trajectory retrieval pool	Extend pool; retrieve each step
RTCF [195]	Successful trajectories	Phase retrieval; fixed test bank
TraceFlow [390]	Success and failure traces	Progress-aligned guidance; rollout admission
RV-ICL [465]	Task; phase; event; clip index	Read-only hierarchical access
(b) Interaction histories and inferred state		
RMA [39]	Recent state–action history	Online dynamics estimation
AMAGO [202]	Trial history and feedback	Append across episodes
ReLIC (navigation) [203]	Multi-episode visual history	Attend across trials
LocoFormer [40]	Recurrent rollout state	Carry across trials
ICWM [41]	Action–effect probes	Probe before execution
(c) Execution memory		
Divide-and-Remember [444]	Selected visual tokens	Recursive $2K \to K$ selection
ECoMEM [445]	Entity; state; event records	Fixed Writer; trained Reader
ECROM [447]	Surface/view evidence; visibility	Opportunity-conditioned belief update
PMTRM [508]	Bounded state–action queue	Re-encode; fixed encoder
MemoryVLA [42]	Perceptual; cognitive tokens	Retrieve; merge
MemER [43]	Historical visual keyframes	Retrieve; predict subtask
MEM [44]	Recent views; semantic history	Recent and long-term
ReMem-VLA [198]	Recurrent memory queries	Frame and chunk updates
MemoryVLA++ [209]	Memory bank; imagined future	Retrieve; predict
μVLA [199]	Recurrent memory tokens	Carry state; reset per episode
EvoMem-VLA [469]	Observed state-change tokens	Ordered frame pairs; episode reset
MarvisNav [466]	Exploration map; route history	Render onto route candidates
EventVLA [200]	Anchors; event keyframes	Event-triggered writes
SimpleMemVLA [45]	Timestamped video window	Resample per decision
AGM [206]	Verified subgoal achievements	Completion-verified update
Workspace Models [395]	Latent history tokens	Saliency-supervised compression
Mimir; OCC4M [392, 393]	Task state; persistent entities	Goal binding; feedback-supported updates
2AM [401]	Agent-side interaction history	Compile current intent into 2D cues
(d) Reusable guidance		
DROC [46]	Distilled correction knowledge	Feedback update; retrieval
ProactiveVLA [474]	Verified exploration rules	Consolidate; freeze for evaluation
RobotUse [470]	Local histories; shared playbook	Summarize locally; retain across attempts
(e) Parameter adaptation and learned memory maintenance		
TaskAnchor [507]	Visual history buffer	Rebuild fast weights; discard
PACE [475]	Transition fast weights	Online gradient; episode reset
SALT [476]	Prior plan tail; projection scales	Gated gradient; episode reset
TTT-RM [471]	Explicit history; residual fast weights	Online gradient; episode reset
RoboTTT [10]	Video and rollout history	Sequential weight updates
WAM-TTT [11]	Human demonstration video	Adapt; freeze
HumanoidTTT [449]	Qualified complete motions	Online learned retention; fixed tracker

Table B3 External-artifact revisions with fixed weights. Bullets: updates; dashes: undocumented. Plan: procedure; guidance: lessons; skills: routines; program: code/settings; oversight: context/verification. ZS: zero-shot video.

System and branch	Plan	Guidance	Skills	Program	Oversight
(a) Constructing and revising the current procedure					
Inner Monologue [186]	•	–	–	–	–
Code as Policies [27]	–	–	•	•	–
ProgPrompt [28]	–	–	–	•	–
VoxPoser (fixed dynamics) [184]	•	–	–	•	–
PRoC3S [214]	–	–	–	•	–
MALMM [213]	•	–	–	•	–
Show-Harness (video, ZS) [31]	•	–	–	–	–
GPT-Policy [94]	–	–	–	•	–
ICPI [9]	–	–	–	•	–
AOR [218]	–	–	–	•	–
ENPIRE (code branch) [89]	–	–	–	•	–
(b) Acquiring examples, guidance, and reusable behavior					
DROC [46]	•	•	–	•	–
ICAL (TEACh, retrieval) [215]	•	•	–	•	–
LRLL [50]	–	–	•	•	•
LYRA [51]	–	–	•	•	–
Uni-Skill (deployment) [217]	–	•	•	•	–
CaP-Agent0 (CaP-X) [166]	–	–	•	•	–
RATs [52]	•	•	•	•	–
RoboRSI (code skills) [347]	•	–	•	•	–
RPG [439]	–	•	•	•	–
InterEvolve [440]	–	–	•	•	–
ASPIRE [47]	–	•	–	•	–
ARCHITECT [48]	–	•	–	•	–
Teach and Grow (pilot) [53]	–	–	•	–	–
Harness VLA (memory) [207]	•	•	–	–	–
Hi-VLA (episode summary) [219]	–	•	–	–	–
(c) Revising context construction and execution oversight					
SHAPER [49]	–	•	–	–	•
Zetta [90]	–	•	•	–	•

Table B4 Data sources and supervision. Human → robot: synthesized views; Est., Sim., Synth.: estimated, simulated, synthesized.

Source	Visual record	Motion supervision	Learning use
Robot and cross-domain recordings			
Open X-Emb. [57]	Robot	Robot control	Robot action targets
DROID [58]	Robot	Robot control	Robot action targets
AgiBot World [60]	Robot	Robot control	Stages; corrections
BridgeData V2 [56]	Robot	Robot control	Robot action targets
RoboSet [243]	Robot	Robot control	Task groups
PrimeBot, XR-2 [244]	Human + robot	Robot; gripper	Separate corpora
RH20T [59]	Human + robot	Robot control	Human–robot pairs
ICRT data [13]	Robot	Robot control	Robot demonstrations
MimicPlay [251]	Human + robot	Robot control	Separate play
Instrumented human recordings			
UMI [62]	Human	Gripper motion	Motion correspondence
YUBI [66]	Human	Gripper motion	Motion correspondence

Table B4 Data resources (continued).

Source	Visual record	Motion supervision	Learning use
HiFi-UMI [223]	Human	Gripper motion	Motion correspondence
ACE-Data-0 [250]	Human	Human motion	Contact; procedure
Human activity video			
Ego4D [68]	Human	No robot actions	Behavior pretraining
Ego-Exo4D [70]	Human	No robot actions	Cross-view
EPIC-KITCHENS-100 [69]	Human	No robot actions	Behavior pretraining
Something-Something [71]	Human	No robot actions	Action clips
EgoVerse [248]	Human	Est. hand motion	Behavior pretraining
Open-AoE [249]	Human	Est. hand motion	Action segments
Synthesis			
MimicGen [74]	Robot	Sim. control	Query synthesis
HumanGen [24]	Human + robot	Robot control	Synth. support
Ego2Robot [222]	Human → robot	Retargeted motion	Synth. query

C Related Survey Perspectives

Table C1 Related surveys by scope, organizing axes, and emphasis. LfD denotes learning from demonstration.

Review focus	Organizing axes	Analytical emphasis
Demonstration and contextual adaptation		
LfD (2009) [122]	Teacher; problem space; policy	Acquiring examples and deriving policies
LfD (2020) [123]	Teaching; policy/reward/plan	Matching teaching and learning to the task
In-context learning [124]	Training, prompting, mechanisms	Examples as language-model context
In-context RL [125]	Pretraining; context; models	History-based adaptation with fixed weights
Nonstationary ICRL [127]	Shift type; timing; evidence	Validity of past evidence after changes
Manipulation ICL [371]	Input; target; update; transfer	Examples and priors for manipulation
Data and embodiment		
Human-video learning [126]	Task; observations; actions	Acquiring skills from video-derived evidence
VLM-based VLA models [372]	Monolithic/hierarchical models	From multimodal features to actions
VLA data infrastructure [129]	Data, benchmarks, data engines	Data and evaluation for robot learning
The embodiment gap [130]	Shared structure; adaptation stage	Reuse and robot-specific requirements
Predictive control and reusable skills		
World models for robotics [128]	Coupling; simulation; form	Prediction for learning and control
World Action Models [178]	Coupling; conditioning; decoding	Coupling predicted states and actions
World-action control [370]	Dynamics; architecture; training	Predictive robot control
Weights or Skills? [369]	Parameters; skills; revision	Capability representation and reuse
ICL for robots (this survey)	Evidence-to-action; transfer	Teaching; physical transfer; reuse